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Nonlocal Elastic Unloading as a Mechanism for Breakdown Work Scaling in Laboratory Earthquakes --Manuscript Draft--

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Abstract:	Understanding how earthquake energy budgets scale from laboratory to natural faults remains a fundamental challenge in earthquake physics. Laboratory experiments report fracture energies consistent with classical friction, yet seismological observations show that breakdown work scales strongly with slip, and the relationship between these two quantities remains unclear. Here we show that this discrepancy reflects two genuinely different physical quantities: the near-front dissipation governed by rupture-tip processes, which we use as a proxy for fracture energy because these small ruptures sustain no K-dominant field, and the system-scale breakdown work dominated by nonlocal elastic unloading which includes the loading column in our experiments. We conducted stick-slip experiments on 15-cm long granite faults under normal stresses of 1-20 MPa in a double-direct-shear configuration. A dense strain-gauge array tracked rupture propagation, enabling near-front dissipation estimates via cohesive zone modeling, while finite element simulations quantified energy partitioning between the fault and loading system. The near-front dissipation remains low (< 10 J/m²) and scales roughly linearly with normal stress, consistent with classical friction. In contrast, total breakdown work scales quadratically with slip, exceeding the near-front dissipation by over an order of magnitude. Finite element simulations confirm that elastic unloading of the compliant loading column dominates this energy release. Normal stress heterogeneity on the lab fault produces additional weakening, after the rupture has reached maximum size, without additional dissipative mechanisms. These results demonstrate that apparent breakdown work scaling in laboratory earthquakes can arise from nonlocal elastic effects rather than intrinsic fault weakening. Our work provides a framework for distinguishing local near-front dissipation from system-scale energy release and breakdown work when extrapolating laboratory results to natural faults.
Response to Reviewers:	Please see the attached point-by-point response to reviewers (uploaded as "Revision Notes") for our detailed replies to all editor and reviewer comments.

Response to Reviewers

Manuscript EPSL-D-26-00315

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Laboratory Earthquakes

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Chun-Yu Ke, Gauss Chang, Chris Marone, Dave Kammer, and Greg McLaskey

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We thank the editor and both reviewers for their careful reading of the manuscript and for their constructive suggestions. These comments have substantially improved the clarity and rigor of the paper. We have addressed all comments point by point below. Reviewer comments are reproduced verbatim in shaded blue boxes. Our responses follow each comment in plain text. Where the manuscript has been revised, the new text is shown in a green box. All line numbers refer to the revised manuscript.

Editor

The reviewers provide several suggestions for improvement which are very clearly presented and I shall not repeat them here. I notice in particular that they recommend you to use a more strict outline of the paper, and that several topics require more detailed discussion. All suggestions for improvement, should be considered for the revision.

Response:

We thank the Editor for the constructive handling of the manuscript. We have addressed the reviewers' suggestions in full, including the two themes the Editor highlights. To provide a stricter outline, we have reorganized the manuscript for a clearer separation of Methods, Results, and Discussion, with Results paragraphs now opening with topic sentences and speculative material moved out of the Conclusions. To address the topics that required more detailed treatment, we have expanded the cohesive-zone model description in the Methods with governing equations and an example fit, added a Discussion paragraph on the thermal energy budget, and revised the discussion of how the experimental scaling relates to natural faults. The figures have been revised throughout for clarity, and a table of all experiments has been added to the Supplementary Material. A point-by-point response to each comment follows below.

Reviewer 1

Summary:

This manuscript investigates scaling relationship between fracture energy and breakdown work with normal stress and slip during stick-slip experiments at the decimeter scale. The authors report both experiments and numerical models that resolve frictional breakdown and elastic unloading. They conclude that fracture energy increases linearly with normal stress whereas breakdown work increases quadratically with normal stress indicating distinct breakdown processes at the rupture tip vs. non-local, system-scale unloading after the frictional breakdown process.

The manuscript describes an interesting series of lab tests that included an innovative strain gage design to track processes during rupture propagation and stick-slip. The laboratory observations and numerical models provide new insight into small scale processes and discuss implications for earthquake energy budgets. The paper is reasonably well written and the lab data is excellent. I believe this work should be published after major revisions.

Response:

We thank Reviewer 1 for the accurate and concise summary of our work, and for the positive assessment and careful reading of the manuscript. We have addressed all detailed comments and suggestions below and believe the revised manuscript is substantially clearer in structure and presentation.

Suggestions for improvements:

Comment R1-S1.

The quality of the manuscript could be significantly improved by clearer structure such as a more careful separation of Method, Results and Discussion. The Method, especially the fracture mechanics model, is insufficiently described. Some results are introduced in the Method section and then repeated. The presentation of the equations should be improved by added labels and moving each equation to a separate line. The Conclusion includes some speculation that is not directly supported by the lab tests. More details are below.

Response:

We thank the reviewer for this overview. Each point is addressed in detail in the responses below: the separation of Methods, Results, and Discussion (Comments R1-5 and R1-9), the expanded description of the fracture mechanics model (R1-9 and R1-12), the removal of results repeated from the Methods (R1-5 and R1-17), the formatting of equations on separate numbered lines (R1-9 and R1-14), and the speculative statements in the Conclusions (R1-26). We note one deliberate exception to a strict separation: the finite-element model

and its results are presented together in the Results and Discussion, because the analysis is an interpretive tool whose setup and outcomes are best understood together.

Detailed Comments:

Comment R1-1.

L 88: A little more explanation on the synchronization would be useful. Given that ruptures lasted for only a fraction of a millisecond, this synchronization would have to be extremely accurate. What was the error in the manual synchronization procedure?

Response:

We thank the reviewer for raising this point. We agree that clarification is needed. The key point is that precise cross-system synchronization is not required, because the two recording systems extract independent physical quantities. The high-speed strain gage signals are used exclusively for rupture speed estimation and cohesive zone model fitting, quantities derived entirely from the relative timing and waveform shapes within that system alone. The slow-sampling channels (normal load, shear load, fault-parallel displacement) yield system-level quantities such as total coseismic slip, stress drop, and loading history, which evolve on timescales far longer than the rupture. Because the interseismic (stick) period between events is many orders of magnitude longer than the rupture duration, individual events are unambiguously identified in both systems by manual inspection. The two systems therefore need not be synchronized to sub-millisecond precision, and we have added text to the manuscript (around L 97) to make this explicit.

Revised text:

... while dynamic strains were recorded separately at 100 kHz ~~and manually synchronized~~. The two systems measure independent quantities. The slow-sampling channels characterize system-level slip and stress while the strain gage array captures the dynamic rupture waveforms used for rupture speed and cohesive zone model fitting. Because the interseismic period between events is orders of magnitude longer than the rupture duration, events are unambiguously identified in both recording systems.

Comment R1-2.

Figure 1 b) explain the different colors. c) What are the horizontal, dotted lines. Why did the first strain gage not show an evidence of rupture? How were the green circles determined?

Response:

We thank the reviewer for these questions. We have revised the caption of Figure 1 to address these points. (b) Gray shading denotes the initial run-in phase, blue shading denotes velocity-step tests (VS_i), and orange shading denotes slide-hold-slide tests (SHS_i). (c) The

horizontal dotted lines mark the zero strain change for each gauge trace, and the horizontal axis is time relative to the event onset. The topmost gauge did not record a rupture signal because the rosette was likely too close to the top edge of the fault and was not in full contact with the fault surface. The green circles mark the onset of rupture at each gauge location; the green dashed line is fitted to these circles, and its slope gives the average rupture propagation speed.

Revised text:

Figure 1. Double-direct shear experiment configuration. (a) Three Barre Gray granite blocks (two side blocks and one center block) are arranged under biaxial loading with bottom support for the two side blocks. Shear load is applied via an interchangeable loading block as a spring at the top of the center block. The right side block is instrumented with 13 strain gauge rosettes along the 150 mm fault. Blue square near the top of the center block marks the location of center block displacement δ measured relative to the bottom of the load frame. (b) Experimental protocol ~~includes an initial run-in phase, followed by alternating tests of velocity steps (VS_i) conducted with a stiff loading spring (steel) and slide-hold-slides (SHS_i) with a compliant loading spring (acrylic), in which.~~ Gray shading indicates the initial run-in phase in which the fault surfaces are conditioned prior to testing, blue shading indicates velocity-step tests (VS_i , stiff steel loading spring), and orange shading indicates slide-hold-slide tests (SHS_i , compliant acrylic loading spring). $\mu = \bar{\tau}/\bar{\sigma}_N$ is the friction coefficient and δ is center block displacement. (c) ~~Rupture speed is estimated through fitting a line to the time of rupture onset tracked from the shear strain ε_{xy} along fault.~~ Dynamic shear strain ε_{xy} recorded at each strain gauge rosette along the fault, plotted as a function of time relative to the event onset. Horizontal dotted lines mark the zero strain change for each gauge trace. The topmost gauge did not record a rupture signal, likely because the rosette was too close to the top edge of the fault and was not in full contact with the fault surface. Green circles mark the onset of rupture at each gauge location; the green dashed line is fitted to these circles, and its slope gives the average rupture propagation speed. (d) A picture of the right side block, aligned to the strain gauge rosette positions shown in (c).

Comment R1-3.

Figure 2a-i: Please mark Δ_{τ_total} , δ_0 , and δ_{min} . How were these quantities determined? What were the errors in stress and slip estimates?

Response:

We thank the reviewer for these questions, which we address in turn.

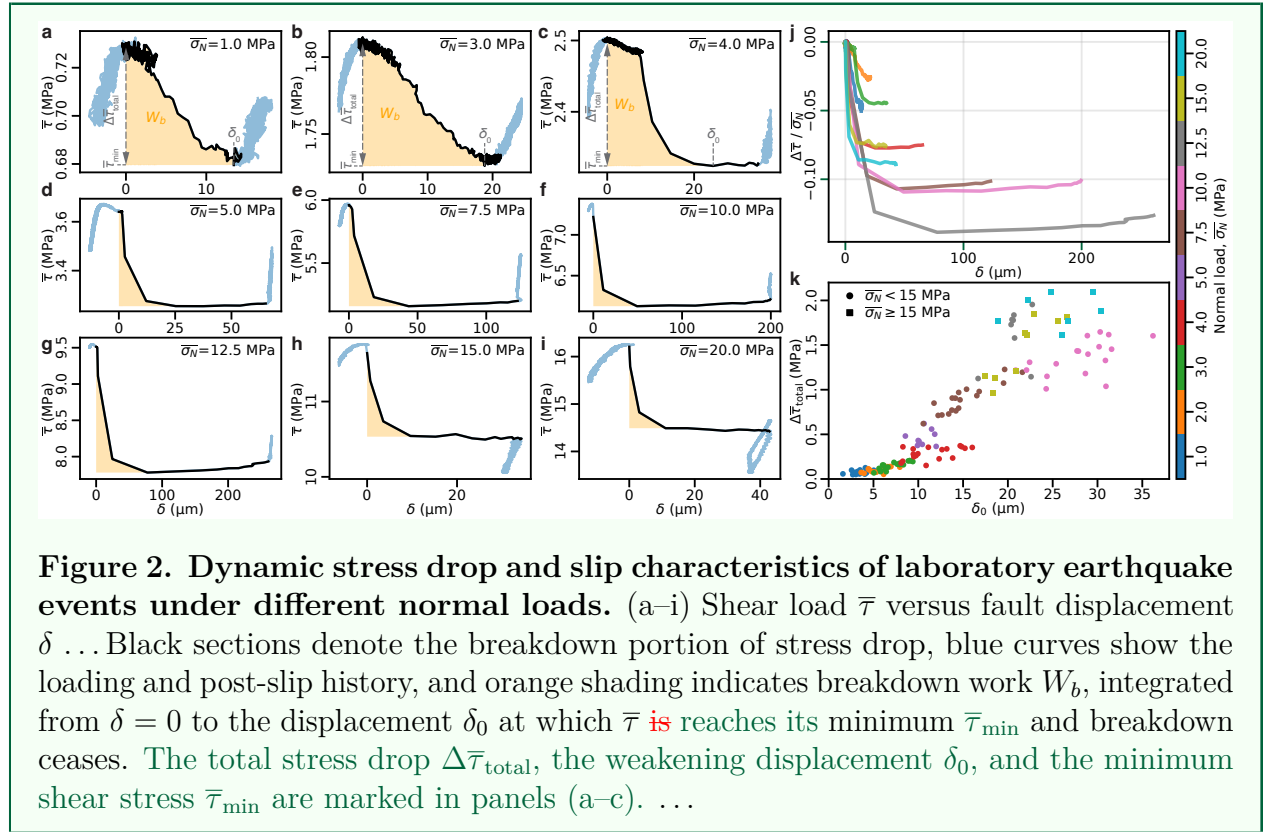
First, regarding δ_{min} , this quantity does not appear in the manuscript, and we suspect it refers to the displacement at which the shear stress reaches its minimum. That displacement is precisely what we denote δ_0 , defined in the original caption as the point “at which $\bar{\tau}$ is minimum and breakdown ceases.” The corresponding minimum shear stress is $\bar{\tau}_{min}$, which enters the breakdown-work integral $W_b = \int_0^{\delta_0} (\bar{\tau} - \bar{\tau}_{min}) d\delta$. We have made these definitions

more prominent in the revised caption to avoid confusion.

Second, $\Delta\bar{\tau}_{\text{total}}$ and δ_0 are already defined in the original caption (δ_0 in the (a-i) caption and $\Delta\bar{\tau}_{\text{total}}$ in panel (k)), as the difference in $\bar{\tau}$ between $\delta = 0$ and $\delta = \delta_0$). To make them easier to locate, we now annotate $\Delta\bar{\tau}_{\text{total}}$, δ_0 , and $\bar{\tau}_{\text{min}}$ directly on panels (a-c).

Third, regarding measurement errors, we note that the data plotted in Fig. 2 are unfiltered, so the measurement noise is directly visible as the scatter of the points in Fig. 2a. From this scatter we estimate the noise in $\bar{\tau}$ to be approximately ± 0.003 MPa (about 0.005 MPa peak-to-peak) and the noise in δ to be approximately ± 1 μm . Fig. 2a shows the 1.0 MPa event, which has the smallest stress drop and therefore represents the worst case for relative uncertainty ($\Delta\bar{\tau}_{\text{total}} \approx 0.045$ MPa, $\delta_0 \approx 13$ μm , i.e. a few percent). For the larger-load events, where stress drops reach several MPa and slips reach hundreds of micrometers, the same absolute scatter gives relative uncertainties well below 1%.

Revised text:



Comment R1-4.

L 109: It would be useful to acknowledge where the experiments were conducted. Did the authors use the Penn State lab? Relatedly, no specific author contributions are provided.

Response:

We confirm that the experiments were conducted at the Rock Mechanics Laboratory at The

Pennsylvania State University. We have added a sentence to this effect at the beginning of the Sample Preparation and Testing Protocol subsection (L 104). Regarding author contributions, a CRediT authorship contribution statement was provided at the end of the submitted manuscript.

Revised text:

All experiments were conducted in the Rock Mechanics Laboratory at The Pennsylvania State University. We conducted experiments at normal loads $\overline{\sigma_N}$ ranging from 1 to 20 MPa, all following the same testing protocol. ...

(existing CRediT statement in the submitted manuscript) **C.-Y. Ke:** Writing - original draft, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Visualization, Supervision, Project administration, Conceptualization, Funding acquisition. **G. T. Chang:** Software, Formal analysis, Data Curation, Visualization. **G. C. McLaskey:** Writing - review & editing, Validation, Resources, Conceptualization. **D. S. Kammer:** Writing - review & editing, Validation, Conceptualization. **C. Marone:** Writing - review & editing, Validation, Investigation, Resources, Supervision, Project administration, Conceptualization, Funding acquisition.

Comment R1-5.

L 135ff and Fig. 3: The clarity of this figure and its descriptions should be improved. Part of the issue is the mixture of methods and results. A clearer structure would certainly help. Results should first be introduced with a topical sentence, then observations, caveats and interpretation. 146ff is the description of rupture front determination which needs to be moved before Fig. 1 and the explanation should be expanded. The determined rupture front at each strain gauge should be marked in Fig. 3 a) and the labeling could be improved.

Response:

We agree, and have restructured this material along the lines suggested.

First, the description of how the rupture front is determined has been moved into the Methods (Section 2.3) so it precedes its use, and the explanation has been expanded. The key point, which was previously implicit, is now stated explicitly. The rupture front is conventionally defined as the peak in ε_{xy} (Paglialunga et al., 2022), but in our experiments a distinct peak is not always present, because our fault length (150 mm) is comparable to the critical nucleation length, so the dynamic rupture often propagates through regions that are still actively nucleating (Fig. 3a). We therefore define the rupture-front arrival as the onset of the rapid acceleration of the ε_{xy} drop (equivalently, the shear stress drop), which is robustly identifiable whether or not a peak is present.

Second, to separate methods from results, the observational description of the specific event (nucleation location and propagation direction) has been moved from the subsection body into the Figure 3 caption, and the subsection now opens with a topic sentence.

Third, we have marked the picked rupture-front arrival at each gauge location in Figure 3a, consistent with this definition. The updated figure is shown under Comment R1-8.

Revised text:

2.3. Rupture Speed Estimation and Cohesive Zone Modeling

~~Fig. 3a shows details of the strain history for one event. Rupture nucleates near $x = 20$ mm and quasi-statically loads the fault, increasing shear strain ε_{xy} at the bottom edge near $x = 150$ mm until dynamic rupture initiates and propagates upward. Note that especially near the top in this example, the dynamic shear strain records lack the distinct peaks commonly observed at rupture fronts in large-scale laboratory earthquakes (Svetlizky et al., 2014) because our fault length (150 mm) is shorter than the critical nucleation length ($h^* \approx 0.5$ m; Ke et al., 2018; McLaskey, 2019). Consequently, when dynamic rupture is triggered, it propagates through regions of the fault that are still actively nucleating, obscuring the typical sharp rupture-front signature at those locations.~~

~~To analyze rupture propagation, we~~ To estimate rupture speed, we first determine the rupture-front arrival time at each gauge location. The rupture front is conventionally identified with the peak in shear strain ε_{xy} (Paglialunga et al., 2022). In our experiments, however, a distinct peak is not always present, because the fault length (150 mm) is comparable to the critical nucleation length, so the dynamic rupture often propagates through regions that are still actively nucleating and the characteristic peak is suppressed at those locations. We therefore define the rupture-front arrival as the onset of the rapid acceleration of the ε_{xy} drop (Ohnaka & Shen, 1999), which coincides with the peak where one is present and remains well defined where it is not. We developed an open-source Python GUI tool (Labquake Explorer; Ke et al., 2024) to pick these arrival times at each ε_{xy} measurement position, ~~defined as the first visible drop in shear strain~~. Arrival times and corresponding fault locations were then fit with linear regression to estimate rupture speed C_f , using positions near the fault center where velocity remained approximately constant (Fig. 1c). ...

Revised Figure 3a caption. Shear strain change $\Delta\varepsilon_{xy}$ time history along the fault for one event with upward dynamic rupture propagation, selected because nucleation and dynamic propagation occur close enough in time to visualize both phases together. Rupture nucleates near $x = 20$ mm and quasi-statically propagates downward, loading the bottom of the fault near $x = 150$ mm, until dynamic rupture initiates at the bottom and propagates upward. Triangles mark the picked rupture-front arrival at each gauge. Note that at some locations the records do not exhibit a distinct peak, because the rupture front passes through regions that are still actively nucleating. ...

Comment R1-6.

All figures include many sub-panels that are insufficiently described and the link between the sub-figures is not always obvious. The manuscript would be clearer with fewer convoluted figures (e.g., separate unrelated sub-panels and increase fontsize.)

Response:

We thank the reviewer for this overarching comment and have reconsidered the organization of every figure. We agree that the most densely packed figure was the former Figure 4, which we have now split into two separate figures (new Figures 4 and 5; shown below). For the remaining figures, the grouping of panels is deliberate, and we briefly justify each arrangement here. We have also expanded the sub-panel descriptions and clarified the links between panels in response to the specific points raised elsewhere in this review (Comments R1-2, R1-3, R1-5, R1-8, R1-21, R1-24, and R1-25).

Figure 1 presents an overview of the experimental setup, so its panels are complementary views of a single apparatus and belong together.

Figure 2 summarizes the dynamic stress drop and slip behavior across all events. Its panels are different views of the same suite of experiments and together provide a single overview, so they are best presented as one figure.

Figure 3 shows two records of the local dynamic shear strain change. Panels (a) and (b–d) could in principle be separated, and we recognize that combining them was a source of confusion. However, panel (a) requires a tall aspect ratio to resolve the strain drop; widening it to stand on its own would stretch the drop horizontally and make it visually less pronounced. Grouping it with panels (b–d) is therefore both more compact and more legible than a single tall panel, and because all panels show time series of the local dynamic shear strain change, the grouping remains physically coherent. We have also clarified the description of the event shown at the top of the figure (see Comment R1-8).

The former Figure 4 grouped seven panels (a)–(g) covering distinct quantities. We have separated it into two figures along a natural boundary. New Figure 4 collects the directly measured scaling relations, namely the rupture speed, total stress drop, and breakdown work (former panels a–d), while new Figure 5 collects the cohesive-zone-model results, namely the rupture-tip stress drop, cohesive zone size, and near-front dissipation (former panels e–g). Both new figures are shown below.

Figure 6 (the former Figure 5) presents the finite-element model results, which form a single coherent analysis and therefore belong together.

Splitting the former Figure 4 also freed space to enlarge the axis and tick labels in the new Figures 4 and 5, which we have done. We have also simplified the Fig.4a vertical axis, which now runs from 0 to 2.0km/s with labeled ticks every 0.5km/s and minor ticks at 0.25km/s, and we changed the dotted LEFM curve to black for legibility. For the other figures we have kept the current font size, which we find legible. In particular, the difficulty in reading the Figure 6 (former Figure 5) tick labels was caused by the labels being spaced too closely together rather than by small font size, and we have increased their spacing (see Comment R1-21). We have also improved label contrast where needed (see Comment R1-8).

Revised text:

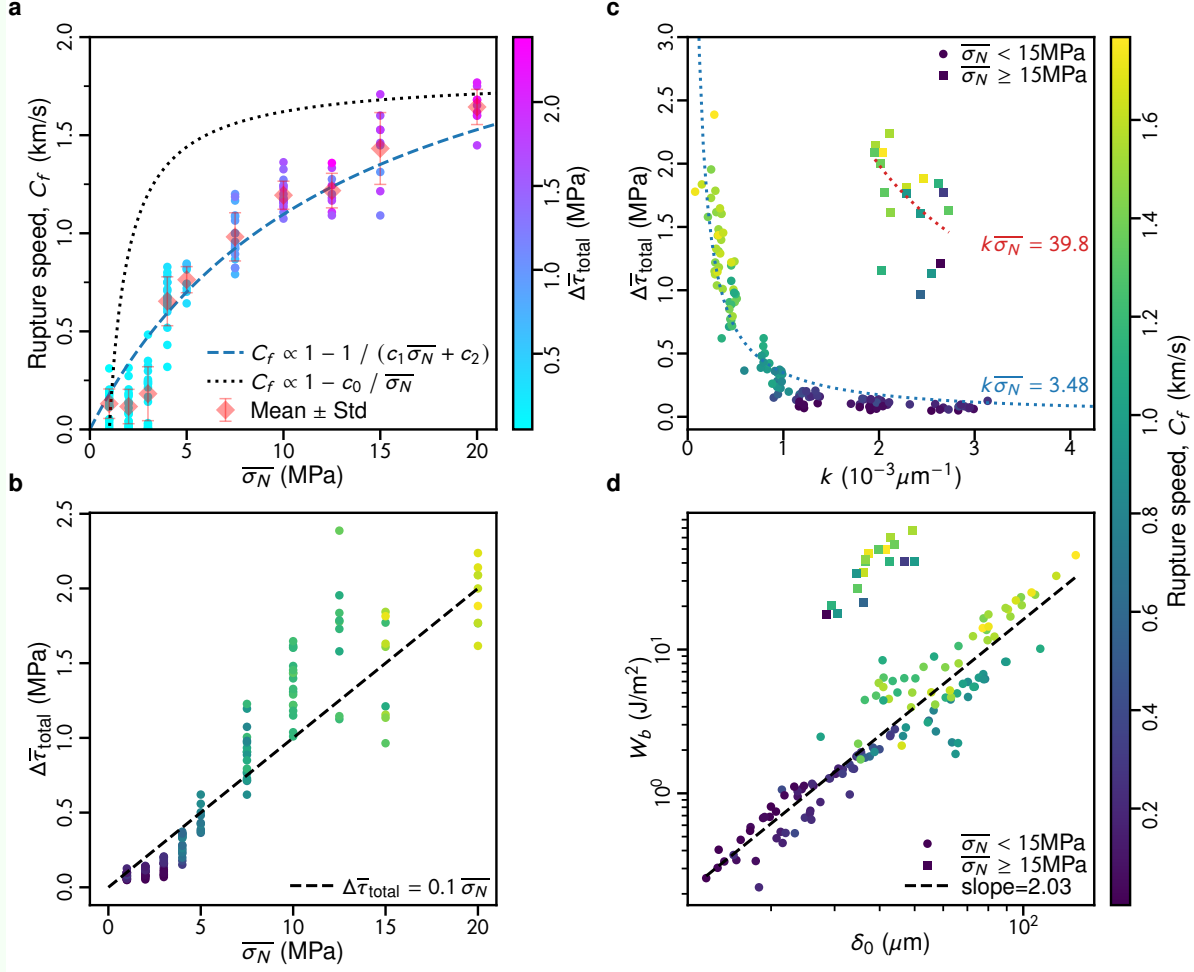


Figure 4. Scaling of rupture speed, stress drop, and breakdown work. (a) Rupture speed C_f increases systematically with normal stress $\bar{\sigma}_N$. The black dotted curve represents the fit $C_f \propto 1 - c_0/\bar{\sigma}_N$, while the dashed curve shows $C_f \propto 1 - 1/(c_1\bar{\sigma}_N + c_2)$. (b) Total stress drop $\Delta\bar{\tau}_{\text{total}}$ increases with normal load, with $\Delta\bar{\tau}_{\text{total}} \approx 0.1\bar{\sigma}_N$. (c) $\Delta\bar{\tau}_{\text{total}}$ versus normalized loading stiffness k shows a trend similar to that reported by Shreedharan et al. (2020), with lower stiffness yielding higher stress drop. The outlier cluster corresponds to events at $\bar{\sigma}_N \geq 15$ MPa conducted with a much higher stiffness. The dotted and dashed curves indicate constant values of the stress-per-displacement stiffness $\bar{k} = k\bar{\sigma}_N$ (in MPa/mm), following the scaling $\Delta\bar{\tau}_{\text{total}} = 0.1\bar{\sigma}_N = 0.1\bar{k}/k$. (d) Breakdown work W_b scales with the weakening displacement δ_0 with an exponent of 2.03 (dashed line).

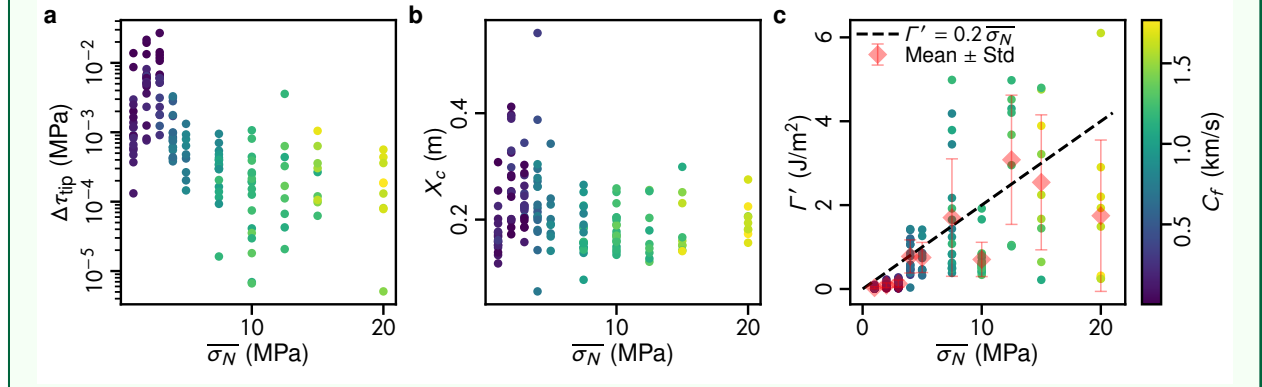


Figure 5. Cohesive-zone-model (near-tip) scaling. (e) (a) Stress drop at the rupture tip $\Delta\tau_{\text{tip}}$, estimated from CZM fits, decreases slightly with $\bar{\sigma}_N$ and is considerably smaller than the total stress drop $\Delta\bar{\tau}_{\text{total}}$. (f) (b) Cohesive zone size X_c shows little dependence on $\bar{\sigma}_N$ but mostly exceeds the 0.15 m fault length. (g) (c) Near-front dissipation Γ' dependence on $\bar{\sigma}_N$ is broadly consistent in magnitude with the fracture-energy scaling $\Gamma \approx 0.2 \mu\text{m} \cdot \bar{\sigma}_N$ reported by Ke et al. (2018, dashed line).

Comment R1-7.

Figure 3a: What is 'nucleation' and in what direction is it propagating relative to the direction of 'dynamic rupture'?

Response:

Nucleation refers to the quasi-static phase of slip growth that precedes dynamic instability. In the event shown in panel (a), slip nucleates near the top of the fault (near $x = 20$ mm) and quasi-statically propagates downward, gradually loading the bottom of the fault; once the bottom is sufficiently loaded, dynamic rupture initiates there and propagates upward, in the opposite direction to the quasi-static nucleation front. This event was specifically selected for panel (a) because nucleation and dynamic propagation occur close enough in time to visualize both phases in a single record. We have clarified this in the revised Figure 3 panel (a) caption (see Comment R1-5), and the updated figure is shown under Comment R1-8.

Revised text:

(see revised Figure 3 panel (a) caption in Comment R1-5) ... Rupture nucleates near $x = 20$ mm and quasi-statically propagates downward, loading the bottom of the fault near $x = 150$ mm, until dynamic rupture initiates at the bottom and propagates upward. ...

Comment R1-8.

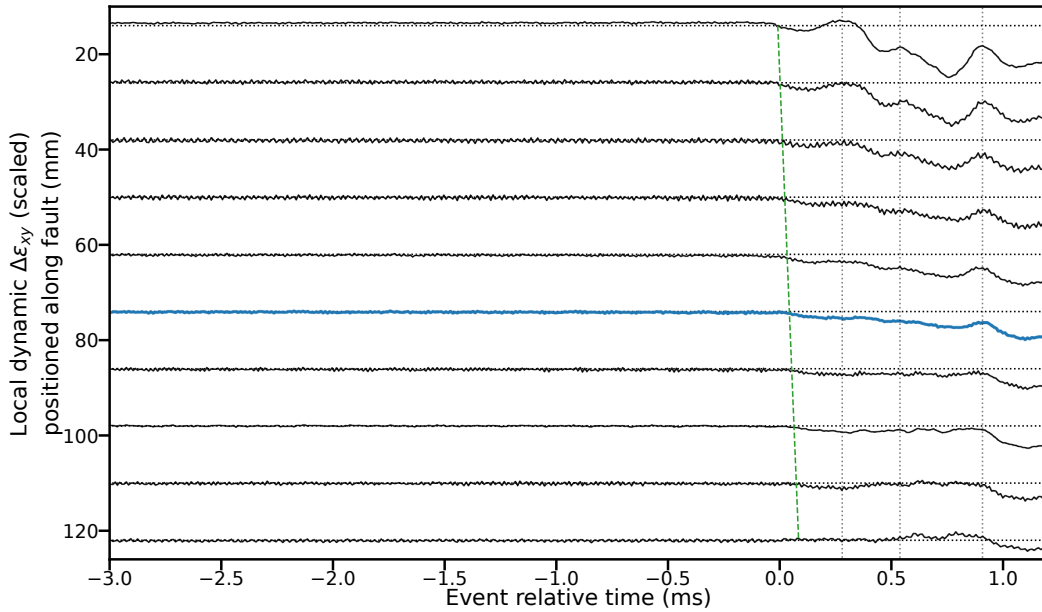
Fig. 3a: change from s to ms. Highlight which portion is shown in Fig. 3d. The time scale in Fig. 3d should be slightly increased to include pre-rupture front strain.

Response:

We thank the reviewer for these suggestions. We have changed the time axis of Fig. 3a from s to ms.

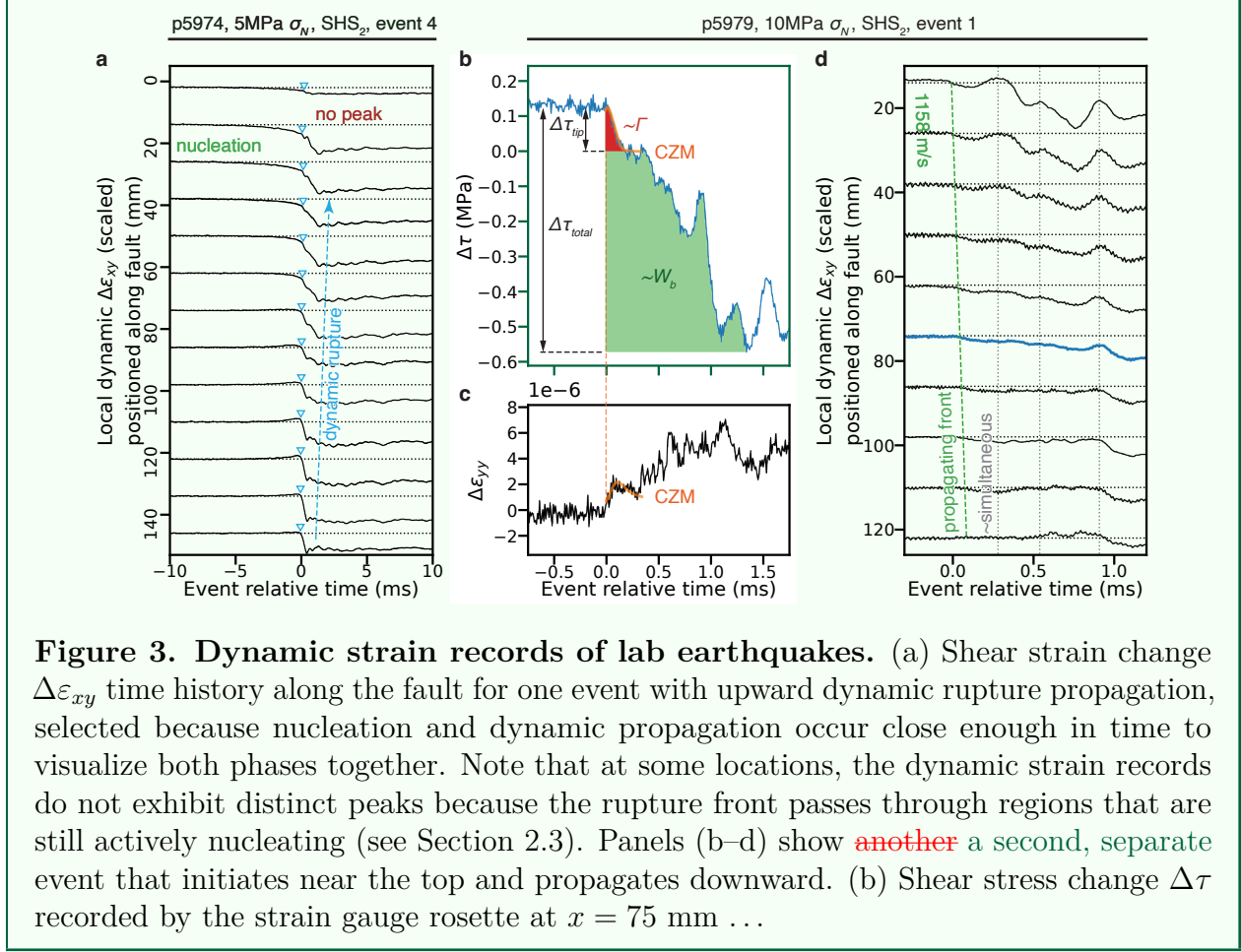
Regarding highlighting the portion of Fig. 3a shown in Fig. 3d, we note that Fig. 3d is not a zoomed-in window of the event in Fig. 3a. Panels (b), (c), and (d) all show the same event, as indicated at the top of the figure, while panel (a) shows a different event that was selected specifically because its nucleation and dynamic rupture phases occur close enough in time to be visualized together. There is therefore no sub-window of Fig. 3a to highlight. To make this grouping clearer, we have changed the panel labels from gray to black to improve their legibility and revised the caption so that the event corresponding to each panel is unambiguous. The revised figure is shown below.

Regarding extending the time scale of Fig. 3d to include the pre-rupture-front strain, we respectfully prefer to keep the current window. The nucleation phase that precedes the dynamic rupture front evolves over a timescale orders of magnitude longer than the dynamic rupture itself, so on the millisecond scale of Fig. 3d the pre-rupture-front strain is essentially flat and carries no additional information about the rupture. Extending the window to capture it would compress the dynamic portion that is the focus of the panel. For the reviewer's reference, the figure below shows the same record over the longer nucleation timescale, where the pre-rupture-front strain is indeed flat right up to the arrival of the dynamic front.



(Figure for the reviewer only; not included in the manuscript.)

Revised text:



Comment R1-9.

L 153ff: Cohesive zone model fitting and fracture estimates are insufficiently described. L160ff seem out of place. Is this part of Results or Discussion?

Response:

We thank the reviewer for this observation. We have added the three governing equations of the CZM to Section 2.3 of the Methods and cite Fig. 3b–c as an exemplary fit; the detailed fitting procedure remains in Supplementary Text S1. Regarding L 160ff (now L 166ff in the revised manuscript), this paragraph motivates the selection of the fitting window by explaining why only the initial weakening phase is fit, and therefore belongs in the Methods section. We have reframed its opening sentence to make this methodological role explicit.

Revised text:

Beyond rupture speed, characterizing the weakening process and fracture energetics requires fitting our dynamic strain measurements using the cohesive zone model, which regularizes the stress singularity inherent in LEFM (Paglialunga et al., 2022; Kammer et

al., 2019; Poliakov et al., 2002; Samudrala et al., 2002). In this formulation, the dynamic stress/strain fields near the tip of a propagating mode II rupture are governed by the rupture speed C_f , fracture energy Γ , cohesive zone size X_c , and off-fault distance y . The near-tip stress change is given by (Freund, 1990)

$$\Delta\tau(\theta, r, C_f) = \frac{K_{II}(C_f)}{\sqrt{2\pi r}} \Sigma_{xy}^{II}(\theta, C_f), \quad (1)$$

where K_{II} is the mode-II stress intensity factor and Σ_{xy}^{II} its angular variation. The fracture energy Γ is related to K_{II} through the dynamic energy release rate (Kammer & McLaskey, 2019)

$$\Gamma = G_{II} = \frac{1 - \nu^2}{E} K_{II}^2 f_{II}(C_f), \quad (2)$$

where $f_{II}(C_f)$ is a known universal function of crack speed. The cohesive zone introduces a linear slip-weakening stress distribution in local coordinates $\xi = x - x_{\text{tip}}$ (Poliakov et al., 2002; Samudrala et al., 2002)

$$\Delta\tau(\xi) = \begin{cases} 0, & \xi < -X_c \\ (1 + \xi/X_c)(\tau_p - \tau_r), & -X_c < \xi < 0, \end{cases} \quad (3)$$

where τ_p and τ_r are the peak and residual shear stresses. Together Eqs. (1)–(3) predict the full $\Delta\tau(t)$ and $\Delta\varepsilon_{yy}(t)$ time histories at off-fault distance y . An example fit is shown in Fig. 3b–c.

~~Applying this model to our measurements requires careful consideration of which portion of the weakening signal reflects local tip processes versus system-scale unloading.~~ The fitting window is chosen to isolate the local tip-weakening phase from the later system-scale unloading, as described below. Because the fault length is limited to 150 mm, the rupture front traverses the fault rapidly ($\sim 150 \text{ mm} \div 1 \text{ km/s} = 0.15 \text{ ms}$), while reduction of τ persists with slip (Fig. 3b). This two-phase behavior reflects distinct energy release mechanisms: initial rapid weakening associated with local energy dissipation at the propagating rupture tip, followed by continued stress reduction driven by nonlocal elastic unloading of the fault blocks and loading column. This distinction is evident in Fig. 3d, where the initial drop in $\Delta\tau$ propagates at a constant rate (green dashed line), whereas subsequent features appear nearly simultaneous across all locations. To isolate the local rupture-tip energy dissipation from system-scale effects, we fit the CZM to only the initial weakening phase in $\Delta\tau$ and $\Delta\varepsilon_{yy}$ (Fig. 3b–c), while excluding the later features that represent nonlocal elastic unloading. For each event, we restricted the fits to the center gauge rosette ($x = 75 \text{ mm}$), where both components are available, thereby enhancing the robustness of the fit. In the fitting procedure, we fix C_f based on estimates from multiple gauge rosettes and set $y = 8 \text{ mm}$ (fault to gauge center distance), then define the time window for the initial $\Delta\tau$ drop and jointly fit $\Delta\tau$ and $\Delta\varepsilon_{yy}$ for X_c and Γ (Supplementary Text S1).

Comment R1-10.

Fig. 3b: How were the different stress drops determined? The stress measurement looks quite noisy. There is a second minimum at ~ 0.8 ms, and another one around 1 ms. How accurate are these measurements?

Response:

We thank the reviewer for these questions, several of which connect to points addressed elsewhere in this review.

Regarding how the stress drops were determined, Fig. 3b reports two distinct quantities. The local total stress drop $\Delta\tau_{\text{total}}$ is the difference between the pre-rupture shear stress and its minimum, using the same definition applied to the quasi-static records (see Comment R1-3). The rupture-tip stress drop $\Delta\tau_{\text{tip}}$ is obtained from the cohesive zone model fit to the initial weakening phase (the red I' region), which we now describe with explicit equations in Section 2.3 (see Comments R1-9 and R1-12).

Regarding accuracy, we would clarify that the large-amplitude oscillations are not measurement noise; they are real signal generated by the dynamic rupture. The genuine measurement noise is the small high-frequency scatter visible in the pre-rupture portion of the traces ($t < 0$ in Fig. 3b). This scatter is approximately ± 0.01 MPa, small compared with $\Delta\tau_{\text{total}} \approx 0.6$ MPa for this event.

Regarding the second minima near 0.8 and 1 ms, as reasoned in the manuscript these reflect the physics of the event rather than measurement artifacts. The initial weakening corresponds to the arrival of the rupture front (the portion fit by the cohesive zone model), while the subsequent oscillations arise from near-rigid-body motion of the center block once the entire fault has begun to slip. Consistent with this interpretation, these later features arrive nearly simultaneously at all gauges (Fig. 3d). Because they are not part of the local rupture-tip weakening, the cohesive zone model is fit only to the initial drop and explicitly excludes them (see Comment R1-9).

Comment R1-11.

L 175ff: $\Delta\epsilon_{xy}$ is not shown in Fig. 3b-c.

Response:

We thank the reviewer for this observation. The reviewer is correct that $\Delta\epsilon_{xy}$ is not directly shown in Fig. 3b-c. Panel (b) shows the derived shear stress $\Delta\tau$, which is related to the measured shear strain by $\Delta\tau = G\Delta\epsilon_{xy}$, where G is the shear modulus. The cohesive zone model fitting is performed in strain space, so the original wording is retained; we have added the relation $\Delta\tau = G\Delta\epsilon_{xy}$ to the caption of panel (b) to make this explicit.

Revised text:

... (b) Shear stress change $\Delta\tau = G\Delta\epsilon_{xy}$ recorded by the strain gauge rosette at $x = 75$ mm.
...

Comment R1-12.

L 181 and supplementary Text: The brief description in the supplement could be moved to the main text since it is less than one page. An equation and exemplary figure would be useful.

Response:

We thank the reviewer for this suggestion. We have added the key governing equations of the cohesive zone model to the Methods section (Section 2.3) and pointed to Fig. 3b–c as an exemplary fit. The detailed fitting procedure remains in Supplementary Text S1 for conciseness. Please see the revised text under Comment R1-9.

Comment R1-13.

L 198ff: Unclear sentence.

Response:

We thank the reviewer for flagging this ambiguity. We agree the sentence was unclear. The “kink” refers to a visible inflection in the stress-slip curve at the very onset of the stress drop, most clearly seen in Fig. 2a–d, where the stress first drops slowly and then accelerates. This two-stage onset reflects the DDS geometry, in which one fault begins to slide slightly before the other. The initial slower drop corresponds to single-fault sliding while the second fault remains briefly locked, after which both faults slip together and the drop steepens. We have revised the sentence to make this explicit.

Revised text:

... ~~In all events, the weakening process shows a kink at the onset of breakdown, caused by one fault sliding slightly while the other remains locked~~ The stress-slip curve shows an inflection at the onset of the stress drop (visible in Fig. 2a–d), reflecting a two-stage weakening in which one fault begins to slide while the other remains locked briefly before both faults slip together (see Supplementary Fig. S4). ...

Comment R1-14.

L 208ff: This paragraph is not very clear. It would be helpful to develop the mathematical expressions more fully and use separate lines and numbers for each equation.

Response:

We agree the derivation is difficult to follow in its current inline form. We have restructured the paragraph to display each key mathematical expression on a separate, numbered line with explicit prose connecting each step. The equation numbers continue from the cohesive-zone-model equations (1)–(3) added to the Methods section (see Comment R1-9).

Revised text:

... For an edge crack in a semi-infinite solid, the mode-II stress intensity factor scales as $K_{II}^S \propto \Delta\tau \propto \overline{\sigma}_N$, implying a static energy release rate

$$G_{II}^S \propto \overline{\sigma}_N^2. \quad (4)$$

Laboratory studies further suggest that fracture energy scales linearly with normal stress (Ke et al., 2018; Brodsky et al., 2020),

$$\Gamma \propto \overline{\sigma}_N. \quad (5)$$

Substituting Eqs. (4) and (5) into the LEFM rupture speed expression (Svetlizky et al., 2017; Kammer et al., 2024),

$$C_f \approx \left(1 - \frac{\Gamma}{G_{II}^S}\right) C_R, \quad (6)$$

yields $C_f \propto 1 - c_0/\overline{\sigma}_N$, indicates

$$C_f \propto 1 - \frac{c_0}{\overline{\sigma}_N}, \quad (7)$$

indicating that rupture speed should follow a hyperbolic approach to C_R as $\overline{\sigma}_N$ increases (dotted curve in Fig. 4a). ...

Comment R1-15.

L220ff: Rewrite: This discrepancy has two potential explanations: 1) An additional mechanisms [...] 2) Alternatively, the 150 mm fault [...]

Response:

We thank the reviewer for this suggestion. We have restructured the paragraph to present the two potential explanations as an explicit numbered list.

Revised text:

... In contrast, our data show only a modest acceleration with $\overline{\sigma}_N$, indicating that rupture speed cannot be explained by the standard LEFM scalings $G_{II}^S \propto \overline{\sigma}_N^2$ and $\Gamma \propto \overline{\sigma}_N$. This discrepancy has two potential explanations: ~~This discrepancy implies an additional mechanism beyond energy release from a homogeneous semi-infinite medium or variations in fracture energy. For instance, if G_{II}^S is treated as a quadratic function of $\overline{\sigma}_N$, the resulting form provides a much closer fit to our data (dashed curve in Fig. 4a). Alternatively, the 150 mm fault length may not provide sufficient distance for ruptures to fully accelerate, particularly for the higher $\overline{\sigma}_N$ events that are expected to have greater potential C_f .~~ 1) An additional mechanism beyond standard energy release: if G_{II}^S is treated as a quadratic function of $\overline{\sigma}_N$, i.e., $G_{II}^S = c_1 \overline{\sigma}_N^2 + c_2 \overline{\sigma}_N$ with fitting constants c_1 and c_2 , the resulting form $C_f \propto 1 - 1/(c_1 \overline{\sigma}_N + c_2)$ provides a much closer fit to our data (dashed curve in Fig. 4a). 2) The 150 mm fault length may not provide sufficient distance for ruptures to fully accelerate, particularly at higher $\overline{\sigma}_N$ where the potential C_f

is greatest. ...

Comment R1-16.

L 240: shows power-law scaling? ...

Response:

We thank the reviewer for this observation. We agree the wording was imprecise. “Scales strongly” did not convey the specific functional form. We have revised the sentence to state the power-law character and exponent explicitly in the opening sentence. Further revisions to the same paragraph are described under Comment R2-2.

Revised text:

~~Breakdown work W_b scales strongly with total slip δ_{total} (Fig. 4d), consistent with patterns observed for natural earthquakes (Abercrombie and Rice, 2005; Viesca and Garagash, 2015; Cocco et al., 2023) and with the laboratory measurements of Selvadurai (2019).~~ Breakdown work W_b follows power-law scaling with the weakening displacement δ_0 , $W_b \propto \delta_0^2$ (Fig. 4d), consistent with patterns observed for natural earthquakes (Abercrombie and Rice, 2005; Viesca and Garagash, 2015; Cocco et al., 2023) and with laboratory measurements (Selvadurai, 2019). A best-fit power law with an exponent of 2 is consistent with scalings inferred for natural earthquakes (exponent range 0.67–2). ...

Comment R1-17.

L256ff: Repetition

Response:

We agree. The cohesive zone model was already introduced and described in the Methods (Section 2.3), making the opening sentence of this subsection redundant. We have removed it and begin directly with the results, adding a parenthetical cross-reference to Section 2.3 at the first mention of CZM.

Revised text:

3.4. Near-Tip Weakening from Cohesive Zone Modeling ~~We developed a cohesive zone model (CZM) to fit our data (Paglialunga et al., 2022; Kammer et al., 2019; also see Supplementary Text S1).~~ The CZM (Section 2.3) fits well for the initial strain drop at the center rosette ($x = 75$ mm) and captures near-tip weakening (Fig. 3e–g**b–c**). ...

Comment R1-18.

L268ff: The reasoning is hard to follow. It sounds like the authors are arguing that the fracture mechanics model produces unphysical results, yet, they still chose to use it?

Response:

We thank the reviewer. The original wording was unclear because it conflated two distinct points, which we now separate.

First, the cohesive zone model is an exact elasticity solution. The near-tip stress and strain fields it predicts (Freund, 1990; Poliakov et al., 2002; Samudrala et al., 2002; following Ida, 1972; Palmer and Rice, 1973) solve a well-posed boundary value problem for any prescribed weakening law, irrespective of the cohesive zone size. Fitting this solution to our measured strain records is therefore valid in itself, and the inferred X_c is a real length scale describing the spatial extent over which the rupture-front stresses evolve. Nothing about the fit is unphysical.

Second, the *interpretation* of the fitted quantities does depend on scale separation. The classical identification of the fitted amplitude with a stress intensity factor, and of the associated energy with a fracture energy, requires a region outside the cohesive zone but still close to the rupture tip where the field is K -dominated (small-scale yielding, $X_c \ll L$). In our experiments X_c is comparable to or larger than the fault length L , so this intermediate K -dominated region does not exist. The fitted amplitude is then only an amplitude parameter of the chosen solution, not a true stress intensity factor, and X_c should not be read as a literal LEFM cohesive zone.

Accordingly, we have removed the over-stated claim that our results call the applicability of LEFM into question, and we no longer present the CZM-derived energy as a strict fracture energy. To make the distinction explicit, we now denote it Γ' , an apparent near-front dissipation scale inferred from the CZM fit, reserving Γ for the strict LEFM fracture energy. Importantly, Γ' is obtained as the area under the local weakening curve, $\int(\tau - \tau_r) d\delta$ over the initial weakening, which is the direct dissipation integral (Palmer and Rice, 1973) and does not require the K -dominated region. Because our gauge sits at $y = 8$ mm while X_c exceeds 150 mm, it lies well inside the cohesive zone ($y \ll X_c$) and reads the local traction evolution directly. The large X_c that invalidates the stress-intensity-factor interpretation is thus also what allows Γ' to be read off the near-field record robustly. The second strain component reinforces this: because we fit not only $\Delta\varepsilon_{xy}$ (Fig. 3b) but also $\Delta\varepsilon_{yy}$ (Fig. 3c), the polarity of $\Delta\varepsilon_{yy}$ provides an independent diagnostic, following the cohesive-zone (tip) solution during the initial weakening window and departing from it afterward, which lets us identify objectively which portion of the record reflects tip behavior.

This does not affect our central result. The breakdown work W_b is obtained from the finite-geometry energy budget, while Γ' is read from the near-field strain record as the area under the initial weakening, and neither relies on the small-scale-yielding assumption that fails here. The comparison $\Gamma' \ll W_b$ stands regardless, because it requires only that Γ' measure near-front dissipation, not that it be a strict fracture energy. We therefore use the CZM as a descriptive parameterization of the near-tip weakening rather than as a strict LEFM model, and have revised the text accordingly.

To avoid any further ambiguity, we have made this terminology consistent throughout the manuscript. We now write Γ' wherever we refer to the measured near-front dissipation, and reserve Γ for the strict LEFM fracture energy and for reported literature values. We also

introduce Γ' as a proxy for fracture energy in the absence of a K -dominant field in both the abstract and the Introduction, so that readers encounter this distinction up front rather than only at the cohesive-zone-modeling results where it is justified in detail.

Revised text:

The cohesive zone size X_c varies only weakly with $\overline{\sigma_N}$ (Fig. 5b), yet consistently exceeds the fault length, ~~indicating a violation of the small-scale yielding condition. This calls into question the applicability of LEFM to our macroscopic measurements when ruptures reach the sample boundaries, although the models fit some aspects of our data reasonably well. Nevertheless, we report Γ from the CZM fits to quantify near-tip energy dissipation and to compare it with the macroscopic breakdown work W_b —not as a rigorous LEFM estimate, but as a model-insensitive estimate of near-tip dissipation, since the area under the τ - δ curve during initial weakening is insensitive to CZM parameters as long as $y \ll X_c$.~~ The cohesive zone model remains a valid elastic description of the near-tip strain fields, and X_c is a real length scale over which the rupture-front stresses evolve. However, because X_c is comparable to or larger than L , there is no region outside the cohesive zone yet close to the tip where the field is K -dominated. The small-scale yielding condition is therefore not satisfied, the fitted amplitude is an amplitude parameter of the solution rather than a true stress intensity factor, and X_c should not be read as a literal cohesive zone. Accordingly, we do not interpret the CZM-derived energy as a strict fracture energy. We instead report it as an apparent near-front dissipation scale Γ' , obtained as the area under the τ - δ curve during initial weakening. This area is constrained directly by the near-field gauge ($y \ll X_c$) and is insensitive to the assumed weakening-law shape, providing a robust measure of near-front dissipation for comparison with the breakdown work W_b . Unlike W_b , which follows directly from the finite-geometry energy budget, Γ' is obtained through the cohesive-zone fit, so its absolute magnitude retains some dependence on that parameterization in the large- X_c regime.

Comment R1-19.

L282: typo

Response:

We thank the reviewer for catching this typo. “it’s” (contraction) has been corrected to “its” (possessive).

Revised text:

... after the rupture has reached ~~it's~~ its final dimensions. ...

Comment R1-20.

Fig. 4: mean rupture speed is not visible.

Response:

We thank the reviewer for pointing this out. The reviewer is correct that the mean rupture speed was difficult to see in the former Fig. 4a. In splitting the former Figure 4 into two less crowded figures (see Comment R1-6), the rupture speed panel now has more space and the mean rupture speed is clearly visible. It is panel (a) of the new Figure 4, shown under Comment R1-6.

Comment R1-21.

L 336: Which figure is showing granite? Which figure is showing acrylic? How about the experiment with the steel forcing block? (the axis tick labels are very hard to read in Fig. 5)

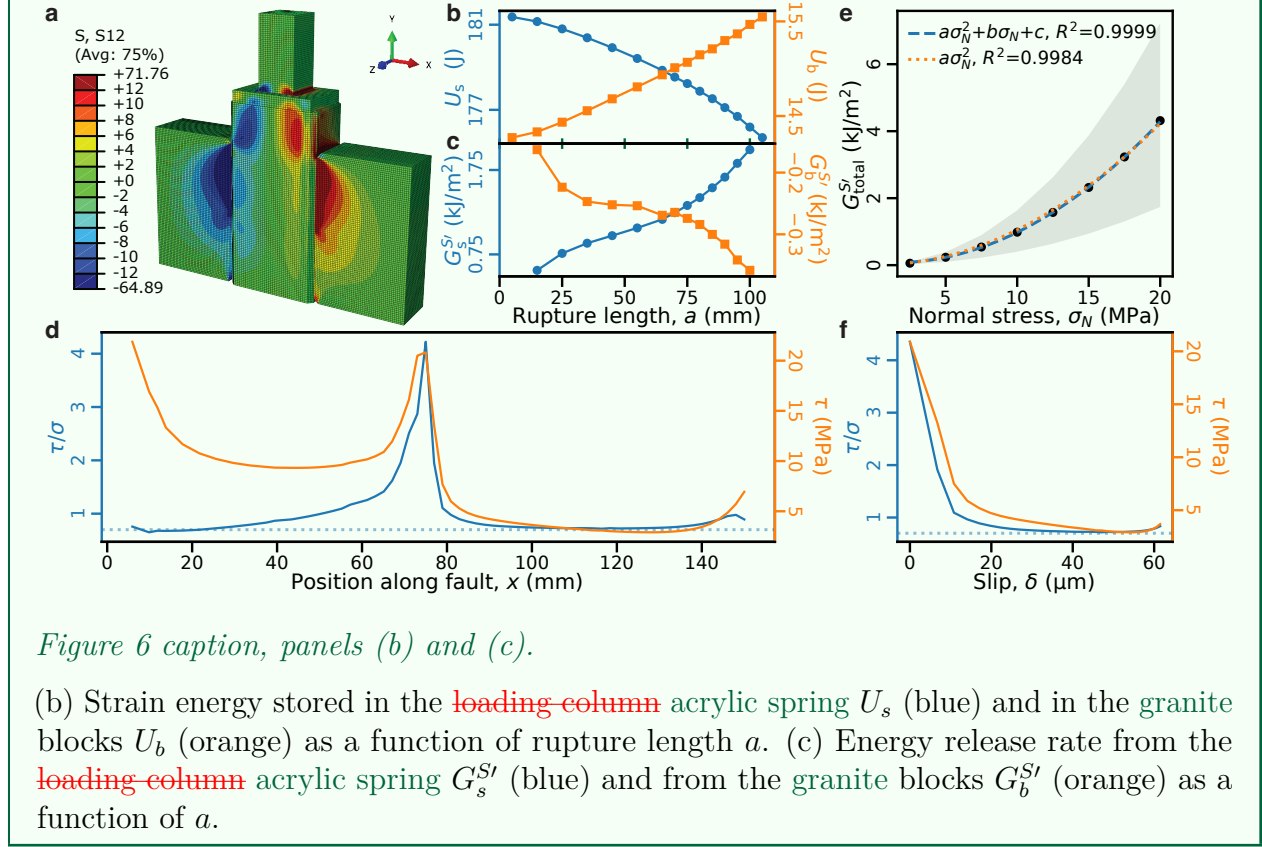
Response:

We thank the reviewer for these questions. We note that, following the split of the former Figure 4 (see Comment R1-6), the finite-element figure that the reviewer refers to as Fig. 5 is now Figure 6; we use the new numbering below. In Fig. 6b and 6c, blue denotes the acrylic spring ($U_s, G_s^{S'}$) and orange denotes the granite blocks ($U_b, G_b^{S'}$). Correspondingly, the subscript s refers to the acrylic spring and the subscript b to the granite blocks. The original caption distinguished the two by somewhat vague names, so we have revised it to name the materials explicitly, replacing “loading column” with “acrylic spring” and labeling the blocks as “granite blocks.”

Regarding the steel forcing block, the stiff steel loading spring is mainly used for the velocity-step tests, not for the stick-slip events (except for $\overline{\sigma_N} > 15$ MPa), and is therefore not entirely relevant to the energy budget modeled here. The finite-element model represents the acrylic spring configuration used during the slide-hold-slide (SHS) tests that produce the labquakes, and the high-stiffness steel case is not modeled separately.

Finally, the tick labels in Fig. 6b and 6c were hard to read because they were spaced too closely together, so we have increased their spacing. Please see the updated figure below.

Revised text:



Comment R1-22.

L 346: Incomplete sentence

Response:

We thank the reviewer for catching this. The sentence was missing a predicate. We have added “is positive” to complete it. To make the contrast with the bottom-up case explicit, we have also added a note in the preceding sentence clarifying that $G_b^{S'} < 0$ in Fig. 6c (the blocks absorb rather than release energy in the bottom-up case), so the reader understands why the positive value in the top-down case is notable.

Revised text:

...Interestingly, the strain energy in the granite blocks increases with rupture area ($A = aw$) rather than releasing energy ($G_b^{S'} < 0$, Fig. 6c). ...In the case with top-down rupture, the energy release rate in the two side blocks is positive (Supplementary Fig. S2), as expected, but remains considerably lower than the energy release rate from the loading column with the acrylic block ($G_s^{S'}$). ...

Comment R1-23.

L 360ff: This again sounds like the wrong model was used. Does that invalidate your conclusion about scaling relations between Gamma, G and Sn?

Response:

We thank the reviewer. This comment is closely related to Comment R1-18. The departure from small-scale yielding does not invalidate our scaling conclusions, for three reasons.

First, the quantity we report from the CZM fits is the apparent near-front dissipation scale Γ' , defined as the area under the initial weakening curve (see Comment R1-18). It is constrained directly by the near-field gauge and is insensitive to the assumed weakening-law shape, and the polarity of the second strain component $\Delta\epsilon_{yy}$ independently confirms tip behavior during the initial weakening window. Because Γ' is measured directly as this area, the observation that it grows linearly with normal stress, $\Gamma' \propto \overline{\sigma}_N$, is an empirical result that does not rest on LEFM or small-scale yielding being valid. It therefore remains a meaningful measurement in our large X_c regime.

Second, the quadratic scaling of the static energy release rate, $G^{S'} \propto \overline{\sigma}_N^2$, is reproduced independently by the finite-element model of our actual experiment, in which small-scale yielding is not satisfied. We compute $G^{S'}$ directly from the finite-geometry energy budget ($G^{S'} = -dU/dA$), which does not assume small-scale yielding. We attach a prime, $G^{S'}$, to mark that this is a finite-geometry energy release rate obtained without a K -dominant field, in parallel with the apparent near-front dissipation Γ' . This is the same finite-geometry energy balance that Palmer and Rice (1973) showed governs rupture propagation without the small-scale-yielding approximation. The simplified analytical prediction of the same scaling was instead derived from standard LEFM, which implicitly assumes small-scale yielding. That the full-geometry energy budget and the LEFM prediction agree indicates the quadratic scaling is robust and not an artifact of the small-scale-yielding assumption. This is more than a consistency check: it shows the $G^{S'} \propto \overline{\sigma}_N^2$ scaling persists even when the process zone is not small compared with the fault.

Third, our central conclusion, that $\Gamma' \ll W_b$ and that the W_b scaling arises from nonlocal elastic unloading, rests on comparing the independently determined Γ' and W_b together with the finite-element energy budget, none of which require LEFM to be strictly valid. The breakdown work W_b in particular follows from the finite-geometry energy budget and is model-independent.

In short, the departure from small-scale yielding limits only the strict LEFM interpretation of the fitted amplitude and the absolute cohesive zone size X_c , not the scaling relations among Γ' , $G^{S'}$, and $\overline{\sigma}_N$. We have revised the relevant paragraph (L396ff) to make this explicit, as shown below (see also the revised text under Comment R1-18).

Revised text:

At a representative rupture length of $a = 75$ mm, the total energy release rate ($G_{\text{total}}^{S'}$) increases quadratically with $\overline{\sigma}_N$, showing no deviation from the simplified LEFM prediction $G_{II}^S \propto \overline{\sigma}_N^2$ (Fig. 6e), despite the geometric and compositional complexity of our

experimental setup. Notably, because $G^{S'}$ is computed directly from the finite-geometry energy budget rather than from a stress intensity factor, this agreement does not rely on small-scale yielding (cf. Palmer and Rice, 1973). We write $G^{S'}$, with a prime as for the apparent near-front dissipation Γ' , to denote a finite-geometry energy release rate determined without a K -dominant field. The quadratic scaling is therefore robust rather than an artifact of the small-scale-yielding assumption underlying the analytical prediction, and it persists beyond the small-scale-yielding regime. ~~This result indicates that first-order modeling is consistent with expectations and LEFM theory but it cannot explain the deviation from $C_f \propto 1 - c_0/\overline{\sigma}_N$ observed in our experiments (Fig. 4a; dotted curve), and suggests that fully dynamic simulations incorporating a friction law may be required.~~ Our finite-element model is static and is not intended to reproduce the rupture speed. It therefore does not capture the modest deviation of C_f from $C_f \propto 1 - c_0/\overline{\sigma}_N$ (Fig. 4a). Reproducing that deviation would likely require fully dynamic simulations with a friction law. Importantly, this deviation concerns only the rupture speed. It does not bear on our energy-budget analysis, from which we conclude that the breakdown-work scaling arises from system-scale elastic unloading.

Comment R1-24.

L 364: Fig. 5d does not show normal stress. Wouldn't it be easier to plot σ_n instead of τ/σ_n if that is the main point? Alternatively, the paragraph could be rephrased.

Response:

We thank the reviewer for this observation. The reviewer is correct that Fig. 6d (the former Fig. 5d) does not plot σ_N directly; it shows the shear stress τ and the shear-to-normal stress ratio τ/σ_N . Plotting σ_N alone is not our aim. The purpose of this paragraph is to caution that experimentalists commonly interpret the measured τ under the implicit assumption that $\overline{\sigma}_N$ is constant. In our finite-geometry setup σ_N is in fact spatially heterogeneous and evolves during slip, so τ and τ/σ_N exhibit markedly different spatial and slip-weakening behavior, with τ developing stress concentrations near the rupture tips while τ/σ_N remains comparatively uniform. As a result, $\tau(\delta)$ shows an apparent secondary weakening that, if interpreted assuming constant σ_N , would be misattributed to enhanced fracture energy at the rupture tip. We have rephrased the paragraph to make this cautionary point explicit and to no longer imply that Fig. 6d shows σ_N directly.

Revised text:

Due to the finite geometry of our experimental setup, σ_N varies spatially along the fault (~~Fig. 5d~~). This heterogeneity causes τ and τ/σ_N to exhibit markedly different spatial patterns, with τ showing pronounced stress concentrations near rupture tips while τ/σ_N remains more uniform (Fig. 6d). ... Consequently, interpreting the measured τ under the implicit assumption of a constant $\overline{\sigma}_N$, as is common in experimental studies, ~~measurements that assume uniform σ_N~~ can overestimate fracture energy Γ by misattributing this

curvature to enhanced dissipation at the rupture tip (Cowie et al., 1992). ...

Comment R1-25.

Fig. 5f, L367: Where were slip and stress measured?

Response:

We thank the reviewer for the question. The data in Fig. 6f (the former Fig. 5f) are taken from all nodes along the fault where $\delta > 0$, at the model step when rupture length = 75 mm, as stated in the caption; we have made the spatial extent more explicit by adding “at all fault nodes.”

Revised text:

... (f) Shear-to-normal stress ratio τ/σ_N (blue) and shear stress τ (orange) versus slip δ for $\delta > 0$ at all fault nodes at a rupture length of 75 mm. ...

Comment R1-26.

L 403ff: Several of the interpretations within the conclusions are speculative and casually extrapolate to natural faults. These statements should be removed or at least moved to a discussion section to separate observations from speculations.

Response:

We thank the reviewer for this suggestion. We agree that several sentences in the second paragraph of the Conclusions are speculative and more appropriate in a Discussion context. We have kept the two sentences directly supported by our data in the Conclusions and moved the remaining speculative content (the asperity analogy and broader implications) to a new subsection at the end of the Results and Discussion section.

Revised text:

3. Results and Discussion (new subsection added at the end)

3.7. Implications for Natural Faults

Our finding that nonlocal elastic unloading governs the breakdown-work scaling invites comparison with natural faults. Strong breakdown-work scaling with earthquake size is inferred from seismology (Abercrombie & Rice, 2005; Tinti et al., 2005) and is commonly attributed to enhanced near-fault weakening such as thermal pressurization (Viesca & Garagash, 2015; Wibberley & Shimamoto, 2005). Our results point to an alternative, in which fault areas surrounding a high-strength asperity, though not in direct contact, store and release the elastic energy that drives this scaling. In this view, a finite-length laboratory fault embedded in a compliant loading system is analogous to a small asperity embedded in a larger elastic medium, where nonlocal unloading rather than enhanced

weakening produces the scaling. Such an effect would be absent from laboratory and numerical models that assume perfectly mated fault interfaces. Our data therefore suggest that accounting for fault topography and contact heterogeneity, alongside frictional properties, may be essential for scaling rupture energetics from the laboratory to tectonic faults.

4. Conclusions

Our laboratory stick-slip experiments reveal that rupture speed C_f , stress drop $\Delta\bar{\tau}_{\text{total}}$, and breakdown work W_b scale systematically with normal load $\bar{\sigma}_N$. While these data are consistent with basic expectations they are not fully aligned with standard assumptions of LEFM. Although the near-front dissipation I' grows approximately linearly with $\bar{\sigma}_N$, it remains an order of magnitude smaller than W_b , indicating that most energy release arises from system-scale unloading, after the rupture has reached the fault boundaries. This later slip is due to energy release of the shear loading column rather than from localized weakening at the rupture tip. Finite-element analysis shows that the static energy release rate $G_{\text{II}}^{S'}$ follows the quadratic scaling predicted by LEFM despite the geometric complexity of the experimental setup, while also revealing that the compliant loading column dominates the energy budget even before full sliding occurs. Cohesive zone modeling further indicates that the cohesive zone size X_c exceeds the sample dimensions, violating the small-scale yielding assumption even though the model fits the dynamic strain data reasonably well.

Our results highlight the role of finite, length-limited fault patches in governing rupture energetics. Our data show that unlocking of such fault patches permits extensive nonlocal unloading, which reproduces the strong W_b scaling inferred in natural earthquakes. ~~Such effects are common in seismic inversions, especially for large ruptures where they are considered subevents of the main fault rupture. Our findings suggest that fault areas surrounding high-strength asperities, though not in direct contact, play a critical role in storing and releasing energy through elastic coupling. In this sense, a finite-length laboratory fault embedded in a compliant loading system behaves analogously to a small asperity embedded in a larger elastic medium, with nonlocal unloading providing a mechanistic explanation for the inferred breakdown work scaling. This effect may be less prominent in some laboratory and numerical models, which typically assume perfectly mated fault interfaces that do not promote nonlocal energy release. However, our data underscore the fact that accounting for the role of fault heterogeneities, in addition to frictional properties, is essential for understanding earthquake physics and scaling from lab to tectonic fault scales.~~ Together, these results identify nonlocal elastic unloading of the loading system, rather than enhanced near-tip dissipation, as the primary control on the breakdown-work scaling we observe.

Comment R1-27.

A table summarizing the results at different normal loads (e.g., stiffness, acrylic/steal, stress drop, slip, etc.) would be useful. This could be moved to the supplement.

Response:

We agree, and have added Table S1 to the Supplementary Material. For each experiment it lists the applied normal load $\bar{\sigma}_N$, the loading spring used for the analyzed slide–hold–slide (SHS) run(s), the loading stiffness k , the total stress drop $\Delta\bar{\tau}_{\text{total}}$, the total slip δ_{total} , the near-front dissipation Γ' , and the rupture speed C_f , together with the sequence of velocity-step (VS) and SHS runs. The reported values are means $\pm$ one standard deviation over the rupture events recorded at each normal load.

Table 1: **List of experiments and summary of results.** For each experiment the applied normal load $\bar{\sigma}_N$ and the loading spring of the analyzed slide–hold–slide (SHS) run(s) are listed, together with the loading-spring stiffness k , total stress drop $\Delta\bar{\tau}_{\text{total}}$, total slip δ_{total} , near-front dissipation Γ' , and rupture speed C_f . Values are reported as mean $\pm$ standard deviation over the rupture events recorded at each normal load.

#	$\bar{\sigma}_N$ (MPa)	Spring	k ($10^{-3} \mu\text{m}^{-1}$)	$\Delta\bar{\tau}_{\text{total}}$ (MPa)	δ_{total} (μm)	Γ' (J/m ²)	C_f (m/s)
p5983	1.0	acrylic	2.78 ± 0.50	0.08 ± 0.02	21 ± 5	0.5 ± 0.2	131 ± 75
p5971	2.0	acrylic	1.94 ± 0.10	0.11 ± 0.03	24 ± 6	1.0 ± 0.4	117 ± 88
p5972	3.0	acrylic	1.24 ± 0.07	0.15 ± 0.04	32 ± 8	1.6 ± 0.6	182 ± 137
p5973	4.0	acrylic	0.95 ± 0.04	0.29 ± 0.07	59 ± 14	4.1 ± 1.7	653 ± 125
p5974 [†]	5.0	acrylic	0.78 ± 0.07	0.46 ± 0.08	75 ± 19	5.9 ± 2.9	763 ± 67
p5975	7.5	acrylic	0.46 ± 0.03	0.89 ± 0.16	45 ± 10	5.3 ± 2.2	982 ± 123
p5979	10.0	acrylic	0.33 ± 0.04	1.36 ± 0.19	73 ± 17	11.1 ± 6.2	1194 ± 72
p5991	12.5	acrylic	0.23 ± 0.07	1.70 ± 0.37	95 ± 27	23.0 ± 10.5	1218 ± 88
p5992	15.0	steel	2.49 ± 0.21	1.46 ± 0.32	37 ± 7	28.9 ± 9.7	1432 ± 183
p5993	20.0	steel	2.11 ± 0.16	1.95 ± 0.20	40 ± 4	49.4 ± 9.5	1644 ± 90

[†]SHS₁ of p5974 was recorded at a low sampling rate and is therefore not usable.

Comment R1-28.

Fig. S4: black and red circles are not described in caption. How were these determined?

Response:

We thank the reviewer for catching this omission. Both the black and red circles mark manually picked rupture-front arrival times at each gauge location. They are distinguished by how they are used in the analysis. The red circles denote the picks included in the linear regression that estimates the rupture speed C_f , while the black circles denote picks excluded from that fit. We restrict the fit to this central segment because the rupture speed is not always constant along the entire fault. The selected segment spans the center gauge rosette, where the speed is approximately constant and where both strain components used for the cohesive zone model fit ($\Delta\varepsilon_{xy}$ and $\Delta\varepsilon_{yy}$) are recorded, providing additional robustness. We have added a description of the circles to the Fig. S4 caption.

Revised text:

(e) Shear strain changes $\Delta\varepsilon_{xy}$ at each gauge location along the fault. Circles mark the manually picked rupture-front arrival at each gauge. Red circles indicate the picks used to estimate the rupture speed C_f by linear regression, spanning the central segment of approximately constant speed around the center gauge rosette, and black circles indicate picks excluded from the fit. At approximately $t = -0.25$ s (relative to dynamic rupture), $\bar{\tau}$ decreases slightly ...

Reviewer 2

Summary:

This is an interesting study about the physical significance of the fracture energy and breakdown work in laboratory earthquakes and their dependence with the final slip distance. By performing sophisticated experiments with a double direct shear apparatus on quartz powders sandwiched by Westerly granite blocks, the authors investigate spontaneous ruptures of the experimental fault interfaces and record strain (with 13 strain gages rosettes), displacements, rupture velocities, etc. at high acquisition rates (up to 100 kHz). The interpretation of the carefully conducted experiments is also supported by numerical models of earthquake energy budgets conducted with the software ABAQUS.

The authors find that the breakdown work scales with normal stress, stress drop and the square of the slip distance, whilst the fracture energy (of the order of 10 J/m²), associated with the onset of slip, has a weak dependence on the normal stress. A similar conclusion was reached by Paglialonga et al., EPSL 2022, as stated in the title of their paper:

"On the scale dependence in the dynamics of frictional rupture: Constant fracture energy versus size-dependent breakdown work"

However, if Paglialonga et al., 2022 attribute the breakdown work to on-fault dissipated energy following slip initiation (see the last statement of their abstract):

"We conclude that the origin of the seismological estimates of breakdown work could be related to the energy dissipated in the long-tailed weakening rather than to the one dissipated near the tip."

in this manuscript breakdown work (and its scaling with normal stress) is proposed to be associated with nonlocal (i.e., mainly from the experimental machine) elastic strain unloading (lines 48-60):

"We interpret this scaling not as a signature of increased fracture resistance and fracture energy, but as a consequence of greater energy release during elastic unloading of the fault and surrounding material including the loading column in our experiments.... The additional energy released as the fault completely unlocks arises from this elastic unloading, without requiring secondary weakening mechanisms (e.g., Gabriel et al., 2024; Lambert et al., 2021; Viesca and Garagash, 2015), and contributes to the observed scaling

of breakdown work (e.g., Cocco et al., 2023; Abercrombie and Rice, 2005)."

Another interesting finding of the study is what, to my knowledge, represents the first estimate of the elastic strain energy stored within the machine and that released by the machine into the experimental fault. The energies stored and released are approximately an order of magnitude greater than those stored and released by the rocks bordering the experimental fault.

Response:

We thank the reviewer for the careful and accurate summary of our work, and in particular for highlighting the significance of our estimate of the elastic strain energy stored in and released by the loading system. We address the two main comments and the minor comments in turn below.

Main comments

Comment R2-1.

The manuscript is generally well written, though I got confused for one main point that I invite the authors to discuss. According to the literature, breakdown work is (Cocco et al., 2023):

"The energy dissipated in the earthquake rupture propagation along a fault is called fracture energy or breakdown work (Cocco et al., 2023)"

The breakdown work is interpreted to be related to dissipative, mainly friction-related, processes occurring behind the rupture tip. This interpretation is consistent with experimental observations, particularly when the experimental fault is capable of accommodating large slip displacements in a rotary shear configuration, both at 'constant slip velocity' (see Nielsen et al., GRL 2021; JOSE 2021), and in abrupt slip pulses (Harbord et al., GRL 2021). The breakdown work increases with fault slip, and so does the temperature in the slip zone (e.g., Aretusini et al., 2021 GRL; Aretusini et al., GRL 2024). This perhaps is not surprising, as most of the mechanical work density is converted into heat while frictional power density into temperature increase (Aretusini et al., 2024 and, though these are not rotary shear experiments, Ortega-Arroyo et al., 2025 AGU Advances). In rotary shear experiments, the velocity functions, "constant" or "pulses", are imposed to the samples, and the amount of elastic strain energy release from the machine is limited.

Here the authors refer to "nonlocal elastic unloading of the fault blocks and loading column". Based on the results of the numerical model, and in line with the elastic rebound model and the physics of the 'spring slider' (as illustrated on pages 15-16) the release of elastic strain energy from the loading column (1) drives the rupture front and, once the rupture has reached the fault boundaries (2) the sliding of the block (lines 165-169, but see also lines 384-386):

"This two-phase behavior reflects distinct energy release mechanisms: initial rapid weak-

ening associated with local energy dissipation at the propagating rupture tip, followed by continued stress reduction driven by nonlocal elastic unloading of the fault blocks and loading column, including cases where the center block of the DDS arrangement undergoes nearly-rigid-body motion."

In fact, the duration of the slip is longer than that of the rupture. If the three blocks of the double direct shear remain in contact, frictional slip and frictional heating must occur along the fault interface. Therefore, ultimately, nonlocal elastic unloading (or breakdown work, in this article) should cause grain fragmentation and frictional heating (Aretusini et al., 2024). However, the magnitude of the fracture work in these experiments is far too low to cause a significant increase in temperature in the slip zone. Based on rough calculations, the mechanical work density for an experiment conducted with a normal stress of 20 MPa and a displacement of 40 micrometres (e.g., Fig. 2i) would be of the order of 600 J/m². Considering the fault surface area of 150 mm × 50 mm, this would result in a mechanical work (ΔW) of approximately 4.5 J. If all this work were converted into heat, given the mass of the quartz powder (0.75 g, see Methods) and the specific heat of quartz (750 J/kg K), this would result in a temperature increase in the slip zone of a fraction of a Kelvin:

$$\Delta W = Q = m c_p \Delta T$$

Other rough estimates of the maximum temperature achieved in the slipping zone could be done if the fault slip velocity is determined (e.g., Carlslaw and Jaeger, 1959):

$$\text{Temperature increase} = (\text{shear stress} * \text{slip rate}) (\text{slip duration})^2 / [\text{rock density} * \text{specific heat} * (3.14 * \text{thermal diffusivity})^2]$$

Also in this case, if we consider slip rates ranging between 0.1-1 m/s for slip durations of tens of ms, probably the temperature increase would be less of 1 K.

In conclusion, where does this released deformation energy end up? There must be heat dissipation along the experimental fault: how significant is it? And how does this energy dissipation relate to the breakdown work (= non-local elastic unloading of the fault blocks and the load column) discussed in the manuscript? How is it possible that this energy is not converted into frictional heating and processes of enhanced dynamic weakening along the fault? Incidentally, this would be at odds with any geological evidence relating to natural faults. In fact, although technically challenging, it would have been useful to measure temperature variations in the vicinity of the slip zone and assess how much of the non-local release of deformation energy is converted into heat on the fault.

Can the authors comment on this in their revised version of the manuscript?

Response:

We thank the reviewer for this thoughtful comment and for the careful temperature estimates. The reviewer asks where the released deformation energy ends up and how the heat dissipated on the fault relates to the breakdown work. The short answer is that most of the released elastic energy is expected to end up as frictional heat on the fault, and the corresponding temperature rise is only a few kelvin. The more fundamental clarification concerns what the

W_b scaling means. The fracture energy Γ measures the energy dissipated at the rupture tip at the onset of slip, the energy the rupture requires to propagate, whereas W_b measures the energy the system releases once the fault unlocks. A growing W_b therefore does not mean that rupture requires more energy. It means that the surrounding system releases more energy, which ends up as heat, radiated waves, and kinetic energy. We develop this with explicit numbers below and in a new paragraph in the Results and Discussion section.

Following the reviewer, we use an event at $\bar{\sigma}_N = 20$ MPa with total slip $\delta \approx 40$ μm (Fig. 2i) as a worked example throughout the calculations below. From the dynamic strain records (Fig. 3b–c), the breakdown and associated slip develop over $t \approx 1$ ms, with the near-tip weakening completing within $\lesssim 0.1$ ms. For a total slip of $\delta \approx 40$ μm this corresponds to a mean slip rate

$$\bar{V} = \frac{\delta}{t} \approx \frac{40 \times 10^{-6} \text{ m}}{1 \times 10^{-3} \text{ s}} \approx 0.04 \text{ m s}^{-1},$$

rising to ~ 0.1 m/s if slip is concentrated within the first half of the window. (For reference, slip rates of 0.1–1 m/s sustained for tens of milliseconds would correspond to millimeters of slip, far exceeding the ~ 40 μm in our events, so the lower values are the relevant ones here.) The work done against friction per unit fault area, taking the measured sliding shear stress $\bar{\tau} \approx 14.2$ MPa at $\bar{\sigma}_N = 20$ MPa (Fig. 2i), is

$$q'' = \bar{\tau} \delta \approx (14.2 \times 10^6 \text{ Pa}) \times (40 \times 10^{-6} \text{ m}) \approx 5.7 \times 10^2 \text{ J m}^{-2},$$

in close agreement with the reviewer’s estimate of ~ 600 J m^{-2} . Over the fault area $A = 0.15 \times 0.05 = 7.5 \times 10^{-3} \text{ m}^2$ this is a total heat $Q = q''A \approx 4.3$ J, again close to the reviewer’s 4.5 J. Deposited in the ~ 0.75 g quartz-gouge layer (~ 250 μm thick, specific heat $c \approx 750$ $\text{J kg}^{-1} \text{K}^{-1}$), the temperature rise is

$$\Delta T = \frac{Q}{mc} \approx \frac{4.3}{(0.75 \times 10^{-3})(750)} \approx 8 \text{ K}.$$

This follows the reviewer’s own mass-based bookkeeping, though we note that the same arithmetic yields several kelvin rather than a fraction of a kelvin, which does not change the conclusion. The estimate is also robust to localization. The thermal diffusion length over the ~ 1 ms event, $\sqrt{\kappa t} \approx 40$ μm (with $\kappa = k/\rho c \approx 1.5 \times 10^{-6} \text{ m}^2 \text{s}^{-1}$, $k \approx 3 \text{ W m}^{-1} \text{K}^{-1}$, $\rho c \approx 2.0 \times 10^6 \text{ J m}^{-3} \text{K}^{-1}$), is smaller than the ~ 250 μm layer thickness, so the heat is retained within the gouge during the event, and even for shear localized onto a thin slip surface the spreading heat gives $\Delta T \approx q''/(\rho c \sqrt{\kappa t}) \approx 7$ K.

With these numbers the energy budget can be stated explicitly. The total work done against friction, $\approx 6 \times 10^2 \text{ J m}^{-2}$, is by far the largest term, and the great majority of it, done at the residual stress level, is expected to be converted to frictional heat on the fault. The breakdown work is the excess above the minimum sliding stress, $W_b = \int_0^{\delta_0} (\bar{\tau} - \bar{\tau}_{\min}) d\delta \approx 0.5 \Delta \bar{\tau}_{\text{total}} \delta_0$, roughly ten percent of the total (Fig. 4d). How W_b itself partitions among additional fault heating, radiated waves, and kinetic energy of the near-rigidly translating center block (L182–187, Fig. 3d) is not directly constrained by our measurements, but all of these sinks are modest and none requires enhanced weakening at the fault. The near-front dissipation $\Gamma' \approx 10 \text{ J m}^{-2}$, the energy dissipated locally during the initial weakening at the onset of slip, is smaller still. Heat dissipation on the fault is therefore energetically dominant yet thermally insignificant.

This also resolves the apparent contradiction with enhanced dynamic weakening. Thermally activated mechanisms such as flash heating require temperature rises of hundreds of kelvin, reached in rotary-shear experiments through imposed slips of centimeters to meters (Nielsen et al., 2021; Aretusini et al., 2021, 2024). Our events slip tens of micrometers, three to five orders of magnitude less, so a few-kelvin rise neither activates such mechanisms nor would be expected to. This is precisely why we interpret the strong W_b scaling as a signature of elastic unloading rather than of thermal weakening. Nor is this at odds with natural faults, where meters of coseismic slip can produce substantial heating. It cautions only that a large inferred breakdown work does not by itself imply a large frictional temperature rise when much of the energy budget is governed by elastic unloading of the surrounding medium.

We agree with the reviewer that direct temperature measurements near the slip zone would be valuable. Although technically challenging, such measurements could directly constrain how much of the released energy is converted to heat on the fault, and we view them as a promising direction for future work. We have added a paragraph to the Results and Discussion section making this energy budget explicit, as shown below.

Revised text:

3.6. Energy partitioning and the fault thermal budget A natural question is where the released elastic energy ends up, and whether the strong scaling of breakdown work implies correspondingly strong frictional heating and thermally enhanced weakening on the fault (e.g., Nielsen et al., 2021; Cocco et al., 2023). While we do not measure temperature directly, mechanical work done against friction is expected to be largely converted to heat (Aretusini et al., 2021, 2024; Ortega-Arroyo et al., 2025), so most of the released energy likely ends up as frictional heat on the fault. Taking an event at $\bar{\sigma}_N = 20$ MPa as an example, the work done against friction is $\int \bar{\tau} d\delta \approx \bar{\tau} \delta \approx 6 \times 10^2 \text{ J m}^{-2}$, and the breakdown work $W_b = \int (\bar{\tau} - \bar{\tau}_{\min}) d\delta$ is the roughly ten percent of this work done above the minimum sliding stress (Fig. 4d). The near-front dissipation $\Gamma' \approx 10 \text{ J m}^{-2}$, the energy dissipated locally during the initial weakening at the onset of slip, is smaller still. How W_b itself partitions among additional fault heating, kinetic energy of the near-rigidly translating center block (Fig. 3d), and radiated waves is not directly constrained by our measurements, but each of these sinks is modest. Spread through the ~ 0.75 g gouge layer, the total frictional heat corresponds to a temperature rise of only $\Delta T = \bar{\tau} \delta A / (m c) \approx 8 \text{ K}$, two orders of magnitude below the rises required to activate flash heating, thermal pressurization, or melting (e.g., Viesca and Garagash, 2015; Harbord et al., 2021), so enhanced thermally activated weakening is neither expected nor observed. The scaling of W_b therefore does not indicate that rupture requires more energy to propagate, which is the role of Γ . It indicates that the surrounding system, including the loading column, releases more energy as the fault unlocks, energy that ends up as modest heat, radiated waves, and kinetic energy.

Comment R2-2.

One of the main results is that the breakdown work scales quadratically with slip distance (Fig. 4d). This is not surprising because of the spring slider model that the direct double shear configuration used here reproduces. The experimental fault has finite and predetermined dimensions, so the slip distance "delta" should scale with stress drop (see discussion in Udias et al. 2014 and references therein). As a consequence, the breakdown work has to scale with δ^2 (see lines 245-247). Then it is written (lines 52-55):

"Our experiments demonstrate this for length-limited faults embedded in a compliant loading system, which is analogous to fault segments embedded in a damage zone or other cases where a small fault patch releases elastic energy stored in a larger volume."

But the strain energy release in natural faults is different from that in the used experimental setup. The "spring" in the experiment releases energy proportionally to the slip on a fault with predetermined dimensions, whereas in faults, the energy release increases not only with slip but also with the elastic volume involved (which grows as the cube of the fracture length L). Since the front (or perimeter) of the rupture grows as L^2 , the strain energy release per unit of rupture front grows with fault length L . In conclusion, since the stress drop, although difficult to measure and highly variable in natural earthquakes, is independent of fault slip (i.e., of the earthquake magnitude), the breakdown work in natural faults should be proportional to the fault slip, not to its square (see discussion in Nielsen et al., 2016 JOSE).

Can the authors comment on this?

Response:

We thank the reviewer for this insightful comment, with which we largely agree. The quadratic dependence $W_b \propto \delta_0^2$ in our experiments is indeed a consequence of the fixed, predetermined fault dimensions. Because the fault size is set, slip is limited by the elastic relaxation of the surrounding system, the stress drop and slip are therefore coupled ($\Delta\bar{\tau}_{\text{total}} \propto \delta_0$, Fig. 2j), and $W_b \approx 0.5 \Delta\bar{\tau}_{\text{total}} \delta_0$ then scales as δ_0^2 . This is the mechanism we state at L266–267. We also agree that self-similar natural ruptures behave differently, because they are not dimension-limited in the way our fault is, so slip and stress drop need not co-vary and the quadratic scaling we measure should not be expected to carry over. We did not intend to claim that our δ^2 scaling represents self-similar earthquake growth, and we do not extrapolate our exponent of two to natural ruptures (Udías et al., 2014; Nielsen et al., 2016).

At the same time, we would like to point out that the reviewer's description of natural faults contains the same physics that our experiments isolate. In the self-similar case the strain energy is drawn from a volume that grows faster than the rupture area, so the energy supply is controlled by the surrounding volume rather than by processes at the rupture tip, even though the local frictional behavior does not change. This is precisely the nonlocal control of breakdown work that our study advocates. In our experiments the energy reservoir is the fixed but highly compliant loading system, while in a self-similar earthquake it is the growing elastic volume around the rupture. The fixed fault dimension then couples stress drop to slip and yields $W_b \propto \delta^2$ in our setup. For natural ruptures the dimension is not fixed and the stress drop is variable, so this coupling need not hold and the same nonlocal energy supply

need not produce a single breakdown-work scaling.

The analogy at L52–55 is meant in this restricted sense, a fault patch of roughly fixed size, such as an asperity or a fault segment embedded in a more compliant damage zone, that releases elastic energy stored in a larger surrounding volume. It is not meant as a model for self-similar rupture growth across magnitudes. We also note that the slip exponents inferred for natural earthquakes span a broad range (≈ 0.67 –2, Abercrombie and Rice, 2005; Cocco et al., 2023), which is consistent with bounded ruptures such as ours approaching an exponent of two while the breakdown-work scaling for self-similar ruptures is far less constrained. We have revised the Introduction to scope the analogy explicitly and revised the corresponding paragraph in the Results and Discussion section (L263–284), as shown below.

Revised text:

Introduction (L52–55):

Our experiments demonstrate this for length-limited faults embedded in a compliant loading system, which is analogous to fault segments embedded in a damage zone or other cases where a small fault patch releases elastic energy stored in a larger volume. This analogy holds only in a restricted sense, for a fault patch of roughly fixed size, such as an asperity or fault segment embedded in a more compliant surrounding volume, for which the fixed dimension makes slip and stress drop co-vary. It is not a model for self-similar rupture growth, where the rupture dimension is not fixed and the stress drop is variable.

Revised text:

Revised paragraph in the Results and Discussion section (L263–284):

... A best-fit power law with an exponent of 2 is consistent with scalings inferred for natural earthquakes (exponent range 0.67–2, Abercrombie and Rice, 2005; Cocco et al., 2023). The $\sim \delta_0^2$ scaling arises because $W_b \sim 0.5 \bar{\tau}_{\text{total}} \Delta \bar{\tau}_{\text{total}} \delta_0$ and $\Delta \bar{\tau}_{\text{total}}$ scales approximately linearly with δ_0 (Fig. 2j). ~~These results demonstrate that laboratory faults with length-limited contacts can reproduce key aspects of natural fault behavior inferred from seismology.~~ This quadratic dependence is a direct consequence of the fixed dimensions of our laboratory fault, where slip is limited by the elastic relaxation of the surrounding compliant system and stress drop and slip therefore co-vary. Natural ruptures are not dimension-limited in the same way, so slip and stress drop need not co-vary as they do here, and the quadratic scaling we measure should not be expected to carry over to them (Udías et al., 2014; Nielsen et al., 2016). In both cases the breakdown work is set by the elastic energy released from the volume surrounding the rupture rather than by frictional processes at the rupture tip. In our length-limited fault the fixed dimension couples slip to stress drop and yields an exponent of two, as it would for a fault patch of roughly fixed size, such as an asperity or fault segment embedded in a more compliant surrounding volume. This is not a model for self-similar rupture growth, where the rupture dimension is not fixed and the stress drop is variable, so this nonlocal mechanism does not imply any single slip exponent, consistent with the broad range of

exponents reported for natural earthquakes. In this restricted sense, laboratory faults with length-limited contacts reproduce key aspects of natural fault behavior inferred from seismology.

Minor comments

Comment R2-M1.

Add a table with a list of experiments.

Response:

We thank the reviewer for this suggestion. Table S1 has been added to the Supplementary Material listing all experiments with key parameters. Please see our response to Comment R1-27 for a detailed description of the table contents.

Comment R2-M2.

What experiment is the one reported in Fig. 3?

Response:

We thank the reviewer for the question. Figure 3 shows two events. Panel (a) is event 4 from experiment p5974 ($\overline{\sigma}_N = 5$ MPa, run SHS₂), selected because its nucleation and dynamic rupture phases are visible together, while panels (b)–(d) show event 1 from experiment p5979 ($\overline{\sigma}_N = 10$ MPa, run SHS₂). These identifiers are also printed at the top of the corresponding panels in the figure for the reader’s reference.

Highlights

Nonlocal Elastic Unloading as a Mechanism for Breakdown Work Scaling in Laboratory Earthquakes

Chun-Yu Ke, Gauss T. Chang, Gregory C. McLaskey, David S. Kammer, Chris Marone

- Breakdown work ~~seals strongly with normal load in laboratory earthquakes due to~~ scaling reflects system-scale unloading, ~~whereas the local fracture energy scales only weakly with fault normal stress~~ not near-front dissipation.
- Finite element ~~analysis~~ model reveals that loading system compliance dominates energy release.
- ~~Our lab data suggests that nonlocal elastic unloading due to unlocking of length-limited fault contacts~~ Nonlocal elastic unloading reproduces breakdown work scaling ~~seen~~ in natural earthquakes.

Nonlocal Elastic Unloading as a Mechanism for Breakdown Work Scaling in Laboratory Earthquakes

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Abstract

Understanding how earthquake energy budgets scale from laboratory to natural faults remains a fundamental challenge in earthquake physics. Laboratory experiments report fracture energies consistent with classical friction, yet seismological observations show that breakdown work scales strongly with slip, and the relationship between these two quantities remains unclear. Here we show that this discrepancy reflects two genuinely different physical quantities: ~~local fracture energy~~ the near-front dissipation governed by rupture-tip processes, ~~and which we use as a proxy for fracture energy because these small ruptures sustain no K -dominant field, and the~~ system-scale breakdown work dominated by nonlocal elastic unloading which includes the loading column in our experiments. We conducted stick-slip experiments on 15-cm long granite faults under normal stresses of 1–20 MPa in a double-direct-shear configuration. A dense strain-gauge array tracked rupture propagation, enabling ~~fracture energy~~ near-front dissipation estimates via cohesive zone modeling, while finite element simulations quantified energy partitioning between the fault and loading system. ~~Local fracture energy~~ The near-front dissipation remains low (<10 J/m²) and scales roughly linearly with normal stress, consistent with classical friction. In contrast, total breakdown work scales quadratically with slip, exceeding ~~fracture energy~~ the near-front

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dissipation by over an order of magnitude. Finite element simulations confirm that elastic unloading of the compliant loading column dominates this energy release. Normal stress heterogeneity on the lab fault produces additional weakening, after the rupture has reached maximum size, without additional dissipative mechanisms. These results demonstrate that apparent breakdown work scaling in laboratory earthquakes can arise from nonlocal elastic effects rather than intrinsic fault weakening. Our work provides a framework for distinguishing local ~~fracture energy~~ near-front dissipation from system-scale energy release and breakdown work when extrapolating laboratory results to natural faults.

Keywords:

Rupture dynamics, Laboratory earthquakes, Breakdown work, Fracture energy

1. Introduction

Earthquake rupture reflects a dynamic redistribution of elastic strain energy within the fault–host rock system, balancing energy release from the surrounding medium against dissipation on and near the fault. This global energy flow governs the seismic stress drop and other source properties, including the rupture propagation speed C_f . Under suitable assumptions, this process can be formulated using linear elastic fracture mechanics (LEFM; Freund, 1990), in which rupture growth satisfies an energy balance $G = \Gamma$ between the dynamic energy release rate G and the fracture energy Γ (Kammer et al., 2024). In the context of LEFM, earthquake ruptures accelerate when the static energy release rate G_{II}^S (mode II) exceeds Γ (Svetlizky et al., 2020; Albertini et al., 2021).

Most experimental, analytical, and numerical studies have focused on the dissipative side of this balance, examining how rupture energy dissipation depends on slip rate, normal stress, fault roughness, and other weakening mechanisms (e.g., Cocco et al., 2006; Okubo et al., 2019; Kammer and McLaskey, 2019; Perry et al., 2020; Paglialunga et al., 2022; Cocco et al., 2023; Morad et al., 2024; Brantut, 2025). Far less attention has been given to the release side, namely, how the surrounding elastic medium unloads and transfers energy toward the rupture front. Yet this nonlocal energy supply can strongly influence rupture properties such as acceleration, total breakdown work, and their apparent dependence on normal stress.

Fault normal stress governs both sides of the fracture energy balance. Increasing normal stress enhances frictional strength and contact area (Dietrich and Kilgore, 1994), raising dissipation, but can also increase the elastic energy stored prior to rupture. In nature, ruptures at depth, where normal stress is high, tend to propagate faster than shallow or near-trench events (Vidale and Houston, 1993; Bilek and Lay, 2002). Laboratory experiments show the same trend: as normal stress increases, rupture speed increases (Johnson and Scholz, 1976; Leeman et al., 2016), approaching the Rayleigh or shear-wave velocity, and transitions to supershear rupture become more common (Xia, 2004; Dong et al., 2023). These observations point to a fundamental coupling between energy storage, release, and dissipation that governs rupture properties including speed.

In this study, we investigate small-scale laboratory earthquakes on a 15-cm long granite fault instrumented with a dense 13-station strain-gauge array. Our experiments enable spatially resolved measurements of rupture dy-

namics akin to those in larger, meter-scale experiments, but with additional
 control on loading rates and history. This allows us to reconstruct rup-
 ture fronts and estimate ~~fracture energy~~ near-front dissipation locally, while
 we estimate breakdown work at the apparatus level, enabling comparison
 between local and system-wide energy dissipation. ~~While inferred fracture~~
~~energy~~ Because these ruptures are small, with a cohesive zone comparable to
the fault length, the near-tip field is not K -dominated; we therefore report
the local near-tip energy not as a strict fracture energy but as an apparent
near-front dissipation Γ' , a proxy for fracture energy obtained directly from
the area under the initial weakening curve (Section 3.4). While the inferred
near-front dissipation remains low, consistent with previous laboratory stud-
 ies on single, relatively simple faults (e.g., Svetlizky et al., 2017; Bayart
 et al., 2015; Ke et al., 2018; Kammer and McLaskey, 2019; Paglialunga et al.,
 2022), total breakdown work increases strongly with normal stress (correlat-
 ing with slip and stress drop), similar to trends observed in natural earth-
 quakes (Cocco et al., 2023; Selvadurai, 2019; Viesca and Garagash, 2015;
 Abercrombie and Rice, 2005). We interpret this scaling not as a signature
 of increased fracture resistance and fracture energy, but as a consequence of
 greater energy release during elastic unloading of the fault and surrounding
 material including the loading column in our experiments. Our experiments
 demonstrate this for length-limited faults embedded in a compliant loading
 system, which is analogous to fault segments embedded in a damage zone
 or other cases where a small fault patch releases elastic energy stored in
 a larger volume. This analogy holds only in a restricted sense, for a fault
patch of roughly fixed size, such as an asperity or fault segment embedded in
a more compliant surrounding volume, for which the fixed dimension makes
slip and stress drop co-vary. It is not a model for self-similar rupture growth,
where the rupture dimension is not fixed and the stress drop is variable. The
 additional energy released as the fault completely unlocks arises from this
 elastic unloading, without requiring secondary weakening mechanisms (e.g.,
 Gabriel et al., 2024; Lambert et al., 2021; Viesca and Garagash, 2015), and
 contributes to the observed scaling of breakdown work (e.g., Cocco et al.,
 2023; Abercrombie and Rice, 2005). These findings suggest that breakdown
 work scaling can be governed by system-scale elastic unloading rather than
 reflecting only the local weakening, offering a new perspective on energy
 partitioning in laboratory and natural earthquakes.

2. Methods

2.1. Experimental Setup

Friction experiments were conducted in the double-direct-shear (DDS) configuration using a biaxial load frame with adjustable loading stiffness (Leeman et al., 2016; Shreedharan et al., 2020). The samples consisted of a center block and two side blocks, all prepared from Barre Gray Granite, forming two 150 mm $\times$ 50 mm frictional interfaces (Fig. 1a). All edges of the center block and the top and bottom edges of the side blocks were chamfered 5 mm to reduce stress concentrations and prevent edge damage during repeated experiments. Throughout each experiment, normal load $\overline{\sigma}_N$ was held constant (force-controlled using a fast-acting servo system), while shear load $\overline{\tau}$ was applied through prescribed load-point velocities (using servo displacement control).

Along one of the 150 mm long faults (Fig. 1a), we installed 13 strain gauge rosettes to measure dynamic shear strain ε_{xy} at equally spaced locations, and normal strain ε_{yy} at 3 selected locations ($x=3, 75$, and 147 mm). The center of each strain gauge was located 8 mm from the fault. The dense array of ε_{xy} measurements enabled us to track the spatiotemporal onset of shear strain drop along the fault and estimate rupture speed (Fig. 1c) of laboratory earthquakes (labquakes). We used a servo-controlled loading sequence to produce 10's to 100's of labquakes in each experiment under the same conditions. Displacements at the load-point δ_{LP} and near the top of the center block δ (Fig. 1a, blue square), together with normal load $\overline{\sigma}_N$ and shear load $\overline{\tau}$, were recorded at 1 kHz, while dynamic strains were recorded separately at 100 kHz ~~and manually synchronized.~~ The two systems measure independent quantities. The slow-sampling channels characterize system-level slip and stress while the strain gage array captures the dynamic rupture waveforms used for rupture speed and cohesive zone model fitting. Because the interseismic period between events is orders of magnitude longer than the rupture duration, events are unambiguously identified in both recording systems.

2.2. Sample Preparation and Testing Protocol

All experiments were conducted in the Rock Mechanics Laboratory at The Pennsylvania State University. We conducted experiments at normal loads $\overline{\sigma}_N$ ranging from 1 to 20 MPa, all following the same testing protocol. Before each experiment, the granite fault surfaces were roughened with #60 grit

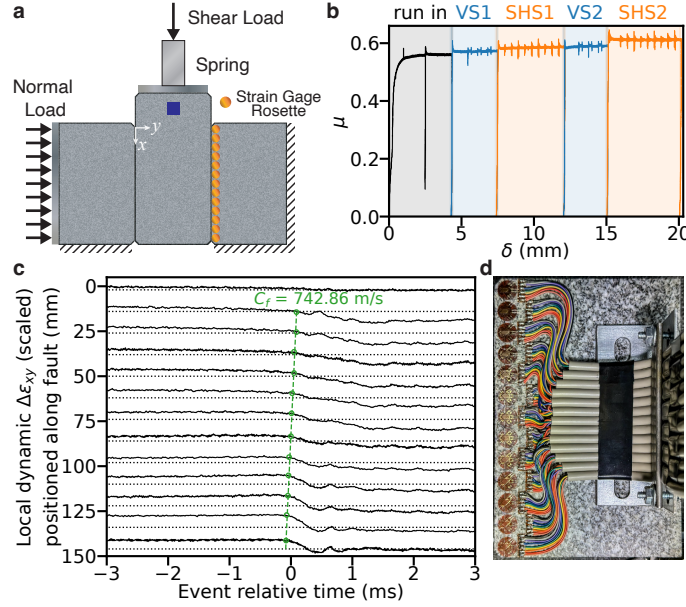


Figure 1: Double-direct shear experiment configuration. (a) Three Barre Gray granite blocks (two side blocks and one center block) are arranged under biaxial loading with bottom support for the two side blocks. Shear load is applied via an interchangeable loading block as a spring at the top of the center block. The right side block is instrumented with 13 strain gauge rosettes along the 150 mm fault. Blue square near the top of the center block marks the location of center block displacement δ measured relative to the bottom of the load frame. (b) Experimental protocol includes an initial run-in phase in which the fault surfaces are conditioned prior to testing, followed by alternating blue shading indicates velocity-step tests of velocity steps (VS*i*) conducted with a stiff steel loading spring (steel), and slide-hold-slide orange shading indicates slide-hold-slide tests (SHS*i*) with a compliant acrylic loading spring (acrylic), in which $\mu = \bar{\tau}/\bar{\sigma}_N$ is the friction coefficient and δ is center block displacement. (c) Rupture speed is estimated through fitting Dynamic shear strain ϵ_{xy} recorded at each strain gauge rosette along the fault, plotted as a line function of time relative to the time of rupture event onset tracked from. Horizontal dotted lines mark the shear zero strain ϵ_{xy} along change for each gauge trace. The topmost gauge did not record a rupture signal, likely because the rosette was too close to the top edge of the fault and was not in full contact with the fault surface. Green circles mark the onset of rupture at each gauge location; the green dashed line is fitted to these circles, and its slope gives the average rupture propagation speed. (d) A picture of the right side block, aligned to the strain gauge rosette positions shown in (c).

110 silicon-carbide (producing an RMS roughness of $\sim 20 \mu\text{m}$) and a thin layer of
 111 moisturized quartz powder (Min-U-Sil 40) was applied to each fault ($\sim 0.75 \text{ g}$;

112 ~ 250 μm thick). This preparation procedure was adopted from Shreedharan
 113 et al. (2020) to achieve steady-state friction conditions and produce long sets
 114 of reproducible labquakes. As shown in Fig. 1b, the protocol included run-in,
 115 velocity-step (VS), and slide-hold-slide (SHS) tests. After applying normal
 116 stress, the center block was sheared to the load-point displacement of ~ 5
 117 mm, with an unload-reload cycle at ~ 3 mm, to run in the faults. VS and
 118 SHS tests were then performed in alternating sequence (Fig. 1b) to assess
 119 cumulative strain effects, denoted VS_i and SHS_i according to their order. To
 120 generate reproducible labquakes with dynamic rupture, we used an acrylic
 121 block as a spring (80-mm long and 40x40 mm cross-sectional area) to reduce
 122 the loading stiffness and match the critical rate of frictional weakening K_c
 123 (e.g., Leeman et al., 2016; Scuderi et al., 2017). The acrylic block was used
 124 only for experiments at < 15 MPa normal stress. It was installed after the
 125 initial run-in and VS tests to produce labquakes during SHS tests.

126 To verify that this sample exhibited frictional behavior comparable to pre-
 127 vious experiments on the same apparatus with Westerly Granite (e.g., Lee-
 128 man et al., 2016), we conducted VS tests to estimate rate-and-state friction
 129 parameters and K_c . The shear rate (load-point velocity) began at 10 $\mu\text{m/s}$,
 130 then alternated between 3 and 30 $\mu\text{m/s}$ for three up-steps, with 500 μm of
 131 shear at each velocity. At normal loads above 7.5 MPa, we omitted VS tests
 132 because the fault did not sustain steady sliding and instead exhibited only
 133 stick-slip, precluding reliable rate-and-state fits. The estimated rate-and-
 134 state parameters (Supplementary Fig. S1) are consistent with those reported
 135 in previous works on Westerly granite and are therefore not discussed further.

136 SHS tests were used to induce dynamic rupture at low normal loads,
 137 where the conditions are otherwise in the stable sliding regime (Leeman et al.,
 138 2016; Pignalberi et al., 2024). Each SHS sequence comprised 3, 30, and 300
 139 s holds, followed in each case by 500 μm of shear at 10 $\mu\text{m/s}$. Repeating
 140 the sequence 3 times produced 9 controlled stick-slip events per test. For
 141 consistency, we focus our detailed analysis on these 9 post-hold events, and
 142 use other events from our series of 15 tests to assess reproducibility and
 143 frictional behavior. At 15 and 20 MPa, we did not use the acrylic loading
 144 block (spring in Fig. 1a) because at these normal stresses the system satisfies
 145 the conditions for instability and exhibits stick-slip motion and labquakes.
 146 For each stick-slip event, we measured the total shear stress drop $\Delta\bar{\tau}_{\text{total}}$,
 147 total slip δ_0 , loading stiffness k taken from the linear segment of $\Delta\mu/\Delta\delta$
 148 prior to failure, and the breakdown work $W_b = \int_0^{\delta_0} (\bar{\tau} - \bar{\tau}_{\text{min}}) d\delta$, computed

149 from the center-block displacement δ , shear load $\bar{\tau}$, and normal load $\bar{\sigma}_N$ (see
150 Fig. 2).

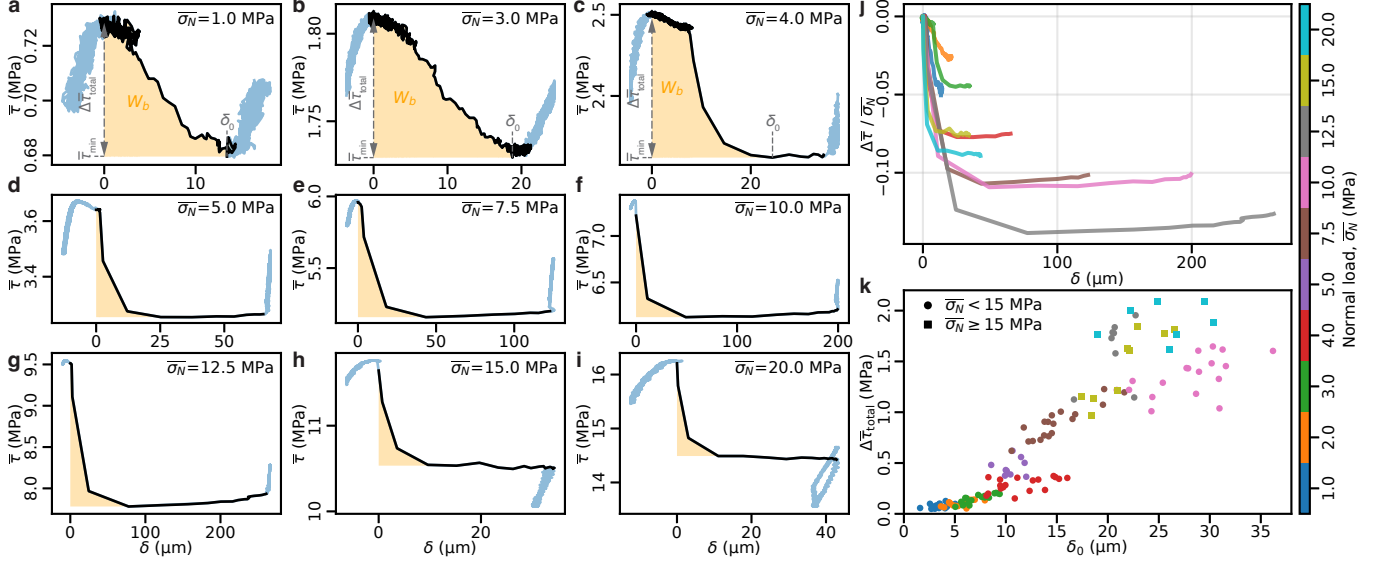


Figure 2: Dynamic stress drop and slip characteristics of laboratory earthquake events under different normal loads. (a–i) Shear load $\bar{\tau}$ versus fault displacement δ for events following SHS tests with 3-second holds, at $\bar{\sigma}_N$ from 1 to 20 MPa (the event at 2 MPa is excluded because of its similarity to the 1 and 3 MPa cases). Black sections denote the breakdown portion of stress drop, blue curves show the loading and post-slip history, and orange shading indicates breakdown work W_b , integrated from $\delta = 0$ to the displacement δ_0 at which $\bar{\tau}$ reaches its minimum $\bar{\tau}_{\text{min}}$ and breakdown ceases. The total stress drop $\Delta\bar{\tau}_{\text{total}}$, the weakening displacement δ_0 , and the minimum shear stress $\bar{\tau}_{\text{min}}$ are marked in panels (a–c). (j) Normalized shear stress changes $\Delta\bar{\tau}$ from events (a–i) overlaid for comparison, including the 2 MPa event for completeness. (k) Total stress drop $\Delta\bar{\tau}_{\text{total}}$ versus weakening displacement δ_0 for all SHS events, where $\Delta\bar{\tau}_{\text{total}}$ is the difference in $\bar{\tau}$ between $\delta = 0$ and $\delta = \delta_0$. $\Delta\bar{\tau}_{\text{total}}$ and δ_0 show an overall linear relationship, with a small break for events with $\bar{\sigma}_N \geq 15$ MPa, where the loading stiffness was increased by removing the acrylic loading block (Fig. 1a).

151 2.3. Rupture Speed Estimation and Cohesive Zone Modeling

152 Fig. 3a shows details of the strain history for one event. Rupture nucleates
153 near $x = 20$ mm and quasi-statically loads the fault, increasing. To estimate
154 rupture speed, we first determine the rupture-front arrival time at each
155 gauge location. The rupture front is conventionally identified with the peak
156 in shear strain ε_{xy} at the bottom edge near $x = 150$ mm until dynamic

rupture initiates and propagates upward. Note that especially near the top in this example, the dynamic shear strain records lack the distinct peaks commonly observed at rupture fronts in large-scale laboratory earthquakes (Svetlizky and Fineberg, 2014) because our (Svetlizky and Fineberg, 2014; Paglialunga et al., 2022). In our experiments, however, a distinct peak is not always present, because the fault length (150 mm) is shorter than comparable to the critical nucleation length ($h^* \approx 0.5$ m in Ke et al., 2018; McLaskey, 2019). Consequently, when dynamic rupture is triggered, it ($h^* \approx 0.5$ m in Ke et al., 2018; McLaskey, 2019), so the dynamic rupture often propagates through regions of the fault that are still actively nucleating, obscuring the typical sharp rupture-front signature and the characteristic peak is suppressed at those locations.

To analyze rupture propagation, we We therefore define the rupture-front arrival as the onset of the rapid acceleration of the ε_{xy} drop (Ohnaka and Shen, 1999), which coincides with the peak where one is present and remains well defined where it is not. We developed an open-source Python GUI tool (Labquake Explorer; Ke, 2024) to pick rupture-front arrival times at each ε_{xy} measurement position, defined as the first visible drop in shear strain (Ohnaka and Shen, 1999). Arrival times and corresponding fault locations were then fit with linear regression to estimate rupture speed C_f , using positions near the fault center where velocity remained approximately constant (Fig. 1c).

Beyond rupture speed, characterizing the weakening process and fracture energetics requires fitting our dynamic strain measurements using the cohesive zone model, which regularizes the stress singularity inherent in LEFM (Paglialunga et al., 2022; Kammer and McLaskey, 2019; Poliakov et al., 2002; Samudrala et al., 2002). In this formulation, the dynamic stress/strain fields near the tip of a propagating mode II (in-plane shear) rupture are governed by the rupture speed C_f , fracture energy Γ , cohesive zone size X_c , and off-fault distance y . The near-tip stress change is given by Freund (1990)

$$\Delta\tau(\theta, r, C_f) = \frac{K_{II}(C_f)}{\sqrt{2\pi r}} \Sigma_{xy}^{II}(\theta, C_f), \quad (1)$$

where K_{II} is the mode-II stress intensity factor and Σ_{xy}^{II} its angular variation. The fracture energy Γ is related to K_{II} through the dynamic energy release rate (Kammer and McLaskey, 2019)

$$\Gamma = G_{II} = \frac{1 - \nu^2}{E} K_{II}^2 f_{II}(C_f), \quad (2)$$

189 where $f_U(C_f)$ is a known universal function of crack speed. The cohesive
 190 zone introduces a linear slip-weakening stress distribution in local coordinates
 191 $\xi = x - x_{\text{tip}}$ (Poliakov et al., 2002; Samudrala et al., 2002)

$$\Delta\tau(\xi) = \begin{cases} 0, & \xi < -X_c \\ (1 + \xi/X_c)(\tau_p - \tau_r), & -X_c < \xi < 0, \end{cases} \quad (3)$$

192 where τ_p and τ_r are the peak and residual shear stresses. Together Eqs. (1)–(3)
 193 predict the full $\Delta\tau(t)$ and $\Delta\varepsilon_{yy}(t)$ time histories at off-fault distance y . An
 194 example fit is shown in Fig. 3b–c.

195 ~~Applying this model to our measurements requires careful consideration~~
 196 ~~of which portion of the weakening signal reflects local tip processes versus~~
 197 ~~The fitting window is chosen to isolate the local tip-weakening phase from the~~
 198 ~~later system-scale unloading, as described below.~~ Because the fault length
 199 is limited to 150 mm, the rupture front traverses the fault rapidly (~ 150
 200 $\text{mm} \div 1 \text{ km/s} = 0.15 \text{ ms}$), while reduction of τ persist with slip (Fig. 3b).
 201 This two-phase behavior reflects distinct energy release mechanisms: initial
 202 rapid weakening associated with local energy dissipation at the propagat-
 203 ing rupture tip, followed by continued stress reduction driven by nonlocal
 204 elastic unloading of the fault blocks and loading column, including cases
 205 where the center block of the DDS arrangement undergoes nearly-rigid-
 206 body motion. This distinction is evident in Fig. 3d, where the initial drop in
 207 shear strain change $\Delta\varepsilon_{xy}$ propagates at a constant rate (green dashed line),
 208 whereas subsequent features appear nearly simultaneous across all locations
 209 in comparison (black dotted lines). To isolate the local rupture-tip energy
 210 dissipation from system-scale effects, we fit the cohesive zone model to only
 211 the initial weakening phase in $\Delta\varepsilon_{xy}$ and $\Delta\varepsilon_{yy}$ (Fig. 3b–c), while excluding
 212 the later features that represent nonlocal elastic unloading. For each event,
 213 we restricted the fits to the center gauge rosette ($x = 75 \text{ mm}$), where both
 214 ε_{xy} and ε_{yy} are available, thereby enhancing the robustness of the fit. In the
 215 fitting procedure, we fix C_f based on estimates from multiple gauge rosettes
 216 and set $y = 8 \text{ mm}$ (fault to gauge center distance), then define the time
 217 window for the initial $\Delta\varepsilon_{xy}$ drop and jointly fit $\Delta\varepsilon_{xy}$ and $\Delta\varepsilon_{yy}$ for X_c and Γ
 218 (Supplementary Text S1).

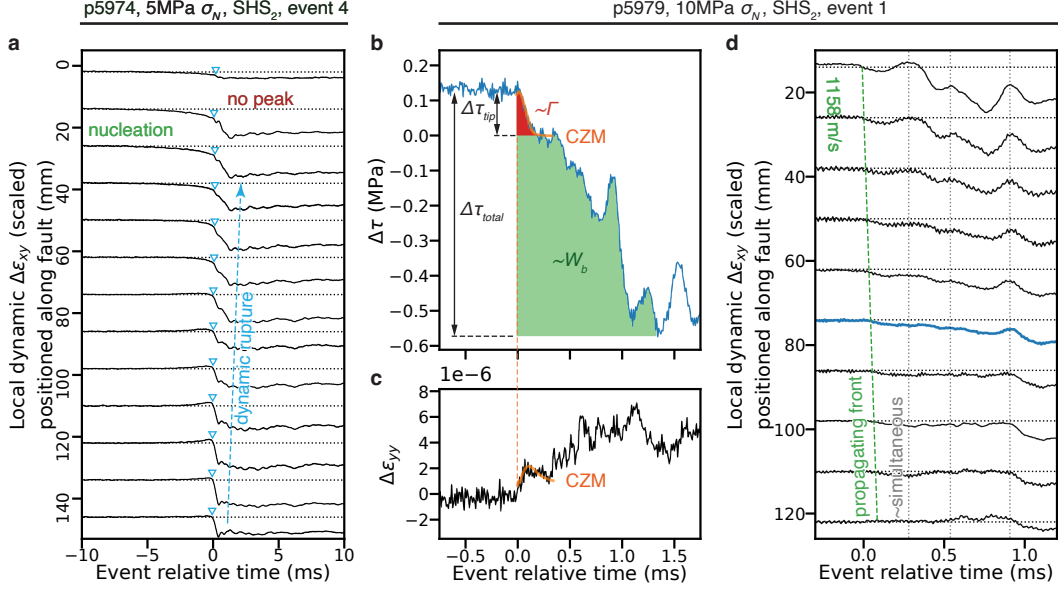


Figure 3: Dynamic strain records of lab earthquakes. (a) Shear strain change $\Delta\epsilon_{xy}$ time history along the fault for one event with upward dynamic rupture propagation, selected because nucleation and dynamic propagation occur close enough in time to visualize both phases together. Rupture nucleates near $x = 20$ mm and quasi-statically propagates downward, loading the bottom of the fault near $x = 150$ mm, until dynamic rupture initiates at the bottom and propagates upward. Triangles mark the picked rupture-front arrival at each gauge. Note that at some locations, the dynamic strain records do not exhibit distinct peaks because the rupture front passes through regions that are still actively nucleating (see Section 2.3). Panels (b–d) show another a second, separate event that initiates near the top and propagates downward. (b) Shear stress change $\Delta\tau - \Delta\tau = G\Delta\epsilon_{xy}$ recorded by the strain gauge rosette at $x = 75$ mm. The trace is annotated with the local total stress drop $\Delta\tau_{total}$ (see panel d) and the stress drop at the rupture tip $\Delta\tau_{tip}$ obtained from the cohesive zone model (CZM, orange curve) fit. The red shaded area corresponds to the fracture energy F near-front dissipation Γ' , while the green area (which includes $F\Gamma'$) corresponds to the breakdown work W_b . (c) Normal strain change $\Delta\epsilon_{yy}$ recorded by the strain gauge rosette at $x = 75$ mm, as in (b). The trace agrees closely with the CZM fit (orange curve) within the fitted window and deviates afterward. Note that the sign of $\Delta\epsilon_{yy}$ is opposite that of $\Delta\epsilon_{xy}$, as expected for this geometry. (d) Shear strain change $\Delta\epsilon_{xy}$ versus time at each gauge location along the fault. The blue curve at $x = 75$ mm corresponds to the trace shown in (b). The onset of shear strain drop propagates downward along the fault (green dashed line), while subsequent strain changes (black dotted lines) appear nearly simultaneous across all locations in comparison, indicating elastic waves generated by near-rigid-body motion of the center block.

3. Results and Discussion

3.1. Breakdown of Frictional Strength During Stick-Slip Events

We observe systematic changes in labquake characteristics over the normal stress range from 1–20 MPa (Fig. 2 and Supplementary Fig. S3). At $\bar{\sigma}_N < 4$ MPa, labquakes occurred only immediately after a hold, as the apparatus stiffness is slightly higher than K_c even with the acrylic loading block in place. In contrast, at $\bar{\sigma}_N \geq 4$ MPa (Fig. 2c–g), labquakes occurred repeatedly and $\Delta\bar{\tau}_{\text{total}}$ increased with δ_0 (Fig. 2k). Following the sudden drop, $\bar{\tau}$ remained at the residual level as δ increased, before rising again. This residual sliding phase appears to lengthen with increasing $\bar{\sigma}_N$ while the acrylic loading block is in place (Fig. 2j), but we note that for $\bar{\sigma}_N < 4$ MPa we do not observe residual, post-event sliding. For events at $\bar{\sigma}_N \geq 15$ MPa (Fig. 2h–i), where the apparatus stiffness is significantly higher, the center block of the DDS arrangement overshoots such that δ exceeds the product of event duration and prescribed load-point shear rate, which caused the servo system to retract and $\bar{\tau}$ to drop further (see blue segment near the end). ~~In all events, the weakening process shows a kink~~ The stress-slip curve shows an inflection at the onset of ~~breakdown, caused by one fault sliding slightly~~ the stress drop (visible in Fig. 2a–d), reflecting a two-stage weakening in which one fault begins to slide while the other remains locked briefly before both faults slip together (see Supplementary Fig. S4). Although this behavior can cause overestimation of W_b , the effect is more pronounced at lower $\bar{\sigma}_N$ and therefore does not significantly affect the derived scaling.

3.2. Experimental Scaling of Rupture Dynamics

Our labquakes reveal clear scaling relationships between fault rupture properties and the normal load $\bar{\sigma}_N$. Rupture propagation speed C_f increases systematically with increasing $\bar{\sigma}_N$, as shown in Fig. ~~??~~4a. At the lowest $\bar{\sigma}_N$ (1–3 MPa), ruptures propagate slowly (a few hundred m/s), whereas at the highest $\bar{\sigma}_N$ (20 MPa), C_f reaches ~ 1.8 km/s, which is still well below the Rayleigh-wave speed limit ($C_R \approx 2.5$ km/s). For an edge crack in a semi-infinite solid, the mode-II stress intensity factor scales as $K_{II}^S \propto \Delta\tau \propto \bar{\sigma}_N$ (Tada et al., 2000; Bayart et al., 2015; Ke et al., 2018), implying a static energy release rate ~~$G_{II}^S \propto \bar{\sigma}_N^2$~~ .

$$\underline{\underline{G_{II}^S \propto \bar{\sigma}_N^2.}} \quad (4)$$

252 Laboratory studies further suggest that fracture energy scales linearly with
 253 normal stress (Ke et al., 2018; Brodsky et al., 2020), $\Gamma \propto \bar{\sigma}_N$ (Ke et al., 2018; Brodsky et al., 2020)
 254 ~~—~~

$$\Gamma \propto \bar{\sigma}_N. \quad (5)$$

255 Substituting these relations into the LEFM rupture speed expression (Svetl-
 256 izky et al., 2017; Kammer et al., 2024), $C_f \approx (1 - \Gamma/G_{II}^S)C_R$, yields $C_f \propto 1 - c_0/\bar{\sigma}_N$,
 257 ~~indicates—~~

$$C_f \approx \left(1 - \frac{\Gamma}{G_{II}^S}\right) C_R, \quad (6)$$

258 ~~yields~~

$$C_f \propto 1 - \frac{c_0}{\bar{\sigma}_N}, \quad (7)$$

259 ~~indicating~~ that rupture speed should follow a hyperbolic approach to C_R
 260 as $\bar{\sigma}_N$ increases (dotted curve in Fig. ??4a). In contrast, our data show
 261 only a modest acceleration with $\bar{\sigma}_N$, indicating that rupture speed cannot
 262 be explained by the standard LEFM scalings $G_{II}^S \propto \bar{\sigma}_N^2$ and $\Gamma \propto \bar{\sigma}_N$. This
 263 discrepancy ~~implies an~~ has two potential explanations: 1) An additional
 264 mechanism beyond ~~energy release from a homogeneous semi-infinite medium~~
 265 ~~or variations in fracture energy.~~ For instance, standard energy release: if G_{II}^S
 266 is treated as a quadratic function of $\bar{\sigma}_N$, ~~i.e., $G_{II}^S = c_1\bar{\sigma}_N^2 + c_2\bar{\sigma}_N + c_3$ with~~
 267 ~~$c_3 = 0$ i.e., $G_{II}^S = c_1\bar{\sigma}_N^2 + c_2\bar{\sigma}_N$ with fitting constants c_1 and c_2 ,~~ the resulting
 268 form ~~$C_f \propto 1 - 1/(c_1\bar{\sigma}_N + c_2)$~~ $C_f \propto 1 - 1/(c_1\bar{\sigma}_N + c_2)$ provides a much closer
 269 fit to our data (dashed curve in Fig. ??a). ~~Alternatively, the 4a).~~ 2) The
 270 150 mm fault length may not provide sufficient distance for ruptures to fully
 271 accelerate, particularly ~~for the at~~ higher $\bar{\sigma}_N$ ~~events that are expected to have~~
 272 ~~greater where the~~ potential C_f is greatest.

273 As shown in Fig. ??4b, the total shear stress drop $\Delta\bar{\tau}_{\text{total}}$ increases with
 274 $\bar{\sigma}_N$, scaling approximately as $0.1\bar{\sigma}_N$, consistent with previous studies (e.g.,
 275 Johnson and Scholz, 1976; Karner and Marone, 2000; Leeman et al., 2016;
 276 Dong et al., 2023; Lyu et al., 2019). Following Shreedharan et al. (2020,
 277 Fig. 1b), who systematically varied shear loading stiffness k , we also find
 278 that $\Delta\bar{\tau}_{\text{total}}$ is larger at lower k (Fig. ??4c). The outlier cluster corresponds
 279 to experiments at $\bar{\sigma}_N$ of 15 and 20 MPa, where the loading stiffness was
 280 significantly higher in the absence of the acrylic loading block. These results
 281 indicate that the stress drop magnitude following complete fault unlocking

is governed by the stiffness of the surrounding medium (which, in our case, includes the apparatus and shear loading column).

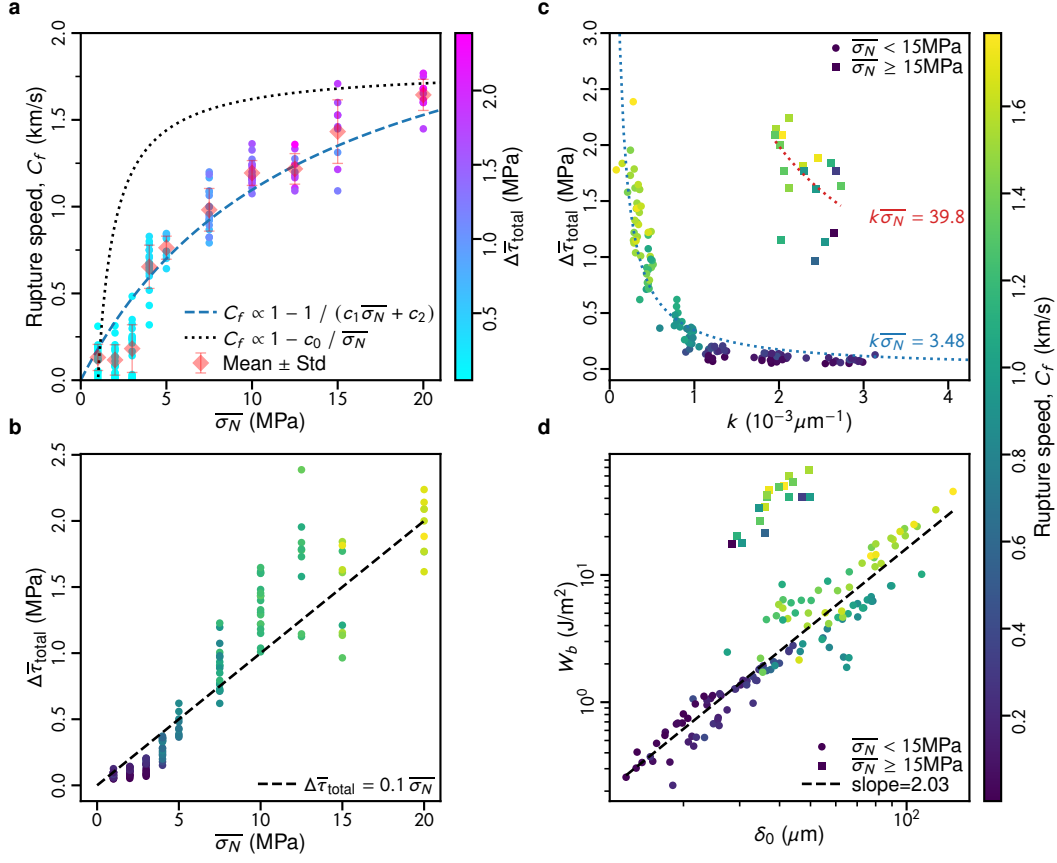


Figure 4: Scaling of rupture speed, stress drop, and breakdown work. (a) Rupture speed C_f increases systematically with normal stress $\bar{\sigma}_N$. The black dotted curve represents the fit $C_f \propto 1 - c_0/\bar{\sigma}_N$, while the dashed curve shows $C_f \propto 1 - 1/(c_1\bar{\sigma}_N + c_2)$. (b) Total stress drop $\Delta\bar{\tau}_{\text{total}}$ increases with normal load, with $\Delta\bar{\tau}_{\text{total}} \approx 0.1\bar{\sigma}_N$. (c) $\Delta\bar{\tau}_{\text{total}}$ versus normalized loading stiffness k shows a trend similar to that reported by Shreedharan et al. (2020), with lower stiffness yielding higher stress drop. The outlier cluster corresponds to events at $\bar{\sigma}_N > 15$ MPa conducted with a much higher stiffness. The dotted and the dashed curves indicate constant values of the stress-per-displacement stiffness $\bar{k} = k\bar{\sigma}_N$ (in MPa/mm), following the scaling $\Delta\bar{\tau}_{\text{total}} = 0.1\bar{\sigma}_N = 0.1\bar{k}/k$. (d) Breakdown work W_b scales with the weakening displacement δ_0 with an exponent of 2.03 (dashed line).

3.3. Breakdown Work Scaling

Breakdown work W_b ~~seals strongly with total slip δ_{total}~~ follows power-law scaling with the weakening displacement δ_0 , $W_b \propto \delta_0^2$ (Fig. ??4d), consistent with patterns observed for natural earthquakes (Abercrombie and Rice, 2005; Viesca and Garagash, 2015; Cocco et al., 2023) and with ~~the laboratory measurements of Selvadurai (2019)~~ laboratory measurements (Selvadurai, 2019). A best-fit power law with an exponent of 2 is consistent with scalings inferred for natural earthquakes ~~(exponent range 0.67–2)~~ (exponent range 0.67–2, Abercrombie and Rice, 2005). The $\sim \delta_0^2$ scaling arises because ~~$W_b \sim 0.5\bar{\tau}_{\text{total}}\delta_0$~~ $W_b \sim 0.5\Delta\bar{\tau}_{\text{total}}\delta_0$ and $\Delta\bar{\tau}_{\text{total}}$ scales approximately linearly with δ_0 (Fig. 2j). ~~These results demonstrate that laboratory~~ This quadratic dependence is a direct consequence of the fixed dimensions of our laboratory fault, where slip is limited by the elastic relaxation of the surrounding compliant system and stress drop and slip therefore co-vary. Natural ruptures are not dimension-limited in the same way, so slip and stress drop need not co-vary as they do here, and the quadratic scaling we measure should not be expected to carry over to them (Udías et al., 2014; Nielsen et al., 2016). In both cases the breakdown work is set by the elastic energy released from the volume surrounding the rupture rather than by frictional processes at the rupture tip. In our length-limited fault the fixed dimension couples slip to stress drop and yields an exponent of two, as it would for a fault patch of roughly fixed size, such as an asperity or fault segment embedded in a more compliant surrounding volume. This is not a model for self-similar rupture growth, where the rupture dimension is not fixed and the stress drop is variable, so this nonlocal mechanism does not imply any single slip exponent, consistent with the broad range of exponents reported for natural earthquakes. In this restricted sense, laboratory faults with length-limited contacts ~~can~~ reproduce key aspects of natural fault behavior inferred from seismology.

Previous larger-scale laboratory earthquake studies focus on rupture dynamics before the rupture spans the entire fault, and thus this post-rupture behavior has not been previously highlighted. In the DDS configuration, large ruptures produce complete unlocking and subsequent rigid block motion, which unloads both the fault-surrounding material and the shear loading column, thereby producing the strong W_b scaling.

3.4. Near-Tip Weakening from Cohesive Zone Modeling

~~We developed a cohesive zone model (CZM) to fit our data (Paglialunga et al., 2022; Kammer and Selvadurai, 2023).~~ The CZM (Section 2.3) fits well for the initial strain drop

321 at the center rosette ($x = 75$ mm) and it captures near-tip weakening
 322 (Fig. 3e-gb-c). The stress drop at the rupture tip $\Delta\tau_{\text{tip}}$ is significantly smaller
 323 than the total stress drop and shows little dependence on $\bar{\sigma}_N$ (Fig. ??e5a),
 324 though $\Delta\tau_{\text{tip}}$ is higher at lower $\bar{\sigma}_N$, likely reflecting increased uncertainty
 325 in distinguishing tip from tail behavior during fitting. The cohesive zone
 326 size X_c varies only weakly with $\bar{\sigma}_N$ (Fig. ??f5b), yet consistently exceeds
 327 the fault length, ~~indicating a violation of the~~. The cohesive zone model
 328 remains a valid elastic description of the near-tip strain fields, and X_c is
 329 a real length scale over which the rupture-front stresses evolve. However,
 330 because X_c is comparable to or larger than L , there is no region outside
 331 the cohesive zone yet close to the tip where the field is K -dominated. The
 332 small-scale yielding condition. This calls into question the applicability of
 333 LEFM to our macroscopic measurements when ruptures reach the sample
 334 boundaries, although the models fit some aspects of our data reasonably
 335 well. Nevertheless, we report Γ from the CZM fits to quantify near-tip
 336 energy dissipation and to compare it with the macroscopic breakdown work
 337 W_b —not as a rigorous LEFM estimate, but as yielding condition is therefore
 338 not satisfied, the fitted amplitude is an amplitude parameter of the solution
 339 rather than a true stress intensity factor, and X_c should not be read as
 340 a literal cohesive zone. Accordingly, we do not interpret the CZM-derived
 341 energy as a model-insensitive estimate of near-tip dissipation, since strict
 342 fracture energy. We instead report it as an apparent near-front dissipation
 343 scale Γ' , obtained as the area under the τ - δ curve during initial weakening is
 344 insensitive to CZM parameters as long as $y \ll X_c$. This area is constrained
 345 directly by the near-field gauge ($y \ll X_c$) and is insensitive to the assumed
 346 weakening-law shape, providing a robust measure of near-front dissipation for
 347 comparison with the breakdown work W_b . The two quantities are obtained
 348 independently, W_b from the finite-geometry energy budget and Γ' from the
 349 near-field strain record, and neither relies on the small-scale-yielding assumption
 350 that fails here.

351 Our data show that ~~fracture energy Γ~~ the near-front dissipation Γ' in-
 352 creases systematically with normal load (Fig. ??g5c). The scaling is roughly
 353 consistent with the fracture energy $\Gamma \approx 0.2 \mu\text{m} \cdot \sigma_N$ reported by Ke et al.
 354 (2018), and our values are comparable in magnitude to previous studies
 355 (Svetlizky et al., 2017; Bayart et al., 2015; Ke et al., 2018; Kammer and
 356 McLaskey, 2019; Paglialunga et al., 2022). However, ~~Γ~~ Γ' remains an order
 357 of magnitude smaller than the breakdown work W_b (Ke et al., 2022), under-
 358 scoring that much of the energy release occurs late in the process, after the

359 rupture front has passed, after local stresses have dropped, and after the
 360 rupture has reached ~~it's~~its final dimensions. Notably, events at $\overline{\sigma}_N \geq 15$
 361 MPa conducted without the compliant loading spring ~~exhibit substantially~~
 362 ~~elevated W_b~~ fall above the $W_b \propto \delta_0^2$ trend (Fig. ~~2k~~4d, squares) ~~because their~~
 363 ~~stress drop is high relative to their slip~~, yet show no corresponding ~~increase in~~
 364 ~~F departure in Γ' at these same normal stresses~~ (Fig. ~~??g~~5c). This ~~provides~~
 365 ~~direct evidence~~indicates that the W_b scaling arises from nonlocal elastic un-
 366 loading of the loading system rather than from enhanced energy dissipation
 367 at the rupture tip.

368 Recent high-velocity experiments show that rupture dynamics, particu-
 369 larly slip acceleration and system stiffness, can strongly influence both tran-
 370 sient weakening and breakdown work (Liao et al., 2014; Chen et al., 2025).
 371 Chen et al. (2025) demonstrated that friction evolution during dynamic rup-
 372 ture systematically deviates from flash-heating predictions, with breakdown
 373 work scaling aligning with natural earthquakes when acceleration effects are
 374 considered. Similarly, Liao et al. (2014) found that slip-pulse loading pro-
 375 duces lower breakdown work than constant-velocity tests despite more in-
 376 tense weakening, attributing this to acceleration-enhanced gouge production.
 377 These studies suggest that system configuration and loading history can fun-
 378 damentally influence energy partitioning during dynamic slip, consistent with
 379 our observation that loading stiffness modulates W_b while leaving ~~F~~ Γ' un-
 380 changed. This indicates that the excess energy release in our experiments
 381 arises from off-fault elastic unloading rather than enhanced dissipation at the
 382 rupture tip through mechanisms such as flash heating (Viesca and Garagash,
 383 2015). These observations motivate finite-element analysis to further eval-
 384 uate how the static energy release rate G_{II}^S is partitioned and how it scales
 385 with $\overline{\sigma}_N$.

386 3.5. Finite Element Modeling for Estimating Energy Release Rate Scaling

387 Typical LEFM applications for lab earthquake experiments (e.g., Kammer
 388 et al., 2015; Bayart et al., 2015; Ke et al., 2018; Paglialunga et al., 2022)
 389 assume that the rupture is embedded in an infinite solid or semi-infinite half-
 390 space. While these approximations provide useful first-order insights, the
 391 finite size of lab fault zones, and in particular the fact that our fault length is
 392 comparable to the nucleation dimension, motivates a more detailed analysis.
 393 To evaluate the energy budget of our labquakes, we developed a finite element
 394 model in ABAQUS (Dassault Systèmes Simulia Corp., 2024), including the
 395 three granite blocks used for DDS, the steel plate on top of the center block

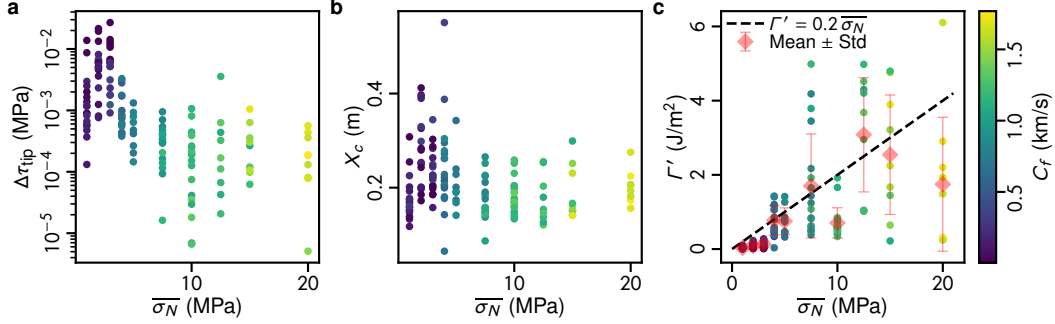


Figure 5: ~~Sealing relations for rupture dynamics.~~ ~~Cohesive-zone-model~~ ~~(a)near-tip~~ Rupture speed C_f increases systematically with normal stress $\bar{\sigma}_N$ scaling. The dotted curve represents the fit $C_f \propto 1 - c_0/\bar{\sigma}_N$, while the dashed curve shows $C_f \propto 1 - 1/(c_1\bar{\sigma}_N + c_2)$. (b) Total stress drop $\Delta\bar{\tau}_{\text{total}}$ increases with normal load, with $\Delta\bar{\tau}_{\text{total}} \approx 0.1\bar{\sigma}_N$. (c) $\Delta\bar{\tau}_{\text{total}}$ versus normalized loading stiffness k shows a trend similar to that reported by Shreedharan et al. (2020), with lower stiffness yielding higher stress drop. The outlier cluster corresponds to events at $\bar{\sigma}_N \geq 15$ MPa conducted with a much higher stiffness. The dotted and the dashed curves indicate constant values of the stress-per-displacement stiffness $\bar{k} = k\bar{\sigma}_N$ (in MPa/mm), following the scaling $\Delta\bar{\tau}_{\text{total}} = 0.1\bar{\sigma}_N = 0.1\bar{k}/k$. (d) Breakdown work W_b scales with total slip δ_0 with an exponent of 2.03 (dashed line). (e) Stress drop at the rupture tip $\Delta\tau_{\text{tip}}$, estimated from CZM fits, decreases slightly with $\bar{\sigma}_N$ and is considerably smaller than the total stress drop $\Delta\bar{\tau}_{\text{total}}$. (f) Cohesive zone size X_c shows little dependence on $\bar{\sigma}_N$ but mostly exceeds the 0.15 m fault length. (g) ~~Fracture energy~~ ~~—~~ Near-front dissipation Γ' dependence on $\bar{\sigma}_N$ is broadly consistent in magnitude with the ~~fracture-energy~~ scaling $\Gamma \approx 0.2\mu \cdot \sigma_N$ reported by Ke et al. (2018, dashed line).

396 for stress distribution, and the acrylic block used to reduced loading stiffness
 397 for lower normal stress (Fig. 6).

398 Because our shear loading rate is negligible relative to the dynamic rup-
 399 ture speed, we prescribe a displacement at the top of the DDS loading column
 400 and gradually increase the rupture area in static steps by varying the rup-
 401 ture length a , assuming the rupture front is uniform across the fault width.
 402 The rupture area is simulated by frictional contact with $\mu = 0.7$, whereas
 403 the unruptured regions are tied together with a zero-relative-displacement
 404 boundary condition. Although this model does not capture the full dynam-
 405 ics of rupture propagation, it provides a first-order framework for evaluating
 406 the validity of LEFM assumptions in our experimental system and for esti-
 407 mating how G_{II}^S scales with σ_N .

408 We ran simulations ranging from $\bar{\sigma}_N$ from 2.5 to 20 MPa, and for each
 409 normal load considered rupture lengths a from 5 to 105 mm, assuming sym-

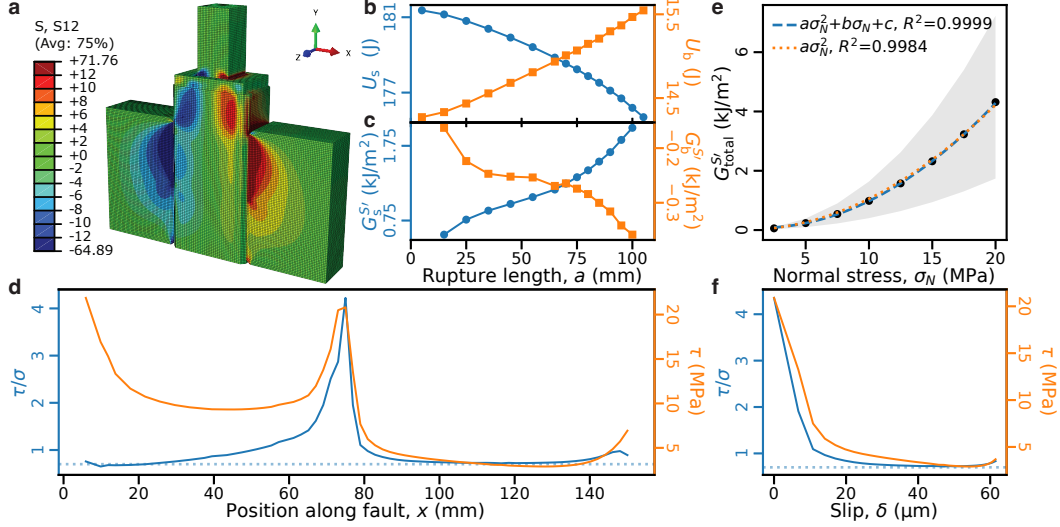


Figure 6: Finite element model results for a bottom-up rupture. (a) Shear stress field along the fault direction τ in the 3D finite element model. S12 denotes shear stress σ_{xy} in the model. (b) Strain energy stored in the loading-column acrylic spring U_s (blue) and in the granite blocks U_b (orange) as a function of rupture length a . (c) Energy release rate from the loading-column G_s^S acrylic spring $G_s^{S'}$ (blue) and from the granite blocks G_b^S $G_b^{S'}$ (orange) as a function of a . (d) Spatial distribution of the shear-to-normal stress ratio τ/σ_N (blue) and shear stress τ (orange) at a rupture length of 75 mm. Dashed line shows the prescribed target friction coefficient. (e) Total static energy release rate G^S $G^{S'}$ as a function of $\bar{\sigma}_N$, overlaid with a quadratic polynomial fit (blue dashed) and a pure quadratic fit (orange dotted). The gray shaded area marks the range of G $G^{S'}$ for rupture lengths $a \in [15, 100]$ mm (as shown in c), and the black dots indicate $a = 75$ mm. (f) Shear-to-normal stress ratio τ/σ_N (blue) and shear stress τ (orange) versus slip δ for $\delta > 0$ at all fault nodes at a rupture length of 75 mm. Dashed line indicates the prescribed target friction coefficient. See Supplementary Fig. S2 for the top-down rupture case.

metric rupture on both frictional interfaces. The energy release rate was computed as $G_{II}^S = -dU/dA$ $G_{II}^{S'} = -dU/dA$ using a central-difference approximation, where U is the total strain energy, $A = aw$ is rupture area, and $w = 50$ mm is the fault width. For comparison with the experiments, we take G_{II}^S $G_{II}^{S'}$ at $a = 75$ mm (rupture front at fault center) as the representative value for each normal stress (Fig. 6e).

The simulations show that the loading column with the acrylic block stores more than an order of magnitude more strain energy than the granite blocks (Fig. 6b). The corresponding energy release rates exhibit the same partitioning, with G_s^S $G_s^{S'}$ denoting energy released from the loading column

(subscript s) and G_b^S from the granite blocks (subscript b) (Fig. 6c). Interestingly, the strain energy in the granite blocks increases with rupture area ($A = aw$) rather than releasing energy ($G_b^{S'} < 0$, Fig. 6c) —*i.e.*, the blocks continue to store energy as rupture grows. This behavior arises because reduced lateral restraint from the side blocks permits greater deformation of the center block as rupture area expands. Thus, even though strain energy is released at the rupture tips, the total strain energy of the blocks increases. In the case with top-down rupture, the energy release rate in the two side blocks is positive (Supplementary Fig. S2), as expected, but remains considerably lower than the energy release rate from the loading column with the acrylic block (G_s^S). These results confirm that elastic energy release from the loading column, rather than localized rupture-tip processes, governs the energy budget in our laboratory earthquakes, and that this energy release begins even before the rupture spans the entire fault and rigid-body motion of the center block commences.

At a representative rupture length of $a = 75$ mm, the total energy release rate (G_{total}^S) increases quadratically with $\bar{\sigma}_N$ (Fig. ??6e), showing no deviation from the simplified LEFM prediction $G_{II}^S \propto \bar{\sigma}_N^2$ (curves in Fig. 6e), despite the geometric and compositional complexity of our experimental setup. ~~This result indicates that first-order modeling is consistent with expectations and LEFM theory but it cannot explain the deviation.~~ Notably, because $G^{S'}$ is computed directly from the finite-geometry energy budget rather than from a stress intensity factor, this agreement does not rely on small-scale yielding (cf. Palmer and Rice, 1973). We write $G^{S'}$, with a prime as for the apparent near-front dissipation Γ' , to denote a finite-geometry energy release rate determined without a K -dominant field. The quadratic scaling is therefore robust rather than an artifact of the small-scale-yielding assumption underlying the analytical prediction, and it persists beyond the small-scale-yielding regime. Our finite-element model is static and is not intended to reproduce the rupture speed. It therefore does not capture the modest deviation of C_f from $C_f \propto 1 - c_0/\bar{\sigma}_N$ ~~observed in our experiments~~ (Fig. ??a; dotted curve), and suggests that ~~4a~~). Reproducing that deviation would likely require fully dynamic simulations ~~incorporating~~ with a friction law ~~may be required~~. Importantly, this deviation concerns only the rupture speed. It does not bear on our energy-budget analysis, from which we conclude that the breakdown-work scaling arises from system-scale elastic unloading.

Beyond the energy release rate scaling, the finite-element model also re-

458 veals how spatial heterogeneity in normal stress affects fault behavior. Due
 459 to the finite geometry of our experimental setup, σ_N varies spatially along
 460 the fault (Fig. 6d). This heterogeneity causes τ and τ/σ_N to exhibit
 461 markedly different spatial patterns, with τ showing pronounced stress con-
 462 centrations near rupture tips while τ/σ_N remains more uniform (Fig. 6d).
 463 During slip, these spatial variations in σ_N evolve, producing different slip-
 464 weakening behavior for the two quantities (Fig. 6f). Notably, $\tau(\delta)$ exhibits
 465 secondary weakening, a gradual stress decrease following the initial sharp
 466 drop, that is much less pronounced in $\tau/\sigma_N(\delta)$. This occurs because the
 467 evolving σ_N distribution produces curved τ - δ behavior even when the fric-
 468 tion coefficient μ remains uniform. Consequently, ~~measurements that assume~~
 469 ~~uniform σ_N interpreting the measured τ under the implicit assumption of a~~
 470 ~~constant $\bar{\sigma}_N$, as is common in experimental studies,~~ can overestimate frac-
 471 ture energy Γ by misattributing this curvature to enhanced dissipation at the
 472 rupture tip (e.g., Cowie and Scholz, 1992). Thus, system-scale elastic unload-
 473 ing and finite-geometry effects can mimic signatures of complex weakening
 474 mechanisms, contributing to the observed breakdown work scaling without
 475 requiring additional energy dissipation near the rupture front.

476 3.6. *Energy partitioning and the fault thermal budget*

477 A natural question is where the released elastic energy ends up, and
 478 whether the strong scaling of breakdown work implies correspondingly strong
 479 frictional heating and thermally enhanced weakening on the fault (e.g., Nielsen et al., 2021; Cocco et
 480 al., 2023). While we do not measure temperature directly, mechanical work done
 481 against friction is expected to be largely converted to heat (Aretusini et al., 2021, 2024; Ortega-Arro
 482 , 2023), so most of the released energy likely ends up as frictional heat on the
 483 fault. Taking an event at $\bar{\sigma}_N = 20$ MPa as an example, the work done
 484 against friction is $\int \bar{\tau} d\delta \approx \bar{\tau} \delta \approx 6 \times 10^2 \text{ J m}^{-2}$, and the breakdown work
 485 $W_b = \int (\bar{\tau} - \bar{\tau}_{\min}) d\delta$ is the roughly ten percent of this work done above the
 486 minimum sliding stress (Fig. 4d). The near-front dissipation $\Gamma' \approx 10 \text{ J m}^{-2}$,
 487 the energy dissipated locally during the initial weakening at the onset of slip,
 488 is smaller still. How W_b itself partitions among additional fault heating,
 489 kinetic energy of the near-rigidly translating center block (Fig. 3d), and
 490 radiated waves is not directly constrained by our measurements, but each
 491 of these sinks is modest. Spread through the ~ 0.75 g gouge layer, the total
 492 frictional heat corresponds to a temperature rise of only $\Delta T = \bar{\tau} \delta A / (m c) \approx 8 \text{ K}$,
 493 where A is the fault area, m is the gouge mass, and $c \approx 750 \text{ J kg}^{-1} \text{ K}^{-1}$
 494 is the specific heat of quartz. This is two orders of magnitude below the

495 rises required to activate flash heating, thermal pressurization, or melting
 496 (e.g., Viesca and Garagash, 2015; Harbord et al., 2021), so enhanced thermally
 497 activated weakening is neither expected nor observed. The scaling of W_b
 498 therefore does not indicate that rupture requires more energy to propagate,
 499 which is the role of Γ . It indicates that the surrounding system, including
 500 the loading column, releases more energy as the fault unlocks, energy that
 501 ends up as modest heat, radiated waves, and kinetic energy.

502 3.7. *Implications for Natural Faults*

503 Our finding that nonlocal elastic unloading governs the breakdown work
 504 scaling invites comparison with natural faults. Strong breakdown work scaling
 505 with earthquake size is inferred from seismology (Abercrombie and Rice, 2005; Tinti et al., 2005)
 506 and is commonly attributed to enhanced near-fault weakening such as thermal
 507 pressurization (Viesca and Garagash, 2015; Wibberley and Shimamoto, 2005).
 508 Our results point to an alternative, in which fault areas surrounding a
 509 high-strength asperity, though not in direct contact, store and release the
 510 elastic energy that drives this scaling. In this view, a finite-length laboratory
 511 fault embedded in a compliant loading system is analogous to a small asperity
 512 embedded in a larger elastic medium, where nonlocal unloading rather than
 513 enhanced weakening produces the scaling. Such an effect would be absent
 514 from laboratory and numerical models that assume perfectly mated fault
 515 interfaces. Our data therefore suggest that accounting for fault topography
 516 and contact heterogeneity, alongside frictional properties, may be essential
 517 for scaling rupture energetics from the laboratory to tectonic faults.

518 4. Conclusions

519 Our laboratory stick-slip experiments reveal that rupture speed C_f , stress
 520 drop $\Delta\bar{\tau}_{\text{total}}$, and breakdown work W_b scale systematically with normal load
 521 $\bar{\sigma}_N$. While these data are consistent with basic expectations they are not fully
 522 aligned with standard assumptions of LEFM. Although ~~fracture energy~~ Γ ~~the~~
 523 near-front dissipation Γ' grows approximately linearly with $\bar{\sigma}_N$, it remains
 524 an order of magnitude smaller than W_b , indicating that most energy release
 525 arises from system-scale unloading, after the rupture has reached the fault
 526 boundaries. This later slip is due to energy release of the shear loading
 527 column rather than from localized weakening at the rupture tip. Finite-
 528 element analysis shows that the static energy release rate G_{II}^S $G_{\text{II}}^{S'}$ follows the
 529 quadratic scaling predicted by LEFM despite the geometric complexity of the

530 experimental setup, while also revealing that the compliant loading column
 531 dominates the energy budget even before full sliding occurs. Cohesive zone
 532 modeling further indicates that the cohesive zone size X_c exceeds the sample
 533 dimensions, violating the small-scale yielding assumption even though the
 534 model fits the dynamic strain data reasonably well.

535 Our results highlight the role of finite, length-limited fault patches in
 536 governing rupture energetics. ~~Such effects are common in seismic inversions,~~
 537 ~~especially for large ruptures where they are considered subevents of the~~
 538 ~~main fault rupture. Our data show that unlocking of such subevent fault~~
 539 ~~patches permits extensive nonlocal unloading, which reproduces the strong~~
 540 ~~W_b scaling inferred in natural earthquakes. Our findings suggest that fault~~
 541 ~~areas surrounding high-strength asperities, though not in direct contact, play~~
 542 ~~a critical role in storing and releasing energy through elastic coupling. In~~
 543 ~~this sense, a finite-length laboratory fault embedded in a compliant loading~~
 544 ~~system behaves analogously to a small asperity embedded in a larger elastic~~
 545 ~~medium, with nonlocal unloading providing a mechanistic explanation for the~~
 546 ~~inferred breakdown-work scaling. This effect may be less prominent in some~~
 547 ~~laboratory and numerical models, which typically assume perfectly mated~~
 548 ~~fault interfaces that do not promote nonlocal energy release. However, our~~
 549 ~~data underscore the fact that accounting for the role of fault heterogeneities,~~
 550 ~~in addition to frictional properties, is essential for understanding earthquake~~
 551 ~~physics and scaling from lab to tectonic fault scales. Together, these results~~
 552 identify nonlocal elastic unloading of the loading system, rather than enhanced
 553 near-tip dissipation, as the primary control on the breakdown-work scaling
 554 we observe.

555 CRediT authorship contribution statement

556 **C.-Y. Ke:** Writing - original draft, Methodology, Software, Validation,
 557 Formal analysis, Investigation, Data Curation, Visualization, Supervision,
 558 Project administration, Conceptualization, Funding acquisition. **G.**
 559 **T. Chang:** Software, Formal analysis, Data Curation, Visualization. **G. C.**
 560 **McLaskey:** Writing - review & editing, Validation, Resources, Conceptual-
 561 ization. **D. S. Kammer:** Writing - review & editing, Validation, Conceptu-
 562 alization. **C. Marone:** Writing - review & editing, Validation, Investigation,
 563 Resources, Supervision, Project administration, Conceptualization, Funding
 564 acquisition.

565 Declaration of competing interest

566 The authors declare that they have no known competing financial inter-
567 ests or personal relationships that could have appeared to influence the work
568 reported in this paper.

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583 Supplementary materials

584 Supplementary material associated with this article can be found, in the
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586 Data availability

587 Data will be made available on request.

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- Finite element model reveals that loading system compliance dominates energy release.
- Nonlocal elastic unloading reproduces breakdown work scaling in natural earthquakes.

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Nonlocal Elastic Unloading as a Mechanism for Breakdown Work Scaling in Laboratory Earthquakes

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Abstract

Understanding how earthquake energy budgets scale from laboratory to natural faults remains a fundamental challenge in earthquake physics. Laboratory experiments report fracture energies consistent with classical friction, yet seismological observations show that breakdown work scales strongly with slip, and the relationship between these two quantities remains unclear. Here we show that this discrepancy reflects two genuinely different physical quantities: the near-front dissipation governed by rupture-tip processes, which we use as a proxy for fracture energy because these small ruptures sustain no K -dominant field, and the system-scale breakdown work dominated by nonlocal elastic unloading which includes the loading column in our experiments. We conducted stick-slip experiments on 15-cm long granite faults under normal stresses of 1–20 MPa in a double-direct-shear configuration. A dense strain-gauge array tracked rupture propagation, enabling near-front dissipation estimates via cohesive zone modeling, while finite element simulations quantified energy partitioning between the fault and loading system. The near-front dissipation remains low (<10 J/m²) and scales roughly linearly with normal stress, consistent with classical friction. In contrast, total breakdown work scales quadratically with slip, exceeding the near-front dissipation by over an order of magnitude. Finite element simulations confirm

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that elastic unloading of the compliant loading column dominates this energy release. Normal stress heterogeneity on the lab fault produces additional weakening, after the rupture has reached maximum size, without additional dissipative mechanisms. These results demonstrate that apparent breakdown work scaling in laboratory earthquakes can arise from nonlocal elastic effects rather than intrinsic fault weakening. Our work provides a framework for distinguishing local near-front dissipation from system-scale energy release and breakdown work when extrapolating laboratory results to natural faults.

Keywords:

Rupture dynamics, Laboratory earthquakes, Breakdown work, Fracture energy

1. Introduction

Earthquake rupture reflects a dynamic redistribution of elastic strain energy within the fault–host rock system, balancing energy release from the surrounding medium against dissipation on and near the fault. This global energy flow governs the seismic stress drop and other source properties, including the rupture propagation speed C_f . Under suitable assumptions, this process can be formulated using linear elastic fracture mechanics (LEFM; Freund, 1990), in which rupture growth satisfies an energy balance $G = \Gamma$ between the dynamic energy release rate G and the fracture energy Γ (Kammer et al., 2024). In the context of LEFM, earthquake ruptures accelerate when the static energy release rate G_{II}^S (mode II) exceeds Γ (Svetlizky et al., 2020; Albertini et al., 2021).

Most experimental, analytical, and numerical studies have focused on the dissipative side of this balance, examining how rupture energy dissipation depends on slip rate, normal stress, fault roughness, and other weakening mechanisms (e.g., Cocco et al., 2006; Okubo et al., 2019; Kammer and McLaskey, 2019; Perry et al., 2020; Paglialunga et al., 2022; Cocco et al., 2023; Morad et al., 2024; Brantut, 2025). Far less attention has been given to the release side, namely, how the surrounding elastic medium unloads and transfers energy toward the rupture front. Yet this nonlocal energy supply can strongly influence rupture properties such as acceleration, total breakdown work, and their apparent dependence on normal stress.

Fault normal stress governs both sides of the fracture energy balance. Increasing normal stress enhances frictional strength and contact area (Dietrich and Kilgore, 1994), raising dissipation, but can also increase the elastic energy stored prior to rupture. In nature, ruptures at depth, where normal stress is high, tend to propagate faster than shallow or near-trench events (Vidale and Houston, 1993; Bilek and Lay, 2002). Laboratory experiments show the same trend: as normal stress increases, rupture speed increases (Johnson and Scholz, 1976; Leeman et al., 2016), approaching the Rayleigh or shear-wave velocity, and transitions to supershear rupture become more common (Xia, 2004; Dong et al., 2023). These observations point to a fundamental coupling between energy storage, release, and dissipation that governs rupture properties including speed.

In this study, we investigate small-scale laboratory earthquakes on a 15-cm long granite fault instrumented with a dense 13-station strain-gauge array. Our experiments enable spatially resolved measurements of rupture

38 dynamics akin to those in larger, meter-scale experiments, but with addi-
 39 tional control on loading rates and history. This allows us to reconstruct
 40 rupture fronts and estimate near-front dissipation locally, while we estimate
 41 breakdown work at the apparatus level, enabling comparison between lo-
 42 cal and system-wide energy dissipation. Because these ruptures are small,
 43 with a cohesive zone comparable to the fault length, the near-tip field is not
 44 K -dominated; we therefore report the local near-tip energy not as a strict
 45 fracture energy but as an apparent near-front dissipation Γ' , a proxy for
 46 fracture energy obtained directly from the area under the initial weakening
 47 curve (Section 3.4). While the inferred near-front dissipation remains low,
 48 consistent with previous laboratory studies on single, relatively simple faults
 49 (e.g., Svetlizky et al., 2017; Bayart et al., 2015; Ke et al., 2018; Kammer and
 50 McLaskey, 2019; Paglialunga et al., 2022), total breakdown work increases
 51 strongly with normal stress (correlating with slip and stress drop), similar to
 52 trends observed in natural earthquakes (Cocco et al., 2023; Selvadurai, 2019;
 53 Viesca and Garagash, 2015; Abercrombie and Rice, 2005). We interpret this
 54 scaling not as a signature of increased fracture resistance and fracture en-
 55 ergy, but as a consequence of greater energy release during elastic unloading
 56 of the fault and surrounding material including the loading column in our
 57 experiments. Our experiments demonstrate this for length-limited faults em-
 58 bedded in a compliant loading system, which is analogous to fault segments
 59 embedded in a damage zone or other cases where a small fault patch releases
 60 elastic energy stored in a larger volume. This analogy holds only in a re-
 61 stricted sense, for a fault patch of roughly fixed size, such as an asperity or
 62 fault segment embedded in a more compliant surrounding volume, for which
 63 the fixed dimension makes slip and stress drop co-vary. It is not a model for
 64 self-similar rupture growth, where the rupture dimension is not fixed and the
 65 stress drop is variable. The additional energy released as the fault completely
 66 unlocks arises from this elastic unloading, without requiring secondary weak-
 67 ening mechanisms (e.g., Gabriel et al., 2024; Lambert et al., 2021; Viesca and
 68 Garagash, 2015), and contributes to the observed scaling of breakdown work
 69 (e.g., Cocco et al., 2023; Abercrombie and Rice, 2005). These findings sug-
 70 gest that breakdown work scaling can be governed by system-scale elastic
 71 unloading rather than reflecting only the local weakening, offering a new
 72 perspective on energy partitioning in laboratory and natural earthquakes.

2. Methods

2.1. Experimental Setup

Friction experiments were conducted in the double-direct-shear (DDS) configuration using a biaxial load frame with adjustable loading stiffness (Leeman et al., 2016; Shreedharan et al., 2020). The samples consisted of a center block and two side blocks, all prepared from Barre Gray Granite, forming two 150 mm $\times$ 50 mm frictional interfaces (Fig. 1a). All edges of the center block and the top and bottom edges of the side blocks were chamfered 5 mm to reduce stress concentrations and prevent edge damage during repeated experiments. Throughout each experiment, normal load $\overline{\sigma_N}$ was held constant (force-controlled using a fast-acting servo system), while shear load $\overline{\tau}$ was applied through prescribed load-point velocities (using servo displacement control).

Along one of the 150 mm long faults (Fig. 1a), we installed 13 strain gauge rosettes to measure dynamic shear strain ε_{xy} at equally spaced locations, and normal strain ε_{yy} at 3 selected locations ($x=3, 75$, and 147 mm). The center of each strain gauge was located 8 mm from the fault. The dense array of ε_{xy} measurements enabled us to track the spatiotemporal onset of shear strain drop along the fault and estimate rupture speed (Fig. 1c) of laboratory earthquakes (labquakes). We used a servo-controlled loading sequence to produce 10's to 100's of labquakes in each experiment under the same conditions. Displacements at the load-point δ_{LP} and near the top of the center block δ (Fig. 1a, blue square), together with normal load $\overline{\sigma_N}$ and shear load $\overline{\tau}$, were recorded at 1 kHz, while dynamic strains were recorded separately at 100 kHz. The two systems measure independent quantities. The slow-sampling channels characterize system-level slip and stress while the strain gage array captures the dynamic rupture waveforms used for rupture speed and cohesive zone model fitting. Because the interseismic period between events is orders of magnitude longer than the rupture duration, events are unambiguously identified in both recording systems.

2.2. Sample Preparation and Testing Protocol

All experiments were conducted in the Rock Mechanics Laboratory at The Pennsylvania State University. We conducted experiments at normal loads $\overline{\sigma_N}$ ranging from 1 to 20 MPa, all following the same testing protocol. Before each experiment, the granite fault surfaces were roughened with #60 grit silicon-carbide (producing an RMS roughness of ~ 20 μm) and a thin layer of

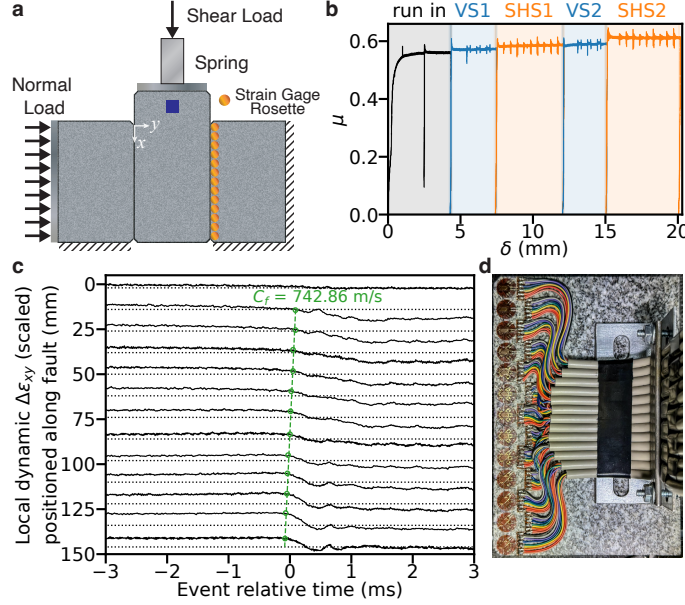


Figure 1: Double-direct shear experiment configuration. (a) Three Barre Gray granite blocks (two side blocks and one center block) are arranged under biaxial loading with bottom support for the two side blocks. Shear load is applied via an interchangeable loading block as a spring at the top of the center block. The right side block is instrumented with 13 strain gauge rosettes along the 150 mm fault. Blue square near the top of the center block marks the location of center block displacement δ measured relative to the bottom of the load frame. (b) Experimental protocol. Gray shading indicates the initial run-in phase in which the fault surfaces are conditioned prior to testing, blue shading indicates velocity-step tests (VS_i, stiff steel loading spring), and orange shading indicates slide-hold-slide tests (SHS_i, compliant acrylic loading spring). $\mu = \bar{\tau}/\bar{\sigma}_N$ is the friction coefficient and δ is center block displacement. (c) Dynamic shear strain ε_{xy} recorded at each strain gauge rosette along the fault, plotted as a function of time relative to the event onset. Horizontal dotted lines mark the zero strain change for each gauge trace. The topmost gauge did not record a rupture signal, likely because the rosette was too close to the top edge of the fault and was not in full contact with the fault surface. Green circles mark the onset of rupture at each gauge location; the green dashed line is fitted to these circles, and its slope gives the average rupture propagation speed. (d) A picture of the right side block, aligned to the strain gauge rosette positions shown in (c).

109 moisturized quartz powder (Min-U-Sil 40) was applied to each fault (~ 0.75 g;
 110 ~ 250 μm thick). This preparation procedure was adopted from Shreedharan
 111 et al. (2020) to achieve steady-state friction conditions and produce long sets
 112 of reproducible labquakes. As shown in Fig. 1b, the protocol included run-in,
 113 velocity-step (VS), and slide-hold-slide (SHS) tests. After applying normal

114 stress, the center block was sheared to the load-point displacement of ~ 5
115 mm, with an unload-reload cycle at ~ 3 mm, to run in the faults. VS and
116 SHS tests were then performed in alternating sequence (Fig. 1b) to assess
117 cumulative strain effects, denoted VS_i and SHS_i according to their order. To
118 generate reproducible labquakes with dynamic rupture, we used an acrylic
119 block as a spring (80-mm long and 40x40 mm cross-sectional area) to reduce
120 the loading stiffness and match the critical rate of frictional weakening K_c
121 (e.g., Leeman et al., 2016; Scuderi et al., 2017). The acrylic block was used
122 only for experiments at < 15 MPa normal stress. It was installed after the
123 initial run-in and VS tests to produce labquakes during SHS tests.

124 To verify that this sample exhibited frictional behavior comparable to pre-
125 vious experiments on the same apparatus with Westerly Granite (e.g., Lee-
126 man et al., 2016), we conducted VS tests to estimate rate-and-state friction
127 parameters and K_c . The shear rate (load-point velocity) began at 10 $\mu\text{m/s}$,
128 then alternated between 3 and 30 $\mu\text{m/s}$ for three up-steps, with 500 μm of
129 shear at each velocity. At normal loads above 7.5 MPa, we omitted VS tests
130 because the fault did not sustain steady sliding and instead exhibited only
131 stick-slip, precluding reliable rate-and-state fits. The estimated rate-and-
132 state parameters (Supplementary Fig. S1) are consistent with those reported
133 in previous works on Westerly granite and are therefore not discussed further.

134 SHS tests were used to induce dynamic rupture at low normal loads,
135 where the conditions are otherwise in the stable sliding regime (Leeman et al.,
136 2016; Pignalberi et al., 2024). Each SHS sequence comprised 3, 30, and 300
137 s holds, followed in each case by 500 μm of shear at 10 $\mu\text{m/s}$. Repeating
138 the sequence 3 times produced 9 controlled stick-slip events per test. For
139 consistency, we focus our detailed analysis on these 9 post-hold events, and
140 use other events from our series of 15 tests to assess reproducibility and
141 frictional behavior. At 15 and 20 MPa, we did not use the acrylic loading
142 block (spring in Fig. 1a) because at these normal stresses the system satisfies
143 the conditions for instability and exhibits stick-slip motion and labquakes.
144 For each stick-slip event, we measured the total shear stress drop $\Delta\bar{\tau}_{\text{total}}$,
145 total slip δ_0 , loading stiffness k taken from the linear segment of $\Delta\mu/\Delta\delta$
146 prior to failure, and the breakdown work $W_b = \int_0^{\delta_0} (\bar{\tau} - \bar{\tau}_{\text{min}}) d\delta$, computed
147 from the center-block displacement δ , shear load $\bar{\tau}$, and normal load $\bar{\sigma}_N$ (see
148 Fig. 2).

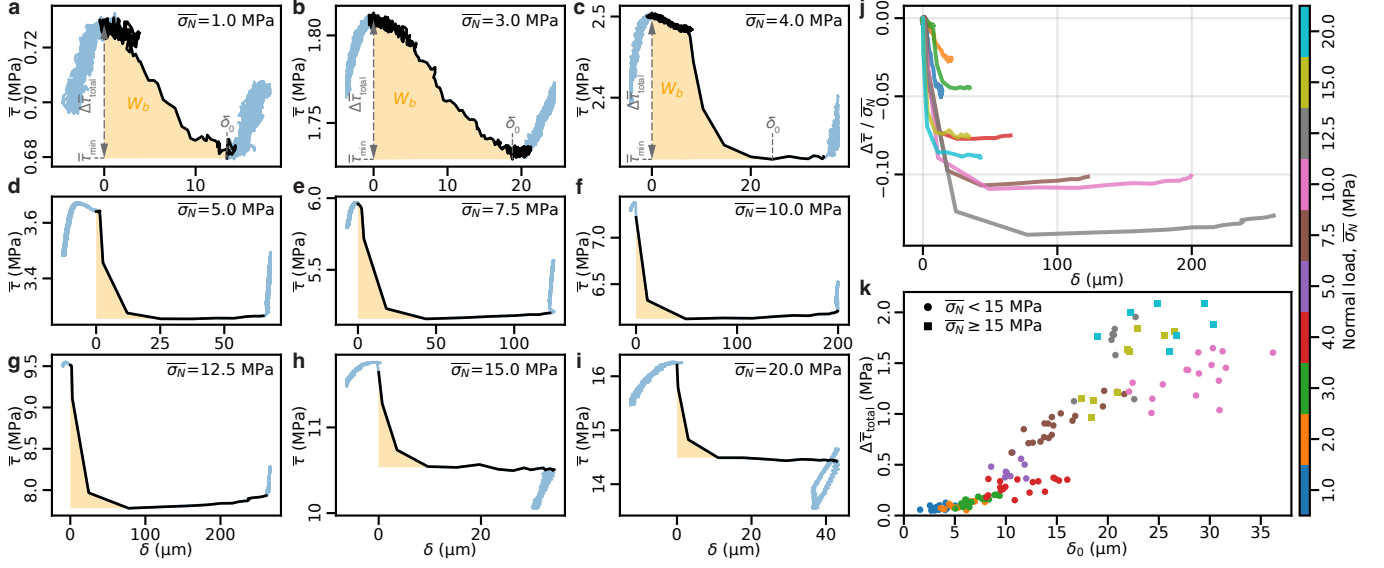


Figure 2: Dynamic stress drop and slip characteristics of laboratory earthquake events under different normal loads. (a–i) Shear load $\bar{\tau}$ versus fault displacement δ for events following SHS tests with 3-second holds, at $\bar{\sigma}_N$ from 1 to 20 MPa (the event at 2 MPa is excluded because of its similarity to the 1 and 3 MPa cases). Black sections denote the breakdown portion of stress drop, blue curves show the loading and post-slip history, and orange shading indicates breakdown work W_b , integrated from $\delta = 0$ to the displacement δ_0 at which $\bar{\tau}$ reaches its minimum $\bar{\tau}_{\min}$ and breakdown ceases. The total stress drop $\Delta\bar{\tau}_{\text{total}}$, the weakening displacement δ_0 , and the minimum shear stress $\bar{\tau}_{\min}$ are marked in panels (a–c). (j) Normalized shear stress changes $\Delta\bar{\tau}$ from events (a–i) overlaid for comparison, including the 2 MPa event for completeness. (k) Total stress drop $\Delta\bar{\tau}_{\text{total}}$ versus weakening displacement δ_0 for all SHS events, where $\Delta\bar{\tau}_{\text{total}}$ is the difference in $\bar{\tau}$ between $\delta = 0$ and $\delta = \delta_0$. $\Delta\bar{\tau}_{\text{total}}$ and δ_0 show an overall linear relationship, with a small break for events with $\bar{\sigma}_N \geq 15$ MPa, where the loading stiffness was increased by removing the acrylic loading block (Fig. 1a).

2.3. Rupture Speed Estimation and Cohesive Zone Modeling

To estimate rupture speed, we first determine the rupture-front arrival time at each gauge location. The rupture front is conventionally identified with the peak in shear strain ε_{xy} (Svetlizky and Fineberg, 2014; Paglialunga et al., 2022). In our experiments, however, a distinct peak is not always present, because the fault length (150 mm) is comparable to the critical nucleation length ($h^* \approx 0.5$ m in Ke et al., 2018; McLaskey, 2019), so the dynamic rupture often propagates through regions that are still actively nucleating and the characteristic peak is suppressed at those locations. We

therefore define the rupture-front arrival as the onset of the rapid acceleration of the ε_{xy} drop (Ohnaka and Shen, 1999), which coincides with the peak where one is present and remains well defined where it is not. We developed an open-source Python GUI tool (Labquake Explorer; Ke, 2024) to pick rupture-front arrival times at each ε_{xy} measurement position. Arrival times and corresponding fault locations were then fit with linear regression to estimate rupture speed C_f , using positions near the fault center where velocity remained approximately constant (Fig. 1c).

Beyond rupture speed, characterizing the weakening process and fracture energetics requires fitting our dynamic strain measurements using the cohesive zone model, which regularizes the stress singularity inherent in LEFM (Paglialunga et al., 2022; Kammer and McLaskey, 2019; Poliakov et al., 2002; Samudrala et al., 2002). In this formulation, the dynamic stress/strain fields near the tip of a propagating mode II (in-plane shear) rupture are governed by the rupture speed C_f , fracture energy Γ , cohesive zone size X_c , and off-fault distance y . The near-tip stress change is given by Freund (1990)

$$\Delta\tau(\theta, r, C_f) = \frac{K_{II}(C_f)}{\sqrt{2\pi r}} \Sigma_{xy}^{II}(\theta, C_f), \quad (1)$$

where K_{II} is the mode-II stress intensity factor and Σ_{xy}^{II} its angular variation. The fracture energy Γ is related to K_{II} through the dynamic energy release rate (Kammer and McLaskey, 2019)

$$\Gamma = G_{II} = \frac{1 - \nu^2}{E} K_{II}^2 f_{II}(C_f), \quad (2)$$

where $f_{II}(C_f)$ is a known universal function of crack speed. The cohesive zone introduces a linear slip-weakening stress distribution in local coordinates $\xi = x - x_{\text{tip}}$ (Poliakov et al., 2002; Samudrala et al., 2002)

$$\Delta\tau(\xi) = \begin{cases} 0, & \xi < -X_c \\ (1 + \xi/X_c)(\tau_p - \tau_r), & -X_c < \xi < 0, \end{cases} \quad (3)$$

where τ_p and τ_r are the peak and residual shear stresses. Together Eqs. (1)–(3) predict the full $\Delta\tau(t)$ and $\Delta\varepsilon_{yy}(t)$ time histories at off-fault distance y . An example fit is shown in Fig. 3b–c.

The fitting window is chosen to isolate the local tip-weakening phase from the later system-scale unloading, as described below. Because the fault

length is limited to 150 mm, the rupture front traverses the fault rapidly ($\sim 150 \text{ mm} \div 1 \text{ km/s} = 0.15 \text{ ms}$), while reduction of τ persist with slip (Fig. 3b). This two-phase behavior reflects distinct energy release mechanisms: initial rapid weakening associated with local energy dissipation at the propagating rupture tip, followed by continued stress reduction driven by nonlocal elastic unloading of the fault blocks and loading column, including cases where the center block of the DDS arrangement undergoes nearly-rigid-body motion. This distinction is evident in Fig. 3d, where the initial drop in shear strain change $\Delta\varepsilon_{xy}$ propagates at a constant rate (green dashed line), whereas subsequent features appear nearly simultaneous across all locations in comparison (black dotted lines). To isolate the local rupture-tip energy dissipation from system-scale effects, we fit the cohesive zone model to only the initial weakening phase in $\Delta\varepsilon_{xy}$ and $\Delta\varepsilon_{yy}$ (Fig. 3b–c), while excluding the later features that represent nonlocal elastic unloading. For each event, we restricted the fits to the center gauge rosette ($x = 75 \text{ mm}$), where both ε_{xy} and ε_{yy} are available, thereby enhancing the robustness of the fit. In the fitting procedure, we fix C_f based on estimates from multiple gauge rosettes and set $y = 8 \text{ mm}$ (fault to gauge center distance), then define the time window for the initial $\Delta\varepsilon_{xy}$ drop and jointly fit $\Delta\varepsilon_{xy}$ and $\Delta\varepsilon_{yy}$ for X_c and Γ (Supplementary Text S1).

3. Results and Discussion

3.1. Breakdown of Frictional Strength During Stick-Slip Events

We observe systematic changes in labquake characteristics over the normal stress range from 1–20 MPa (Fig. 2 and Supplementary Fig. S3). At $\overline{\sigma}_N < 4 \text{ MPa}$, labquakes occurred only immediately after a hold, as the apparatus stiffness is slightly higher than K_c even with the acrylic loading block in place. In contrast, at $\overline{\sigma}_N \geq 4 \text{ MPa}$ (Fig. 2c–g), labquakes occurred repeatedly and $\Delta\bar{\tau}_{\text{total}}$ increased with δ_0 (Fig. 2k). Following the sudden drop, $\bar{\tau}$ remained at the residual level as δ increased, before rising again. This residual sliding phase appears to lengthen with increasing $\overline{\sigma}_N$ while the acrylic loading block is in place (Fig. 2j), but we note that for $\overline{\sigma}_N < 4 \text{ MPa}$ we do not observe residual, post-event sliding. For events at $\overline{\sigma}_N \geq 15 \text{ MPa}$ (Fig. 2h–i), where the apparatus stiffness is significantly higher, the center block of the DDS arrangement overshoots such that δ exceeds the product of event duration and prescribed load-point shear rate, which caused the servo system to retract and $\bar{\tau}$ to drop further (see blue segment near the end). The

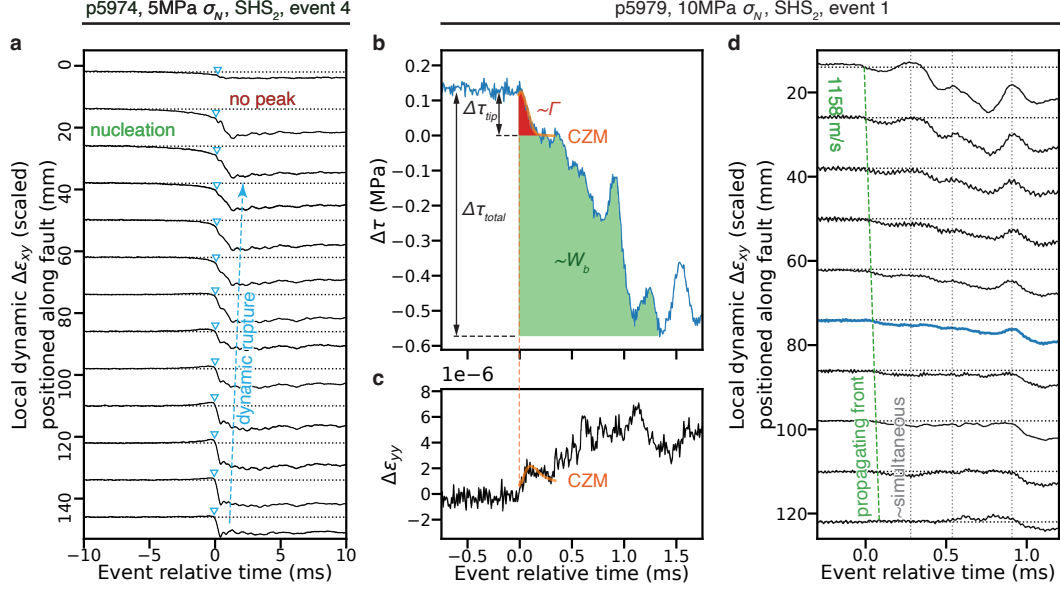


Figure 3: Dynamic strain records of lab earthquakes. (a) Shear strain change $\Delta\epsilon_{xy}$ time history along the fault for one event with upward dynamic rupture propagation, selected because nucleation and dynamic propagation occur close enough in time to visualize both phases together. Rupture nucleates near $x = 20$ mm and quasi-statically propagates downward, loading the bottom of the fault near $x = 150$ mm, until dynamic rupture initiates at the bottom and propagates upward. Triangles mark the picked rupture-front arrival at each gauge. Note that at some locations, the dynamic strain records do not exhibit distinct peaks because the rupture front passes through regions that are still actively nucleating (see Section 2.3). Panels (b–d) show a second, separate event that initiates near the top and propagates downward. (b) Shear stress change $\Delta\tau = G\Delta\epsilon_{xy}$ recorded by the strain gauge rosette at $x = 75$ mm. The trace is annotated with the local total stress drop $\Delta\tau_{\text{total}}$ (see panel d) and the stress drop at the rupture tip $\Delta\tau_{\text{tip}}$ obtained from the cohesive zone model (CZM, orange curve) fit. The red shaded area corresponds to the near-front dissipation I' , while the green area (which includes I') corresponds to the breakdown work W_b . (c) Normal strain change $\Delta\epsilon_{yy}$ recorded by the strain gauge rosette at $x = 75$ mm, as in (b). The trace agrees closely with the CZM fit (orange curve) within the fitted window and deviates afterward. Note that the sign of $\Delta\epsilon_{yy}$ is opposite that of $\Delta\epsilon_{xy}$, as expected for this geometry. (d) Shear strain change $\Delta\epsilon_{xy}$ versus time at each gauge location along the fault. The blue curve at $x = 75$ mm corresponds to the trace shown in (b). The onset of shear strain drop propagates downward along the fault (green dashed line), while subsequent strain changes (black dotted lines) appear nearly simultaneous across all locations in comparison, indicating elastic waves generated by near-rigid-body motion of the center block.

stress-slip curve shows an inflection at the onset of the stress drop (visible in Fig. 2a–d), reflecting a two-stage weakening in which one fault begins to slide while the other remains locked briefly before both faults slip together (see Supplementary Fig. S4). Although this behavior can cause overestimation of W_b , the effect is more pronounced at lower $\overline{\sigma}_N$ and therefore does not significantly affect the derived scaling.

3.2. Experimental Scaling of Rupture Dynamics

Our labquakes reveal clear scaling relationships between fault rupture properties and the normal load $\overline{\sigma}_N$. Rupture propagation speed C_f increases systematically with increasing $\overline{\sigma}_N$, as shown in Fig. 4a. At the lowest $\overline{\sigma}_N$ (1–3 MPa), ruptures propagate slowly (a few hundred m/s), whereas at the highest $\overline{\sigma}_N$ (20 MPa), C_f reaches ~ 1.8 km/s, which is still well below the Rayleigh-wave speed limit ($C_R \approx 2.5$ km/s). For an edge crack in a semi-infinite solid, the mode-II stress intensity factor scales as $K_{II}^S \propto \Delta\tau \propto \overline{\sigma}_N$ (Tada et al., 2000; Bayart et al., 2015; Ke et al., 2018), implying a static energy release rate

$$G_{II}^S \propto \overline{\sigma}_N^2. \quad (4)$$

Laboratory studies further suggest that fracture energy scales linearly with normal stress (Ke et al., 2018; Brodsky et al., 2020),

$$\Gamma \propto \overline{\sigma}_N. \quad (5)$$

Substituting these relations into the LEFM rupture speed expression (Svetlizky et al., 2017; Kammer et al., 2024),

$$C_f \approx \left(1 - \frac{\Gamma}{G_{II}^S}\right) C_R, \quad (6)$$

yields

$$C_f \propto 1 - \frac{c_0}{\overline{\sigma}_N}, \quad (7)$$

indicating that rupture speed should follow a hyperbolic approach to C_R as $\overline{\sigma}_N$ increases (dotted curve in Fig. 4a). In contrast, our data show only a modest acceleration with $\overline{\sigma}_N$, indicating that rupture speed cannot be explained by the standard LEFM scalings $G_{II}^S \propto \overline{\sigma}_N^2$ and $\Gamma \propto \overline{\sigma}_N$. This discrepancy has two potential explanations: 1) An additional mechanism beyond standard energy release: if G_{II}^S is treated as a quadratic function of $\overline{\sigma}_N$,

248 i.e., $G_{II}^S = c_1 \overline{\sigma}_N^2 + c_2 \overline{\sigma}_N$ with fitting constants c_1 and c_2 , the resulting form
 249 $C_f \propto 1 - 1/(c_1 \overline{\sigma}_N + c_2)$ provides a much closer fit to our data (dashed curve
 250 in Fig. 4a). 2) The 150 mm fault length may not provide sufficient distance
 251 for ruptures to fully accelerate, particularly at higher $\overline{\sigma}_N$ where the potential
 252 C_f is greatest.

253 As shown in Fig. 4b, the total shear stress drop $\Delta \overline{\tau}_{\text{total}}$ increases with
 254 $\overline{\sigma}_N$, scaling approximately as $0.1 \overline{\sigma}_N$, consistent with previous studies (e.g.,
 255 Johnson and Scholz, 1976; Karner and Marone, 2000; Leeman et al., 2016;
 256 Dong et al., 2023; Lyu et al., 2019). Following Shreedharan et al. (2020,
 257 Fig. 1b), who systematically varied shear loading stiffness k , we also find
 258 that $\Delta \overline{\tau}_{\text{total}}$ is larger at lower k (Fig. 4c). The outlier cluster corresponds
 259 to experiments at $\overline{\sigma}_N$ of 15 and 20 MPa, where the loading stiffness was
 260 significantly higher in the absence of the acrylic loading block. These results
 261 indicate that the stress drop magnitude following complete fault unlocking
 262 is governed by the stiffness of the surrounding medium (which, in our case,
 263 includes the apparatus and shear loading column).

264 3.3. Breakdown Work Scaling

265 Breakdown work W_b follows power-law scaling with the weakening dis-
 266 placement δ_0 , $W_b \propto \delta_0^2$ (Fig. 4d), consistent with patterns observed for nat-
 267 ural earthquakes (Abercrombie and Rice, 2005; Viesca and Garagash, 2015;
 268 Cocco et al., 2023) and with laboratory measurements (Selvadurai, 2019). A
 269 best-fit power law with an exponent of 2 is consistent with scalings inferred
 270 for natural earthquakes (exponent range 0.67–2, Abercrombie and Rice, 2005;
 271 Cocco et al., 2023). The $\sim \delta_0^2$ scaling arises because $W_b \sim 0.5 \Delta \overline{\tau}_{\text{total}} \delta_0$ and
 272 $\Delta \overline{\tau}_{\text{total}}$ scales approximately linearly with δ_0 (Fig. 2j). This quadratic depen-
 273 dence is a direct consequence of the fixed dimensions of our laboratory fault,
 274 where slip is limited by the elastic relaxation of the surrounding compliant
 275 system and stress drop and slip therefore co-vary. Natural ruptures are not
 276 dimension-limited in the same way, so slip and stress drop need not co-vary
 277 as they do here, and the quadratic scaling we measure should not be expected
 278 to carry over to them (Udías et al., 2014; Nielsen et al., 2016). In both cases
 279 the breakdown work is set by the elastic energy released from the volume
 280 surrounding the rupture rather than by frictional processes at the rupture
 281 tip. In our length-limited fault the fixed dimension couples slip to stress drop
 282 and yields an exponent of two, as it would for a fault patch of roughly fixed
 283 size, such as an asperity or fault segment embedded in a more compliant
 284 surrounding volume. This is not a model for self-similar rupture growth,

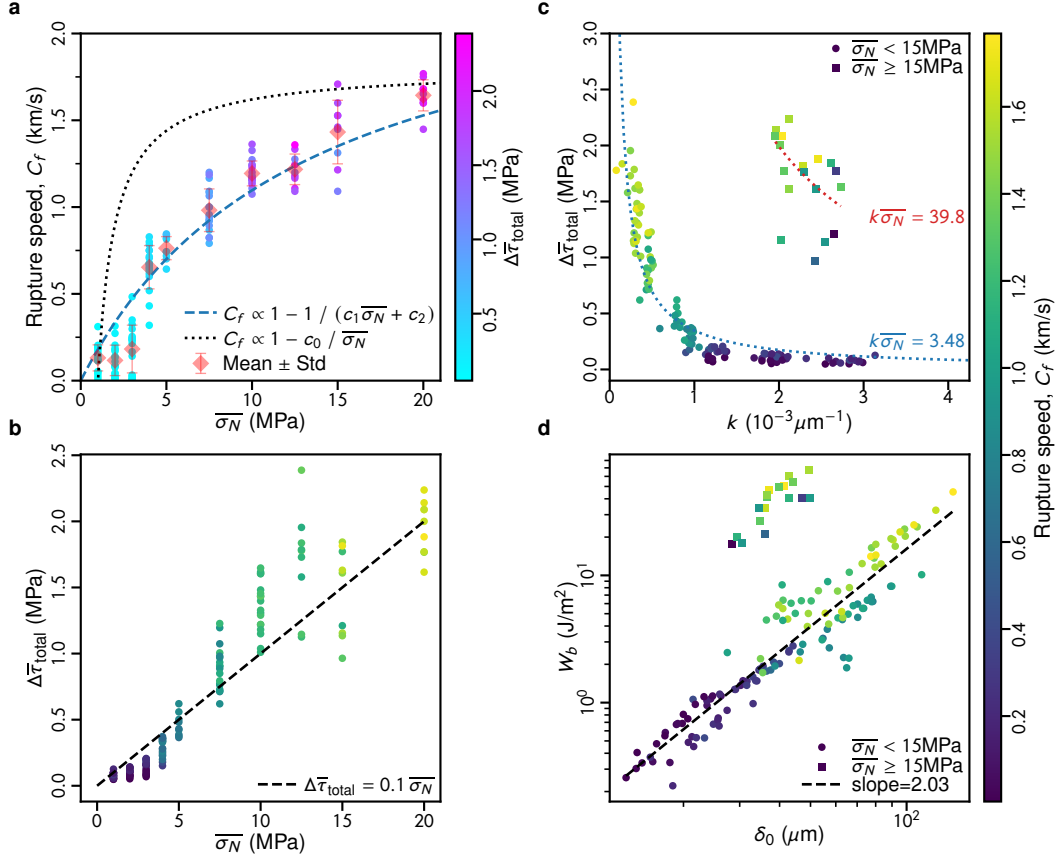


Figure 4: Scaling of rupture speed, stress drop, and breakdown work. (a) Rupture speed C_f increases systematically with normal stress $\bar{\sigma}_N$. The black dotted curve represents the fit $C_f \propto 1 - c_0/\bar{\sigma}_N$, while the dashed curve shows $C_f \propto 1 - 1/(c_1\bar{\sigma}_N + c_2)$. (b) Total stress drop $\Delta\bar{\tau}_{total}$ increases with normal load, with $\Delta\bar{\tau}_{total} \approx 0.1\bar{\sigma}_N$. (c) $\Delta\bar{\tau}_{total}$ versus normalized loading stiffness k shows a trend similar to that reported by Shreedharan et al. (2020), with lower stiffness yielding higher stress drop. The outlier cluster corresponds to events at $\bar{\sigma}_N \geq 15$ MPa conducted with a much higher stiffness. The dotted and the dashed curves indicate constant values of the stress-per-displacement stiffness $\bar{k} = k\bar{\sigma}_N$ (in MPa/mm), following the scaling $\Delta\bar{\tau}_{total} = 0.1\bar{\sigma}_N = 0.1\bar{k}/k$. (d) Breakdown work W_b scales with the weakening displacement δ_0 with an exponent of 2.03 (dashed line).

285 where the rupture dimension is not fixed and the stress drop is variable, so
 286 this nonlocal mechanism does not imply any single slip exponent, consistent
 287 with the broad range of exponents reported for natural earthquakes. In this
 288 restricted sense, laboratory faults with length-limited contacts reproduce key

289 aspects of natural fault behavior inferred from seismology.

290 Previous larger-scale laboratory earthquake studies focus on rupture dy-
291 namics before the rupture spans the entire fault, and thus this post-rupture
292 behavior has not been previously highlighted. In the DDS configuration,
293 large ruptures produce complete unlocking and subsequent rigid block mo-
294 tion, which unloads both the fault-surrounding material and the shear loading
295 column, thereby producing the strong W_b scaling.

296 3.4. Near-Tip Weakening from Cohesive Zone Modeling

297 The CZM (Section 2.3) fits well for the initial strain drop at the center
298 rosette ($x = 75$ mm) and it captures near-tip weakening (Fig. 3b–c). The
299 stress drop at the rupture tip $\Delta\tau_{\text{tip}}$ is significantly smaller than the total
300 stress drop and shows little dependence on $\bar{\sigma}_N$ (Fig. 5a), though $\Delta\tau_{\text{tip}}$ is
301 higher at lower $\bar{\sigma}_N$, likely reflecting increased uncertainty in distinguishing
302 tip from tail behavior during fitting. The cohesive zone size X_c varies only
303 weakly with $\bar{\sigma}_N$ (Fig. 5b), yet consistently exceeds the fault length. The
304 cohesive zone model remains a valid elastic description of the near-tip strain
305 fields, and X_c is a real length scale over which the rupture-front stresses
306 evolve. However, because X_c is comparable to or larger than L , there is no
307 region outside the cohesive zone yet close to the tip where the field is K -
308 dominated. The small-scale yielding condition is therefore not satisfied, the
309 fitted amplitude is an amplitude parameter of the solution rather than a true
310 stress intensity factor, and X_c should not be read as a literal cohesive zone.
311 Accordingly, we do not interpret the CZM-derived energy as a strict fracture
312 energy. We instead report it as an apparent near-front dissipation scale Γ' ,
313 obtained as the area under the τ – δ curve during initial weakening. This area
314 is constrained directly by the near-field gauge ($y \ll X_c$) and is insensitive to
315 the assumed weakening-law shape, providing a robust measure of near-front
316 dissipation for comparison with the breakdown work W_b . The two quantities
317 are obtained independently, W_b from the finite-geometry energy budget and
318 Γ' from the near-field strain record, and neither relies on the small-scale-
319 yielding assumption that fails here.

320 Our data show that the near-front dissipation Γ' increases systematically
321 with normal load (Fig. 5c). The scaling is roughly consistent with the frac-
322 ture energy $\Gamma \approx 0.2 \mu\text{m} \cdot \sigma_N$ reported by Ke et al. (2018), and our values
323 are comparable in magnitude to previous studies (Svetlizky et al., 2017; Ba-
324 yart et al., 2015; Ke et al., 2018; Kammer and McLaskey, 2019; Paglialunga
325 et al., 2022). However, Γ' remains an order of magnitude smaller than the

breakdown work W_b (Ke et al., 2022), underscoring that much of the energy release occurs late in the process, after the rupture front has passed, after local stresses have dropped, and after the rupture has reached its final dimensions. Notably, events at $\overline{\sigma_N} \geq 15$ MPa conducted without the compliant loading spring fall above the $W_b \propto \delta_0^2$ trend (Fig. 4d, squares) because their stress drop is high relative to their slip, yet show no corresponding departure in Γ' at these same normal stresses (Fig. 5c). This indicates that the W_b scaling arises from nonlocal elastic unloading of the loading system rather than from enhanced energy dissipation at the rupture tip.

Recent high-velocity experiments show that rupture dynamics, particularly slip acceleration and system stiffness, can strongly influence both transient weakening and breakdown work (Liao et al., 2014; Chen et al., 2025). Chen et al. (2025) demonstrated that friction evolution during dynamic rupture systematically deviates from flash-heating predictions, with breakdown work scaling aligning with natural earthquakes when acceleration effects are considered. Similarly, Liao et al. (2014) found that slip-pulse loading produces lower breakdown work than constant-velocity tests despite more intense weakening, attributing this to acceleration-enhanced gouge production. These studies suggest that system configuration and loading history can fundamentally influence energy partitioning during dynamic slip, consistent with our observation that loading stiffness modulates W_b while leaving Γ' unchanged. This indicates that the excess energy release in our experiments arises from off-fault elastic unloading rather than enhanced dissipation at the rupture tip through mechanisms such as flash heating (Viesca and Garagash, 2015). These observations motivate finite-element analysis to further evaluate how the static energy release rate G_{II}^S is partitioned and how it scales with $\overline{\sigma_N}$.

3.5. Finite Element Modeling for Estimating Energy Release Rate Scaling

Typical LEFM applications for lab earthquake experiments (e.g., Kammer et al., 2015; Bayart et al., 2015; Ke et al., 2018; Paglialonga et al., 2022) assume that the rupture is embedded in an infinite solid or semi-infinite half-space. While these approximations provide useful first-order insights, the finite size of lab fault zones, and in particular the fact that our fault length is comparable to the nucleation dimension, motivates a more detailed analysis. To evaluate the energy budget of our labquakes, we developed a finite element model in ABAQUS (Dassault Systèmes Simulia Corp., 2024), including the three granite blocks used for DDS, the steel plate on top of the center block

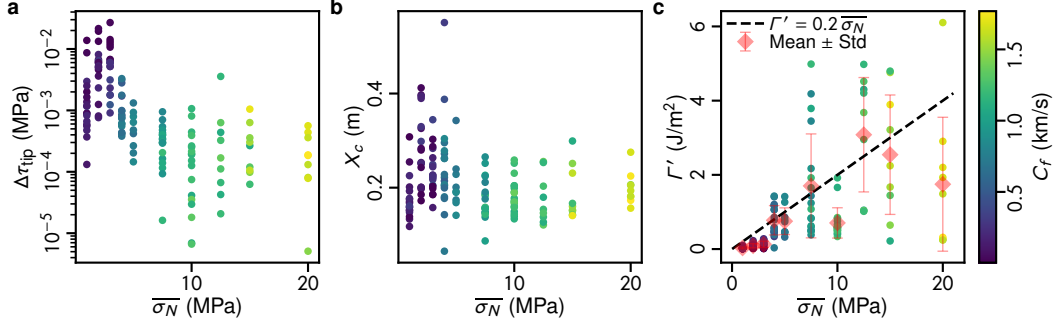


Figure 5: Cohesive-zone-model (near-tip) scaling. (a) Stress drop at the rupture tip $\Delta\tau_{\text{tip}}$, estimated from CZM fits, decreases slightly with $\bar{\sigma}_N$ and is considerably smaller than the total stress drop $\Delta\bar{\tau}_{\text{total}}$. (b) Cohesive zone size X_c shows little dependence on $\bar{\sigma}_N$ but mostly exceeds the 0.15 m fault length. (c) Near-front dissipation Γ' dependence on $\bar{\sigma}_N$ is broadly consistent in magnitude with the fracture-energy scaling $\Gamma \approx 0.2 \mu\text{m} \cdot \sigma_N$ reported by Ke et al. (2018, dashed line).

for stress distribution, and the acrylic block used to reduced loading stiffness for lower normal stress (Fig. 6).

Because our shear loading rate is negligible relative to the dynamic rupture speed, we prescribe a displacement at the top of the DDS loading column and gradually increase the rupture area in static steps by varying the rupture length a , assuming the rupture front is uniform across the fault width. The rupture area is simulated by frictional contact with $\mu = 0.7$, whereas the unruptured regions are tied together with a zero-relative-displacement boundary condition. Although this model does not capture the full dynamics of rupture propagation, it provides a first-order framework for evaluating the validity of LEFM assumptions in our experimental system and for estimating how G_{II}^S scales with σ_N .

We ran simulations ranging from $\bar{\sigma}_N$ from 2.5 to 20 MPa, and for each normal load considered rupture lengths a from 5 to 105 mm, assuming symmetric rupture on both frictional interfaces. The energy release rate was computed as $G^{S'} = -dU/dA$ using a central-difference approximation, where U is the total strain energy, $A = aw$ is rupture area, and $w = 50$ mm is the fault width. For comparison with the experiments, we take $G^{S'}$ at $a = 75$ mm (rupture front at fault center) as the representative value for each normal stress (Fig. 6e).

The simulations show that the loading column with the acrylic block

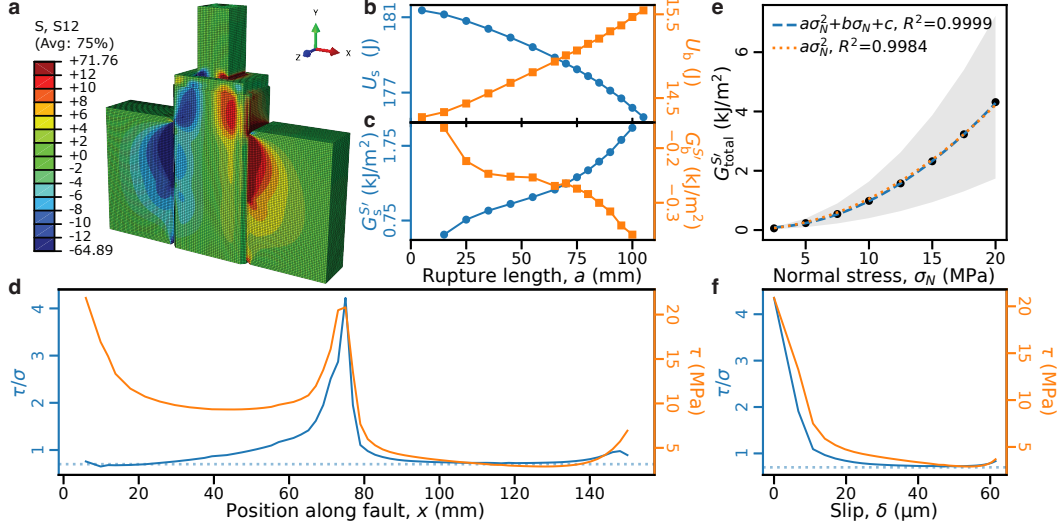


Figure 6: Finite element model results for a bottom-up rupture. (a) Shear stress field along the fault direction τ in the 3D finite element model. S12 denotes shear stress σ_{xy} in the model. (b) Strain energy stored in the acrylic spring U_s (blue) and in the granite blocks U_b (orange) as a function of rupture length a . (c) Energy release rate from the acrylic spring $G_s^{S'}$ (blue) and from the granite blocks $G_b^{S'}$ (orange) as a function of a . (d) Spatial distribution of the shear-to-normal stress ratio τ/σ_N (blue) and shear stress τ (orange) at a rupture length of 75 mm. Dashed line shows the prescribed target friction coefficient. (e) Total static energy release rate $G^{S'}$ as a function of $\overline{\sigma_N}$, overlaid with a quadratic polynomial fit (blue dashed) and a pure quadratic fit (orange dotted). The gray shaded area marks the range of $G^{S'}$ for rupture lengths $a \in [15, 100]$ mm (as shown in c), and the black dots indicate $a = 75$ mm. (f) Shear-to-normal stress ratio τ/σ_N (blue) and shear stress τ (orange) versus slip δ for $\delta > 0$ at all fault nodes at a rupture length of 75 mm. Dashed line indicates the prescribed target friction coefficient. See Supplementary Fig. S2 for the top-down rupture case.

384 stores more than an order of magnitude more strain energy than the granite
385 blocks (Fig. 6b). The corresponding energy release rates exhibit the same
386 partitioning, with $G_s^{S'}$ denoting energy released from the loading column
387 (subscript s) and $G_b^{S'}$ from the granite blocks (subscript b) (Fig. 6c). Inter-
388 estingly, the strain energy in the granite blocks increases with rupture area
389 ($A = aw$) rather than releasing energy ($G_b^{S'} < 0$, Fig. 6c) —*i.e.*, the blocks
390 continue to store energy as rupture grows. This behavior arises because re-
391 duced lateral restraint from the side blocks permits greater deformation of
392 the center block as rupture area expands. Thus, even though strain energy
393 is released at the rupture tips, the total strain energy of the blocks increases.

394 In the case with top-down rupture, the energy release rate in the two side
 395 blocks is positive (Supplementary Fig. S2), as expected, but remains consid-
 396 erably lower than the energy release rate from the loading column with the
 397 acrylic block ($G_s^{S'}$). These results confirm that elastic energy release from
 398 the loading column, rather than localized rupture-tip processes, governs the
 399 energy budget in our laboratory earthquakes, and that this energy release
 400 begins even before the rupture spans the entire fault and rigid-body motion
 401 of the center block commences.

402 At a representative rupture length of $a = 75$ mm, the total energy release
 403 rate ($G_{\text{total}}^{S'}$) increases quadratically with $\overline{\sigma}_N$ (Fig. 6e), showing no deviation
 404 from the simplified LEFM prediction $G_{II}^S \propto \overline{\sigma}_N^2$ (curves in Fig. 6e), despite
 405 the geometric and compositional complexity of our experimental setup. No-
 406 tably, because $G^{S'}$ is computed directly from the finite-geometry energy bud-
 407 get rather than from a stress intensity factor, this agreement does not rely on
 408 small-scale yielding (cf. Palmer and Rice, 1973). We write $G^{S'}$, with a prime
 409 as for the apparent near-front dissipation I' , to denote a finite-geometry
 410 energy release rate determined without a K -dominant field. The quadratic
 411 scaling is therefore robust rather than an artifact of the small-scale-yielding
 412 assumption underlying the analytical prediction, and it persists beyond the
 413 small-scale-yielding regime. Our finite-element model is static and is not
 414 intended to reproduce the rupture speed. It therefore does not capture the
 415 modest deviation of C_f from $C_f \propto 1 - c_0/\overline{\sigma}_N$ (Fig. 4a). Reproducing that
 416 deviation would likely require fully dynamic simulations with a friction law.
 417 Importantly, this deviation concerns only the rupture speed. It does not bear
 418 on our energy-budget analysis, from which we conclude that the breakdown-
 419 work scaling arises from system-scale elastic unloading.

420 Beyond the energy release rate scaling, the finite-element model also re-
 421 veals how spatial heterogeneity in normal stress affects fault behavior. Due
 422 to the finite geometry of our experimental setup, σ_N varies spatially along
 423 the fault. This heterogeneity causes τ and τ/σ_N to exhibit markedly differ-
 424 ent spatial patterns, with τ showing pronounced stress concentrations near
 425 rupture tips while τ/σ_N remains more uniform (Fig. 6d). During slip, these
 426 spatial variations in σ_N evolve, producing different slip-weakening behavior
 427 for the two quantities (Fig. 6f). Notably, $\tau(\delta)$ exhibits secondary weakening,
 428 a gradual stress decrease following the initial sharp drop, that is much less
 429 pronounced in $\tau/\sigma_N(\delta)$. This occurs because the evolving σ_N distribution
 430 produces curved τ - δ behavior even when the friction coefficient μ remains
 431 uniform. Consequently, interpreting the measured τ under the implicit as-

432 sumption of a constant $\overline{\sigma_N}$, as is common in experimental studies, can overes-
 433 timate fracture energy Γ by misattributing this curvature to enhanced dissi-
 434 pation at the rupture tip (e.g., Cowie and Scholz, 1992). Thus, system-scale
 435 elastic unloading and finite-geometry effects can mimic signatures of complex
 436 weakening mechanisms, contributing to the observed breakdown work scaling
 437 without requiring additional energy dissipation near the rupture front.

438 3.6. Energy partitioning and the fault thermal budget

439 A natural question is where the released elastic energy ends up, and
 440 whether the strong scaling of breakdown work implies correspondingly strong
 441 frictional heating and thermally enhanced weakening on the fault (e.g., Nielsen
 442 et al., 2021; Cocco et al., 2023). While we do not measure temperature di-
 443 rectly, mechanical work done against friction is expected to be largely con-
 444 verted to heat (Aretusini et al., 2021, 2024; Ortega-Arroyo et al., 2025), so
 445 most of the released energy likely ends up as frictional heat on the fault. Tak-
 446 ing an event at $\overline{\sigma_N} = 20$ MPa as an example, the work done against friction is
 447 $\int \bar{\tau} d\delta \approx \bar{\tau} \delta \approx 6 \times 10^2 \text{ J m}^{-2}$, and the breakdown work $W_b = \int (\bar{\tau} - \bar{\tau}_{\min}) d\delta$
 448 is the roughly ten percent of this work done above the minimum sliding stress
 449 (Fig. 4d). The near-front dissipation $\Gamma' \approx 10 \text{ J m}^{-2}$, the energy dissipated
 450 locally during the initial weakening at the onset of slip, is smaller still. How
 451 W_b itself partitions among additional fault heating, kinetic energy of the
 452 near-rigidly translating center block (Fig. 3d), and radiated waves is not di-
 453 rectly constrained by our measurements, but each of these sinks is modest.
 454 Spread through the ~ 0.75 g gouge layer, the total frictional heat corresponds
 455 to a temperature rise of only $\Delta T = \bar{\tau} \delta A / (m c) \approx 8 \text{ K}$, where A is the fault
 456 area, m is the gouge mass, and $c \approx 750 \text{ J kg}^{-1} \text{ K}^{-1}$ is the specific heat of
 457 quartz. This is two orders of magnitude below the rises required to activate
 458 flash heating, thermal pressurization, or melting (e.g., Viesca and Garagash,
 459 2015; Harbord et al., 2021), so enhanced thermally activated weakening is
 460 neither expected nor observed. The scaling of W_b therefore does not indi-
 461 cate that rupture requires more energy to propagate, which is the role of
 462 Γ . It indicates that the surrounding system, including the loading column,
 463 releases more energy as the fault unlocks, energy that ends up as modest
 464 heat, radiated waves, and kinetic energy.

465 3.7. Implications for Natural Faults

466 Our finding that nonlocal elastic unloading governs the breakdown work
 467 scaling invites comparison with natural faults. Strong breakdown work scal-

ing with earthquake size is inferred from seismology (Abercrombie and Rice,
 2005; Tinti et al., 2005) and is commonly attributed to enhanced near-fault
 weakening such as thermal pressurization (Viesca and Garagash, 2015; Wib-
 berley and Shimamoto, 2005). Our results point to an alternative, in which
 fault areas surrounding a high-strength asperity, though not in direct con-
 tact, store and release the elastic energy that drives this scaling. In this view,
 a finite-length laboratory fault embedded in a compliant loading system is
 analogous to a small asperity embedded in a larger elastic medium, where
 nonlocal unloading rather than enhanced weakening produces the scaling.
 Such an effect would be absent from laboratory and numerical models that
 assume perfectly mated fault interfaces. Our data therefore suggest that ac-
 counting for fault topography and contact heterogeneity, alongside frictional
 properties, may be essential for scaling rupture energetics from the laboratory
 to tectonic faults.

4. Conclusions

Our laboratory stick-slip experiments reveal that rupture speed C_f , stress
 drop $\Delta\bar{\tau}_{\text{total}}$, and breakdown work W_b scale systematically with normal load
 $\bar{\sigma}_N$. While these data are consistent with basic expectations they are not
 fully aligned with standard assumptions of LEFM. Although the near-front
 dissipation Γ' grows approximately linearly with $\bar{\sigma}_N$, it remains an order of
 magnitude smaller than W_b , indicating that most energy release arises from
 system-scale unloading, after the rupture has reached the fault boundaries.
 This later slip is due to energy release of the shear loading column rather
 than from localized weakening at the rupture tip. Finite-element analysis
 shows that the static energy release rate $G_{\text{II}}^{S'}$ follows the quadratic scaling
 predicted by LEFM despite the geometric complexity of the experimental
 setup, while also revealing that the compliant loading column dominates the
 energy budget even before full sliding occurs. Cohesive zone modeling fur-
 ther indicates that the cohesive zone size X_c exceeds the sample dimensions,
 violating the small-scale yielding assumption even though the model fits the
 dynamic strain data reasonably well.

Our results highlight the role of finite, length-limited fault patches in
 governing rupture energetics. Together, these results identify nonlocal elastic
 unloading of the loading system, rather than enhanced near-tip dissipation,
 as the primary control on the breakdown-work scaling we observe.

503 **CRedit authorship contribution statement**

504 **C.-Y. Ke:** Writing - original draft, Methodology, Software, Validation,
505 tion, Formal analysis, Investigation, Data Curation, Visualization, Supervision,
506 Project administration, Conceptualization, Funding acquisition. **G.**
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513 **Declaration of competing interest**

514 The authors declare that they have no known competing financial interests
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532 **Supplementary materials**

533 Supplementary material associated with this article can be found, in the
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535 **Data availability**

536 Data will be made available on request.

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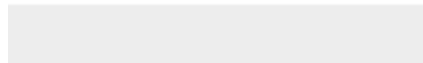




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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: